

# New Models for Epistemic Modal Logic in Simplicial Complexes and Their Applications

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## Abstract

In this thesis we set out to develop some novel models for epistemic modal logic in simplicial complexes. The thesis is largely broken down into three standalone papers. In the first paper, Chapter 2, we develop a novel semantics for belief using simplicial complexes. In order to ensure that the axiom **T** is not sound, we allow that each agent  $a$  have their own simplicial complex  $S_a$ , and use this to determine the agents beliefs at a given world or facet. We then show that this model for belief has advantages over previous approaches.[11, 14]

In the second paper, Chapter 3, we modify the semantics for belief from Chapter 2 so that atomic formulae are assigned to the nodes of the complex, rather than the facets. This is more in line with what we call the “perspective-perspective”, which we take to the epistemic underpinning for simplicial semantics. Since perspectives map onto the nodes of a simplicial complexes, possible worlds, which are sets of perspectives, are thus themselves “made of” perspectives. We then show that simplicial models offer an excellent model for “Belief Revision”, as originally presented by David Lewis. [24] This is because if worlds are “made of” perspectives, we can say that two worlds are more or less “similar” from each other depending on how many perspectives they have in common. We offer some examples of this notion of revision in practice, as well as independently conceptually and philosophically motivate this notion of “similarity”

In the third paper, Chapter 4, we explore some of the more concrete applications of simplicial semantics to the realm of distributed computing. In particular, we give a definition for “action models”, in the sense of [5], fit for our semantics. Using these definitions of actions, we then prove for a simplicial model (for knowledge)  $I$ , and an action  $I[A]$ , there is a function on the faces of  $I$ , called  $\Xi$ , to sets of faces in  $I[A]$  which defines a protocol in the sense of [23] We then give a notion of action which applies to the models for belief presented in Chapter 3. With a slight, but well motivated, modification to the definition of protocol, which we call a “belief protocol”, we can show that actions in the belief setting give rise to belief protocols. We conclude by showing that incorporating the notion of revision from Chapter 3 into the definition of action also gives rise to a “belief protocol”.

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# 1 Introduction

Imagine you are attempting to figure out whether or not your friend is certain about the truth of some fact  $P$ . How would you go about trying to imagine this, and resolving it? Presumably, you would do something akin to “placing yourself in their shoes”, that is, you would try to imagine what the world looks like from their point-of-view, or “perspective”. More specifically, you would attempt to discern if there’s a way-the-world-is which they are considering where  $P$  is true, and/or a way-the-world-is which they are considering where  $P$  is false. In particular, maybe you are trying to determine whether or not they have seen  $P$  directly. This leaves you with three possible options: Either your friend is considering a single world possible where  $P$  is true, a single world where  $P$  is false, or one of each kind.

For the modal logician, there are two key insights from a story like the above. The first is to note that we typically seem to consider *perspectives* before we consider *possible worlds*. In the above example, we introduce worlds, or “ways-the-world-is”, as a means of accounting for the content of your friend’s perspective. In this way, perspectives *constrain* possible worlds. When modeling a particular scenario, the modal logician tends to introduce worlds in order to account for the content of possible perspectives. This informal approach to modal logic, which takes perspectives as primary, we will call the “perspective-perspective”. We flesh this idea out more fully in Chapter 3.

The second key insight concerns the idea that your friend could be observing whether or not  $P$  is the case in a direct sense, such as if they are observing for themselves whether or not  $P$  is the case, or they could have a belief about the truth of  $P$  in a more indirect sense. Suppose your friend were observing the truth of  $P$  for themselves. As we articulated before, this leads the modeler to the three choices described above: Create a single world where  $P$  is true, a single world where  $P$  is false, or one of each kind. However, suppose instead you are imagining that your friend is not in a position, and never was, to observe the truth value of  $P$  directly. This does not actually impact your options as the modeler. You’re still left with the same three choices. Hence, possible worlds on their own cannot cache out the following distinction: Your friend believes  $P$  because they’ve observed it directly, or your friend believes  $P$  because of some secondary information which confirmed  $P$  for them. As we hope to demonstrate in this thesis, some formal interpretations of the “perspective-perspective” can in fact not only make sense of this distinction, but leverage it. Again, we will see such content leveraged in Chapter 3.

We believe that the setting of “Simplicial Semantics” for modal logic affords the modeler tools which account for these two insights. Interpreting multi-agent epistemic logic using simplicial complexes is a recent and thriving area of research. [11, 14, 18, 17, 21, 19, 20, 28] This research largely has its origins in the computer science literature, and attempts to connect existing models for distributed protocols that use combinatorial topology, such as those given in [23], with models for protocols given in graphs or frames which interpret epistemic modal logic [16]. This is done by interpreting an uncertainty relation directly on the simplicial model, whereby an agent cannot tell apart two “worlds” if they have the same “perspective” in each of those worlds.

For those aware of the formalism already, we can say a few things. There are many definitions in the literature that apply combinatorial topology to distributed computing which already appeal to the epistemic intuitions afforded by the simplicial semantics. For example, all of the various definitions of “protocol” given in [23] appeal to an agent being unable to “tell apart” two worlds, or in particular, two “facets”. The inability for a particular agent to tell apart two facets is identified in these definitions with the agent having the same perspective in or at both facets. This is, of course, the exact mechanism behind the “uncertainty” relation in simplicial semantics, though the actual details of the semantics are not given in this textbook. [23] In the Chapter 3, we hope to give a philosophical story that grounds these intuitions in the “perspective-perspective”.

This thesis is composed primarily of three standalone papers, being chapters 2, 3, and 4, and will proceed as follows. In Chapter 2, “Belief in Simplicial Complexes”, we offer a modified semantics for epistemic modal logic that allows simplicial semantics to compellingly

model “belief” as opposed to “knowledge”. In particular, most previous models of epistemic modal logic have made sound the axiom of “factivity”, or **T**. [14, 21, 19, 20, 28] This axiom states that if an agent knows  $\varphi$  to be true, then  $\varphi$  is true. If one replaces “knowledge” with “belief”, however, this axiom becomes unreasonable. Agents of course can have false beliefs. Our attempt at modifying simplicial semantics to model belief is not the first. [11, 14] However, in Chapter 2, we show that our attempt has certain conceptual and formal advantages over previous models.

Chapter 3, “Simplicial Semantics for Belief Revision”, will then attempt to leverage this model of belief in novel ways. We begin by articulating the conceptual foundation for simplicial models, that is, the “perspective-perspective”. Once this philosophical story is understood, the modeler can then see that, in simplicial semantics, the formal object associated with worlds, namely “facets”, is itself composed of agent perspectives.<sup>1</sup> If worlds are made of perspectives, so to speak, one can then say that two worlds are more “similar” if they share more perspectives. As an intuition, we are not the first to hint at this. Again, the textbook “Distributed Computing for Combinatorial Topology” suggests that one can intuitively appeal to this notion of “similarity”, though does not leverage it in any specific way.<sup>2</sup> We appeal to this notion of “similarity” to give a new semantics for “Belief Revision”, as defined in David Lewis’ seminal paper “Probabilities of Conditionals and Conditional Probabilities”. [24] We believe that this novel notion of “nearness” solves one of the chief difficulties in applying belief revision. Namely, it is often unclear which worlds should be nearer or further from each other in a practical modeling situation. As such, the fact that the perspective-perspective gives the modeler a notion of nearness “for-free” is a clear and specific advantage of modeling uncertainty in this setting.

Chapter 4, “Simplicial Actions”, concludes by exploring some of the more concrete applications of simplicial semantics. We first define “action models” both in the context of simplicial models for knowledge and our simplicial models for belief. While, again, this is not necessarily the first attempt at defining action models for simplicial models, there are two ways in which our approach will be novel. Firstly, our models of knowledge are more general in a few ways, and so our action models will need to be more general as well to account for this. More significantly, however, no attempt has yet been made to connect a simplicial model *for belief* and action models. Once these definitions of actions are given, we then prove that an action model, either for belief or knowledge, specifies a “protocol” in the sense of [23]. We also present examples of known, simple distributed protocols and show how one can translate them into simplicial action models. We then conclude by showing how one can incorporate the notion of revision from Chapter 3 into these action models, giving us a notion of protocol where agents learn via revision rather than via conditioning. We also prove that such action models are protocols in the sense of [23].

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<sup>1</sup>In our model, as is typical, we assume exactly that a facet contains one perspective for each agent.

<sup>2</sup>And, of course, they do not connect this notion of similarity to simplicial semantics itself, as they do not entertain the latter formalism at all.

## 2 Belief In Simplicial Complexes

This first paper was coauthored between Prof. Adam Bjorndahl and Philip Sink. It has been accepted for publication by the journal “Advances in Modal Logic”.

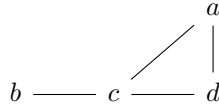
### Abstract

We provide novel semantics for belief using simplicial complexes. In our framework, belief is a KD45 modality that satisfies “knowledge implies belief” (“If you know  $\phi$ , then you believe  $\phi$ ”); in addition, we adopt the (standard) assumption that each facet in our simplicial models contains exactly one vertex for each agent. No existing model of belief in simplicial complexes that we are aware of is able to satisfy all of these conditions without trivializing belief to coincide with knowledge. We establish a truth-preserving correspondence between our simplicial framework and standard relational models for knowledge and belief (including, notably, improper relational models), and use this to provide a simple axiomatization.

### 2.1 Introduction

Interpreting multi-agent epistemic logic using simplicial complexes is a recent and thriving area of research. [11, 14, 18, 17, 21, 19, 20, 28] Our goal in this chapter is to extend this approach to accommodate not only knowledge but also *belief*; this turns out to involve some subtleties, the resolutions of which are interesting in their own right.

To set the stage, we begin by briefly reviewing the basic notion of a simplicial complex and the intended epistemic interpretation. Informally, a *simplex* can be pictured as a multi-dimensional triangle; that is, a collection of vertices where every vertex is connected to every other vertex by an edge. So with three vertices we obtain a standard triangle, four gives a tetrahedron, and in general,  $n$  many vertices gives an  $(n - 1)$ -dimensional triangle. A *simplicial complex* can be thought of as a collection of simplices with some perhaps sharing vertices, edges, or larger faces. The simple example below depicts a 1-dimensional simplex ( $bc$ ) and a 2-dimensional simplex ( $acd$ ) glued together at a single vertex ( $c$ ):



Formally, a **simplicial complex** is a set  $S$  closed under subsets; that is, if  $X \subseteq Y \in S$ , then  $X \in S$ . Intuitively,  $S$  is the set of all triangles in the simplicial complex. Closure under subsets corresponds to the idea that any triangle (of any dimension) contains all of the lower-dimensional triangles that make it up; for example, a 2-dimensional triangle contains its edges (which are 1-dimensional triangles), and these edges contain their endpoint vertices (which are 0-dimensional triangles). Accordingly, every element  $X \in S$  is called a *face*, the elements in such an  $X$  are called *vertices*, and those faces  $X \in S$  that are maximal with respect to set inclusion are called *facets*. For example, the diagram above corresponds to the simplicial complex

$$S = \{\{b, c\}, \{a, c, d\}, \{a, c\}, \{a, d\}, \{c, d\}, \{b\}, \{a\}, \{c\}, \{d\}, \emptyset\},$$

which has facets  $\{b, c\}$  and  $\{a, c, d\}$ .

The basic idea behind interpreting epistemic logic using simplicial complexes is to associate vertices with agents and to view facets as analogous to possible worlds [14, 19]. Intuitively, a vertex associated to agent  $a$  represents a particular “perspective” of  $a$ , with the simplicial structure encoding which agent perspectives are epistemically compatible with which others. Thus, whereas in standard relational semantics  $a$ ’s knowledge is given by quantifying over those worlds that  $a$  “considers possible” (via an accessibility relation), in the simplicial setting this quantification is instead over those facets that share a vertex associated to agent  $a$ . We formalize all the details in Section 2.2.

As noted, the goal of this chapter is to generalize this semantics to provide a model for multi-agent belief instead of (or in addition to) knowledge. A key obstacle to this endeavor is that simplicial semantics hardcodes *factivity* ( $K\varphi \rightarrow \varphi$ ) in a way that relational semantics does not. Naturally, models for belief ought not validate the corresponding principle ( $B\varphi \rightarrow \varphi$ ), since agents can have false beliefs.

In relational semantics, knowledge is typically weakened to belief by passing to a subset of those worlds considered possible. More precisely, if  $R(w)$  denotes the set of all worlds epistemically accessible from  $w$ , then a reasonable candidate for a relation  $Q$  representing *doxastic* accessibility (i.e., beliefs) is one where  $Q \subseteq R$  (so also  $Q(w) \subseteq R(w)$ ). This ensures that knowledge implies belief ( $K\varphi \rightarrow B\varphi$ ) and allows agents to potentially believe falsehoods even though they cannot know falsehoods (because  $Q$  need not be reflexive even though  $R$  is assumed to be).

Somewhat surprisingly, the obvious analogous approach does not work out in the simplicial setting. Suppose our model begins with a simplicial complex  $S$  in the background (representing knowledge, as usual), and adds a subcomplex  $S' \subseteq S$  to represent the beliefs of each agent—so to determine an agent’s beliefs we would only quantify over facets in  $S'$ . Then, as desired, for a given facet  $X$  in  $S$ , it might be the case that  $\varphi$  is true at  $X$  yet at all facets  $X'$  in  $S'$  which are  $a$ -accessible from  $X$ ,  $\varphi$  is false. This produces a counterexample to factivity for belief. However, at all facets  $X'$  in  $S'$ , every agent *will* have factive beliefs, and as a consequence this model does not allow agents to consider it possible that any *other* agents have false beliefs; this is formalized in Proposition 2.1.

Our solution to this issue is to enrich the model with multiple “belief” subcomplexes—one for each agent. This solves the problem outlined above, and leads naturally to the question of how these enriched simplicial models are related to standard relational models for knowledge and belief. This question, too, is not entirely straightforward, since the standard translation between simplicial and relational frameworks requires a “properness” condition that turns out to be quite restrictive when it comes to representing beliefs. We discuss this translation, and its restrictiveness, in Section 2.3, and then leverage (a slight generalization of) the work in [10] to resolve the issue.

We are not the first to propose a representation of belief using simplicial complexes. Both “Knowledge and Simplicial Complexes” [14] as well as “Simplicial Belief” [11] tackle the same idea. While each of these approaches is interesting in its own right, they differ substantially from our models both in formal implementation and in the overall logic of belief that they validate. We outline these frameworks and discuss their relationship to our models in Section 2.5.

The rest of the paper is organized as follows. In Section 2.2, we provide the core definitions, introduce our new model, and prove that in this framework multiple complexes are needed to allow the right kind of inter-agent uncertainty. In Section 2.3 we establish a translation between standard relational models for knowledge and belief and our simplicial representation, with particular attention to the properness condition. Section 2.4 applies this translation to prove soundness and completeness. Section 2.5 concludes with a discussion of related and future work.

## 2.2 Semantics for Belief

The literature on simplicial semantics divides roughly into two approaches to defining valuations for propositional atoms. Loosely speaking, the first approach assigns truth values directly to the facets [20, 18, 21], while the second “vertex-based” approach assigns them (in a partial way) to the vertices and then “lifts” them to facets. [28, 14, 19, 17] While we feel that the latter approach is more interesting from a logical and epistemic perspective, the former is much more technically straightforward and will allow us to better illustrate the specific novelty of the belief models we wish to present in this work, so that is the approach we adopt here.

Let  $Ag$  be a finite set of agents. A **simplicial frame** is a tuple  $(N, V, S)$  where  $N$  is a set of *nodes*,  $V : N \rightarrow Ag$  assigns an agent to each node (this is sometimes called the *coloring function*), and  $S \subseteq 2^N$  is a simplicial complex. Let  $\mathcal{F}(S)$  denote the set of facets of  $S$ . As in much of the previous literature [14, 19], we henceforth restrict attention to simplicial frames that satisfy the **uniquely colored facets (UCF)** condition: for each  $X \in \mathcal{F}(S)$  and each  $a \in Ag$ ,



there is exactly one node  $n \in X$  such that  $V(n) = a$ . Note that this implies that all facets of  $S$  contain exactly  $|Ag|$  nodes, one for each agent.

Let  $\mathfrak{P}$  be a countable set of propositional atoms. A **simplicial model**  $\mathcal{M}$  is a simplicial frame  $(N, V, S)$  together with a *valuation function*  $L : \mathfrak{P} \rightarrow 2^{\mathcal{F}(S)}$ ; intuitively,  $L(P)$  tells us which facets  $P$  is true at. In this setting, the language  $\mathcal{L}_K(Ag)$  recursively defined by

$$\varphi ::= P \mid \perp \mid \varphi \rightarrow \psi \mid K_a \varphi,$$

where  $P \in \mathfrak{P}$  and  $a \in Ag$ , can be interpreted in simplicial models as follows:

$$\begin{aligned} \mathcal{M}, X &\models P \text{ iff } X \in L(P) \\ \mathcal{M}, X &\not\models \perp \\ \mathcal{M}, X &\models \varphi \rightarrow \psi \text{ iff } \mathcal{M}, X \models \varphi \text{ implies } \mathcal{M}, X \models \psi \\ \mathcal{M}, X &\models K_a \varphi \text{ iff } \forall Y \in \mathcal{F}(S) \text{ if } \pi_a(Y) = \pi_a(X) \text{ then } \mathcal{M}, Y \models \varphi, \end{aligned}$$

where  $\pi_a(Z) = V^{-1}(a) \cap Z$ , that is,  $\pi_a$  maps each facet to the unique  $a$ -colored vertex it contains. The other Boolean connectives are defined in the usual way.

This is the standard simplicial semantics for multi-agent knowledge: agent  $a \in Ag$  knows  $\varphi$  at facet  $X$  just in case  $\varphi$  is true at all facets that share an  $a$ -colored node with  $X$ . Note that the factivity of knowledge follows immediately: if  $X \models K_a \varphi$ , then since trivially  $\pi_a(X) = \pi_a(X)$ , the semantic clause for  $K_a$  forces us to conclude that  $X \models \varphi$ .

To incorporate belief—which of course we don’t want to be factive—we add in additional simplicial complexes. A **simplicial belief model** is a tuple  $(N, V, S, \{S_a\}_{a \in Ag}, L)$ , where  $(N, V, S, L)$  is a simplicial model and each  $S_a$  is a nonempty simplicial complex contained in  $S$ ; we further assume that each complex  $S_a$  satisfies the UCF condition. Note that this guarantees that  $\mathcal{F}(S_a) \subseteq \mathcal{F}(S)$ . Expand the language recursively to include unary belief modalities  $B_a$  for each agent  $a \in Ag$  and call the result  $\mathcal{L}_{KB}(Ag)$ . Given a simplicial belief model  $\mathcal{M}$ , we can then enrich the semantics with the following clause:

$$\mathcal{M}, X \models B_a \varphi \text{ iff } \forall Y \in \mathcal{F}(S_a) \text{ if } \pi_a(Y) = \pi_a(X) \text{ then } \mathcal{M}, Y \models \varphi.$$

So this is just like the clause for knowledge, except for each agent  $a$  we quantify only over facets in  $S_a$  rather than all the facets of  $S$ . Because of this, belief is not in general forced to be factive, since at any  $X \in \mathcal{F}(S) \setminus \mathcal{F}(S_a)$  we can have  $X \models B_a P$  but  $X \not\models P$  (for example, if  $L(P) = \mathcal{F}(S_a)$ ).

As discussed in Section 2.1, a natural question that arises is why we need a different simplicial complex  $S_a$  for each agent  $a$ . After all, simplicial models for multi-agent knowledge make do with just one. The issue is made clear in the following proposition.

**Proposition 2.1.** *Let  $\mathcal{M}$  be a simplicial belief model with  $S_a = S_b$ . Then  $\mathcal{M} \models B_a(B_b \varphi \rightarrow \varphi)$ .*

*Proof.* Let  $X \in \mathcal{F}(S)$ , and consider any  $Y \in \mathcal{F}(S_a)$  such that  $\pi_a(Y) = \pi_a(X)$ . If  $Y \models B_b \varphi$ , then by definition for all  $Z \in \mathcal{F}(S_b)$  such that  $\pi_b(Z) = \pi_b(Y)$  we must have  $Z \models \varphi$ . Since  $\mathcal{F}(S_b) = \mathcal{F}(S_a)$  and  $Y \in \mathcal{F}(S_a)$  by assumption, we know that  $Y \in \mathcal{F}(S_b)$ , so in fact we must have  $Y \models \varphi$ . This shows that  $Y \models B_b \varphi \rightarrow \varphi$  and thus  $X \models B_a(B_b \varphi \rightarrow \varphi)$ , as desired.  $\square$

In other words, when  $S_a$  and  $S_b$  coincide, agent  $a$  becomes incapable of considering agent  $b$  fallible with respect to their beliefs (and, of course, vice-versa). This is not an issue for multi-agent knowledge because knowledge is not fallible at all (so, naturally, agents know this).

## 2.3 Connection to Relational Semantics

It is well-known [14] that a broad class of relational models for multi-agent knowledge can be transformed into simplicial models in a truth-preserving manner. We briefly review this transformation here in order to illustrate how it works—and where it fails—when belief is added into the mix.

Recall that a **relational model (over  $Ag$ )** is a tuple  $\mathcal{N} = (W, (R_a)_{a \in Ag}, v)$  where  $W$  is a nonempty set of *possible worlds*, each  $R_a$  is a binary relation on  $W$  called an *accessibility relation*, and  $v : \mathfrak{P} \rightarrow 2^W$  is a *valuation*. Semantics are given in the usual way:

$$\begin{aligned} \mathcal{N}, w &\models P \text{ iff } w \in v(P) \\ \mathcal{N}, w &\not\models \perp \\ \mathcal{N}, w &\models \varphi \rightarrow \psi \text{ iff } \mathcal{N}, w \models \varphi \text{ implies } \mathcal{N}, w \models \psi \\ \mathcal{N}, w &\models K_a \varphi \text{ iff } \forall w' \in R_a(w) (\mathcal{N}, w' \models \varphi), \end{aligned}$$

where  $R_a(w) = \{w' \in W : w R_a w'\}$ . A **relational frame** is a model without the valuation.

We restrict our attention to relational models where each  $R_a$  is an equivalence relation—such models validate factivity ( $K_a \varphi \rightarrow \varphi$ ), positive introspection ( $K_a \varphi \rightarrow K_a K_a \varphi$ ), and negative introspection ( $\neg K_a \varphi \rightarrow K_a \neg K_a \varphi$ )—and refer to these as **relational models for introspective knowledge**. Since these principles are also validated by all simplicial models, we must impose them on relational structures in order to have any hope of a truth-preserving transformation. We write  $[w]_a$  to denote the equivalence class of  $w$  with respect to  $R_a$ .

Given a relational model  $\mathcal{N}$  as above, we define a simplicial frame  $(N_{\mathcal{N}}, V_{\mathcal{N}}, S_{\mathcal{N}})$  as follows:

- $N_{\mathcal{N}} = \{([w]_a, a) : w \in W, a \in Ag\}$
- $V_{\mathcal{N}}([w]_a, a) = a$
- $S_{\mathcal{N}} = \{F \subseteq N_{\mathcal{N}} : \bigcap_{n \in F} e(n) \neq \emptyset\},$

where  $e : N_{\mathcal{N}} \rightarrow 2^W$  is projection to the first coordinate (returning the equivalence class associated with each node). Intuitively, each equivalence class  $[w]_a$  captures a certain epistemic “perspective” of agent  $a$ , and thus we take our nodes to be all such equivalence classes (labelled by the agent in question). The simplicial structure is then determined by grouping precisely those nodes that correspond to equivalence classes with nonempty intersection, since these are the perspectives that can all obtain simultaneously (i.e., at the worlds in the intersection). Thus, each face of  $S_{\mathcal{N}}$  corresponds to nonempty intersection of equivalence classes from distinct agents.

**Proposition 2.2.** *The simplicial frame  $(N_{\mathcal{N}}, V_{\mathcal{N}}, S_{\mathcal{N}})$  satisfies UCF.*

*Proof.* We must show that each facet contains exactly one node for each agent. First observe that no face in  $S_{\mathcal{N}}$  can contain more than one  $a$ -colored node, since if  $([w]_a, a)$  and  $([w']_a, a)$  are distinct, then  $[w]_a$  and  $[w']_a$  must also be distinct and thus have empty intersection.

Now suppose that for some  $a \in Ag$ ,  $F \in S_{\mathcal{N}}$  does not contain an  $a$ -colored node. By definition, there exists some  $w \in \bigcap_{n \in F} e(n)$ ; it follows immediately that  $[w]_a \cap \bigcap_{n \in F} e(n) \neq \emptyset$ , and thus  $F \cup \{([w]_a, a)\} \in S_{\mathcal{N}}$ . This shows that  $F$  is not a facet (since it’s not maximal), which in turn implies that any facet must contain an  $a$ -colored node for every  $a \in Ag$ .  $\square$

The latter half of the proof above suggests a natural way to associate worlds in  $\mathcal{N}$  with facets in the simplicial frame we have constructed: namely, define  $f : W \rightarrow \mathcal{F}(S_{\mathcal{N}})$  by

$$f(w) = \{([w]_a, a) : a \in Ag\}.$$

It is not hard to see that  $f$  is surjective, but unfortunately, it may not be injective:  $f(w) = f(w')$  precisely when, for all  $a \in Ag$ ,  $[w]_a = [w']_a$ . In this case there is no clear way to define a valuation on  $\mathcal{F}(S_{\mathcal{N}})$  that “simulates” the valuation in  $\mathcal{N}$ , since two worlds that disagree about the truth value of  $P$ , say, may be associated to the same facet.

This motivates the following definition that has become standard in the literature [14]: a relational model is called **proper** if, for all  $w \in W$ ,  $|\bigcap_{a \in Ag} [w]_a| = 1$ . From this it easily follows that  $f$  is injective. Define  $L_{\mathcal{N}}(P) = f[v(P)] = \{f(w) : w \in v(P)\}$  and let  $M_{\mathcal{N}}$  be the simplicial model  $(N_{\mathcal{N}}, V_{\mathcal{N}}, S_{\mathcal{N}}, L_{\mathcal{N}})$ . We then have:

**Theorem 2.3.** *Let  $\mathcal{N}$  be a proper relational model. For all formulas  $\varphi \in \mathcal{L}_K(Ag)$ , for all  $w \in W$ ,*

$$\mathcal{N}, w \models \varphi \text{ iff } M_{\mathcal{N}}, f(w) \models \varphi.$$

*Proof.* See Appendix 2.6. □

The existing literature has largely taken properness to be a technical condition, introduced in order to facilitate translations from relational to simplicial models. [14, 28, 20, 19] Whether it is a *reasonable* restriction, in the sense of being epistemically motivated or defensible, is debatable (and actively debated by the authors of this chapter!).<sup>3</sup> What is less contentious is that once we incorporate belief into the models, properness is far too strong a requirement. A concrete example will help to illustrate this.

A **relational model for introspective knowledge and belief (over  $Ag$ )** is just a relational model for introspective knowledge supplemented with new relations  $Q_a \subseteq R_a$ , one for each  $a \in Ag$ ; we require that each  $Q_a$  be serial and constant on  $R_a$ -equivalence classes: that is, if  $wR_aw'$  then  $Q_a(w) = Q_a(w')$ . These new relations are used to interpret belief modalities in the language  $\mathcal{L}_{KB}(Ag)$  in the usual way, namely:

$$w \models B_a\varphi \text{ iff } \forall w' \in Q_a(w)(w' \models \varphi).$$

Under these semantics, the properties of  $Q_a$  listed above ensure that our models validate knowledge implies belief ( $K_a\varphi \rightarrow B_a\varphi$ ), consistency of belief ( $B_a\varphi \rightarrow \neg B_a\neg\varphi$ ), strong positive introspection for belief ( $B_a\varphi \rightarrow K_aB_a\varphi$ ), and strong negative introspection for belief ( $\neg B_a\varphi \rightarrow K_a\neg B_a\varphi$ ).

Consider the model depicted in Figure 1 for three agents  $Ag = \{a, b, c\}$ ; the red, blue, and green arrows display the relations  $Q_a$ ,  $Q_b$ , and  $Q_c$ , respectively. In particular, at all worlds  $w$ , agent  $a$  considers only  $w_1$  and  $w_2$  possible, agent  $b$  is sure that the true world is  $w_1$ , and agent  $c$  is sure that the true world is  $w_2$ . Note that we have not explicitly depicted the relations  $R_a$ ,  $R_b$ , or  $R_c$ , but because each of these must be an equivalence relation containing the corresponding belief relation, there is only one possibility: they are all the complete relation. Thus, the corresponding relational model for introspective knowledge is not proper and so cannot be translated into a simplicial model.

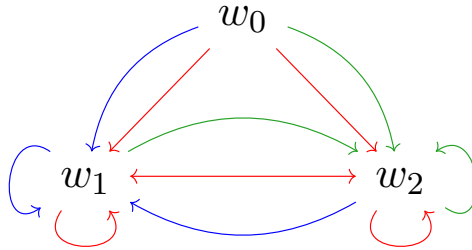


Figure 1: A 3-agent relational frame for introspective knowledge and belief

However, there does not seem to be anything trivial or redundant about this model of belief; as such, an inability to translate it into the simplicial setting is a substantive restriction. Fortunately, while this model itself cannot be so translated, there does exist an equivalent (in the sense of bisimilarity) model that can be. This follows from a modest extension of the following result:

**Theorem 2.4** ([10], Proposition 2.4). *Let  $\mathcal{N} = (W, (R_a)_{a \in Ag}, v)$  be a relational model. Then there is a relational model  $\tilde{\mathcal{N}}$  which is proper and a surjective, bounded morphism from  $\tilde{\mathcal{N}}$  to  $\mathcal{N}$ . Furthermore, if  $\mathcal{N}$  is transitive (respectively, Euclidean, serial, reflexive, symmetric), then  $\tilde{\mathcal{N}}$  is also transitive (respectively, Euclidean, serial, reflexive, symmetric).*

<sup>3</sup>One could argue that it eliminates a kind of “redundancy of perspectives”, though this depends on a further philosophical supposition that every relevant fact about a world is captured in some agent’s perspective.

Roughly speaking, by expanding this result to apply also to belief, we can translate the model in Figure 1 into a proper model that “simulates” it via an appropriate surjective, bounded morphism. This will yield the model depicted in Figure 2.

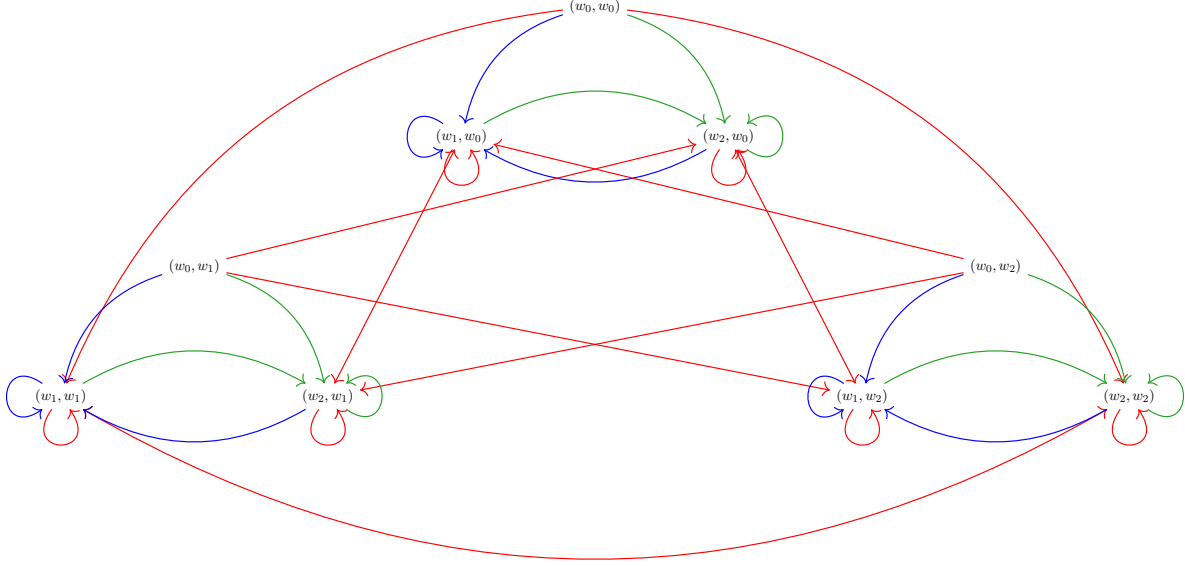


Figure 2: A translation of the model from Figure 1 into a proper model

In this case, the promised bounded morphism maps  $(w_i, w_j)$  to  $w_i$ , thus ensuring that these points always have the same theory and preserving the logical information we started with. Moreover, because the relational model in Figure 2 is proper, we are able to transform it into a simplicial model using (a slight generalization of) Theorem 2.3. The result of this transformation is shown in Figure 3.

In the next section we fill in the details omitted above—namely, generalizing Theorems 2.4 and 2.3 to relational models for knowledge *and* belief—and use these results to establish a novel completeness result.

Before turning to the technical details, it is worth pausing to reflect on the properness condition in light of this example. Intuitively, the model shown in Figure 1 encodes a single perspective for each agent: the set of doxastically accessible worlds is constant for each of  $Q_a$ ,  $Q_b$ , and  $Q_c$ , after all. However, with only one perspective for every agent, simplicial semantics only has the resources to produce a single facet, which is not enough to capture this scenario. As such, any technique for translating relational models into simplicial models must sometimes proliferate agential perspectives; indeed, Figure 2 has 9 worlds instead of 3 and intuitively includes 3 “perspectives” for each agent, each of which is a “copy” of that agent’s original, single perspective (up to formula satisfaction). In Section 2.5, we discuss how the more general notion of *simplicial sets* may be a useful framework for avoiding this redundancy of representation.

## 2.4 Soundness and Completeness

Recall that relational models for introspective knowledge and belief validate knowledge implies belief ( $K_a\varphi \rightarrow B_a\varphi$ ), strong positive introspection ( $B_a\varphi \rightarrow K_aB_a\varphi$ ), and strong negative introspection ( $\neg B_a\varphi \rightarrow K_a\neg B_a\varphi$ ). Let FULL be the axiom system obtained by adding these three axiom schemes to the system S5 for knowledge combined with KD45 for belief.<sup>4</sup>

<sup>4</sup>For an overview of these standard axiom systems, see for example [9].

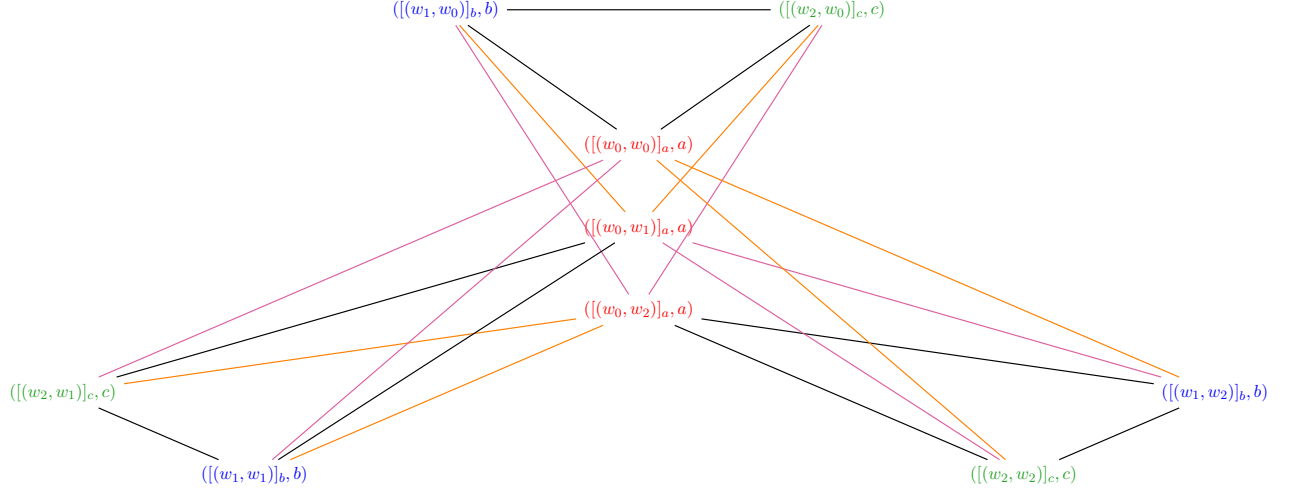


Figure 3: A translation of the model from Figure 2 into a simplicial model (purple facets belong to  $S_a$  and  $S_b$ , orange facets belong to  $S_a$  and  $S_c$ , while black facets belong to only  $S$ )

**Theorem 2.5.** *FULL is sound and complete with respect to the class of all simplicial belief models.*

*Proof.* We prove the result in much the same way as has been done for comparable results in previous literature. [14, 19] Soundness is straightforward so we focus on completeness. We first generate a canonical relational model for introspective knowledge and belief, using maximal FULL-consistent sets in the usual way. Next we transform this canonical model into an equivalent, proper model using a generalization of the result from [10]. Finally, we translate this proper model into a simplicial belief model using essentially the same technique as in Section 2.3.

Let  $W$  be the collection of maximal FULL-consistent sets. For arbitrary  $a \in Ag$ , we define  $R_a$  as follows:  $wR_a w'$  if and only if for all formulas  $\varphi$ , if  $K_a \varphi \in w$ , then  $\varphi \in w'$ . Similarly, let  $Q_a$  be given by  $wQ_a w'$  if and only if for all formulas  $\varphi$ , if  $B_a \varphi \in w$ , then  $\varphi \in w'$ . Lastly, define  $v(P) = \{w \in W : P \in w\}$ . Set  $\mathcal{N} = (W, (R_a)_{a \in Ag}, (Q_a)_{a \in Ag}, v)$ .

It is well known that with the above definitions for relations, and the fact that the S5 axioms for knowledge are included in FULL, each  $R_a$  is an equivalence relation. Indeed, it is straightforward to check that  $\mathcal{N}$  is a relational model for introspective knowledge and belief. For example, the fact that  $K_a \varphi \rightarrow B_a \varphi \in \text{FULL}$  forces  $Q_a \subseteq R_a$ . We will show here in detail that each  $Q_a$  is constant on  $R_a$ -equivalence classes, and leave the remaining, easy checks to the reader. Suppose  $wR_a w'$ . Suppose further that  $wQ_a u$ , and  $B_a \varphi \in w'$ . From strong positive introspection it follows that  $K_a B_a \varphi \in w'$ . Since  $wR_a w'$ , we have that  $w'R_a w$ , and therefore  $B_a \varphi \in w$ . Since  $wQ_a u$ , it follows that  $\varphi \in u$ . Since  $\varphi$  was arbitrary, this shows that  $w'Q_a u$ , which suffices to show that  $Q_a$  is constant on  $R_a$ -equivalence classes.

It is easy to show, in the standard way, that  $\mathcal{N}, w \models \varphi$  if and only if  $\varphi \in w$ . Thus  $\mathcal{N}$  refutes every non-theorem of FULL. Our next step is to find a *proper* relational model for introspective knowledge and belief that does the same; for this we adapt the technique from [10].

Since our language is countable, it follows that  $W$  is of size continuum. Let  $g: W \rightarrow \mathbb{R}$  be a bijection. Define  $\tilde{W} = W^2$  and fix a distinguished  $b \in Ag$ ; for all  $a \neq b$ , define

$$\begin{aligned} (w, u)\tilde{R}_a(w', u') & \text{ iff } u = u' \text{ and } wR_a w' \\ (w, u)\tilde{Q}_a(w', u') & \text{ iff } u = u' \text{ and } wQ_a w' \end{aligned}$$

and set

$$\begin{aligned} (w, u)\tilde{R}_b(w', u') &\text{ iff } g(u) - g(w) = g(u') - g(w) \text{ and } wR_bw' \\ (w, u)\tilde{Q}_b(w', u') &\text{ iff } g(u) - g(w) = g(u') - g(w) \text{ and } wQ_bw' \end{aligned}$$

Finally, define  $\tilde{v}(P) = \{(x, y) \in \tilde{W} : x \in v(P)\}$ , and let  $\tilde{\mathcal{N}} = (\tilde{W}, (\tilde{R}_a)_{a \in Ag}, (\tilde{Q}_a)_{a \in Ag}, \tilde{v})$ . The rough idea here is that we are making  $|W|$ -many copies of the model  $\mathcal{N}$ , but for one special agent,  $b$ , we “skew” their accessibility relations so they connect worlds across copies rather than within copies. This guarantees that (except for reflexive edges), agent  $b$ ’s accessibility relations never coincide with any other agent’s, which in turn ensures the model is proper. For a more detailed exposition of the intuition behind this construction, see [10].

Since  $Q_a \subseteq R_a$  for all  $a \in Ag$ , the definitions above guarantee that also  $\tilde{Q}_a \subseteq \tilde{R}_a$  for all  $a \in Ag$ . It’s also easy to check that for all  $a \in Ag$ ,  $\tilde{R}_a$  is an equivalence relation and  $\tilde{Q}_a$  is serial and constant on  $\tilde{R}_a$ -equivalence classes. The main result of [10] now guarantees  $\tilde{\mathcal{N}}$  is proper (with respect to the relations  $\tilde{R}_a$ ) and moreover, that the map  $\rho : \tilde{W} \rightarrow W$  given by projection to the first coordinate is a surjective, bounded morphism (again, with respect to the  $\tilde{R}_a$  relations). It is easy to check that  $\rho$  is also a bounded morphism with respect to the  $\tilde{Q}_a$  relations. It follows that every formula refuted in  $\mathcal{N}$  is also refuted in  $\tilde{\mathcal{N}}$ , as desired.

The final step is to transform  $\tilde{\mathcal{N}}$  into an equivalent simplicial belief model. Because  $\tilde{\mathcal{N}}$  is proper and each  $\tilde{R}_a$  is an equivalence relation, we can define  $N_{S_{\tilde{\mathcal{N}}}}, V_{S_{\tilde{\mathcal{N}}}}, S_{S_{\tilde{\mathcal{N}}}}$ , and  $L_{S_{\tilde{\mathcal{N}}}}$  as we did above, in the lead-up to Theorem 2.3. What’s missing are the belief subcomplexes, which we define as follows:

$$\mathcal{F}(S_{\tilde{\mathcal{N}},a}) = \{F \in \mathcal{F}(S_{\tilde{\mathcal{N}}}) : f^{-1}(F)\tilde{Q}_af^{-1}(F)\},$$

recalling that  $f : \tilde{W} \rightarrow \mathcal{F}(S_{\tilde{\mathcal{N}}})$  is the bijective correspondence between worlds in  $\tilde{\mathcal{N}}$  and facets in  $S_{S_{\tilde{\mathcal{N}}}}$ . Specifying the facets of  $S_{\tilde{\mathcal{N}},a}$  of course uniquely determines the simplex itself; it also guarantees that  $S_{\tilde{\mathcal{N}},a}$  is a UCF subcomplex of  $S_{\tilde{\mathcal{N}}}$ . Of course, given any proper relational model for introspective knowledge and belief,  $\mathcal{N}$ , we can use the above definitions to produce a simplicial belief model  $\mathcal{M}_{\mathcal{N}}$ . We now extend Theorem 2.3 in this context to the full language including belief modalities.

**Lemma 2.6.** *Let  $\mathcal{N}$  be a proper relational model for introspective knowledge and belief (over  $Ag$ ). For all formulas  $\varphi \in \mathcal{L}_{KB}(Ag)$ , for all  $w \in W$ ,*

$$\mathcal{N}, w \models \varphi \text{ iff } \mathcal{M}_{\mathcal{N}}, f(w) \models \varphi.$$

*Proof.* The proof is identical to that of Theorem 2.3 except for the inductive step for the belief modalities, which must be added. So suppose inductively, as in that proof, that for all  $w \in W$ ,  $\mathcal{N}, w \models \varphi$  iff  $\mathcal{M}_{\mathcal{N}}, f(w) \models \varphi$ . We wish to show that for all  $w \in W$ ,  $\mathcal{N}, w \models B_a\varphi$  iff  $\mathcal{M}_{\mathcal{N}}, f(w) \models B_a\varphi$ .

Fix  $w \in W$  and suppose  $\mathcal{N}, w \models B_a\varphi$ . Let  $F \in \mathcal{F}(S_{\mathcal{N},a})$  be such that  $\pi_a(F) = \pi_a(f(w))$ . It follows that  $f^{-1}(F)Q_af^{-1}(F)$  and  $wR_af^{-1}(F)$ . Because  $Q_a$  is constant on  $R_a$  equivalence classes,  $Q_a(w) = Q_a(f^{-1}(F))$ ; thus,  $wQ_af^{-1}(F)$ , and so  $\mathcal{N}, f^{-1}(F) \models \varphi$ . By the inductive hypothesis, we have that  $\mathcal{M}_{\mathcal{N}}, F \models \varphi$ . It follows that  $\mathcal{M}_{\mathcal{N}}, f(w) \models B_a\varphi$ , as desired.

Suppose now that  $\mathcal{M}_{\mathcal{N}}, f(w) \models B_a\varphi$ . Let  $u \in W$  be such that  $wQ_au$ . By the Euclidean property we have that  $uQ_au$ . It follows that  $f(u) \in \mathcal{F}(S_{\mathcal{N},a})$ . Moreover,  $wQ_au$  implies that  $wR_au$ , and so  $\pi_a(f(u)) = \pi_a(f(w))$ . It follows that  $\mathcal{M}_{\mathcal{N}}, f(u) \models \varphi$ . By the inductive hypothesis, we have that  $\mathcal{N}, u \models \varphi$ . Therefore,  $\mathcal{N}, w \models B_a\varphi$ , as desired.  $\square$

We can now establish completeness. Suppose for simplicial belief models we have  $\Gamma \models \varphi$ , and suppose for contradiction that  $\Gamma \not\models_{\text{FULL}} \varphi$ . Let  $w$  be a maximal FULL-consistent set extending  $\Gamma \cup \{\neg\varphi\}$  (obtained, as usual, via Lindenbaum’s Lemma). Then  $\mathcal{N}, w \models \Gamma \cup \{\neg\varphi\}$ . It then follows from the above that (for any  $u$ ),  $\mathcal{M}_{\tilde{\mathcal{N}}}, f((w, u)) \models \Gamma \cup \{\neg\varphi\}$ , a contradiction.  $\square$

## 2.5 Related and Future Work

The simplicial representation of belief we introduce in this chapter stands in contrast to two existing approaches. In “Knowledge and Simplicial Complexes” [14], standard UCF simplicial models  $(N, V, S, L)$  are augmented with idempotent functions  $f_a: V^{-1}(a) \rightarrow V^{-1}(a)$ , one for each agent  $a \in Ag$ . These functions are used to define belief via the following clause:

$$\mathcal{M}, X \models B_a \varphi \text{ iff } \forall Y \in \mathcal{F}(S) \text{ if } \pi_a(Y) = f_a(\pi_a(X)) \text{ then } \mathcal{M}, Y \models \varphi.$$

Loosely speaking, whereas a node  $n \in V^{-1}(a)$  represents a “perspective” of agent  $a$  for the purposes of evaluating knowledge modalities at facets containing  $n$ , for the purposes of evaluating *belief* modalities at facets containing  $n$ , this semantics instead uses the node  $f_a(n)$ . In other words, agent  $a$ ’s beliefs at  $X$  are given by her knowledge at  $f_a(\pi_a(X))$ . Because each  $f_a$  is idempotent, it is easy to check that the  $B_a$  modalities satisfy the KD45 axioms.

However, this notion of belief has essentially no relationship to knowledge; in particular,  $K_a \varphi \rightarrow B_a \varphi$  is not valid. A natural solution to this might be to strengthen the semantics so as to validate this formula. Unfortunately, this results in triviality, for it requires the following to hold for all  $X \in \mathcal{F}(S)$ :

$$\{Y \in \mathcal{F}(S) : f_a(\pi_a(X)) = \pi_a(Y)\} \subseteq \{Y \in \mathcal{F}(S) : \pi_a(X) = \pi_a(Y)\};$$

under the UCF assumption, this containment does not hold in general unless  $f_a$  is the identity function. In this case, of course, belief and knowledge coincide. Thus, this model for belief cannot satisfy both UCF and the validity of “knowledge implies belief” without trivializing belief.

We find a quite different approach to defining belief in simplicial complexes in “Simplicial Belief”. [11] The idea here depends on breaking the UCF assumption so that facets may contain one *or more*  $a$ -colored nodes, and then using the number of nodes a given facet contains to determine whether or not it is “accessible” for the purposes of evaluating belief. Specifically, given  $X \in \mathcal{F}(S)$ , let  $m_a(X) := |\{n \in X : V(n) = a\}|$  count the number of  $a$ -colored nodes in  $X$ , and for any  $X, Y \in \mathcal{F}(S)$ , define

$$X \leq_a Y \text{ iff } m_a(X) \leq m_a(Y).$$

Write  $X \sim_a Y$  if and only if  $\exists n \in X \cap Y$  such that  $V(n) = a$ , and define  $\leq_a = \leq_a \cap \sim_a$ . These relations are used to define belief via the following clause:

$$\mathcal{M}, X \models B_a \varphi \text{ iff } \forall Y \in \mathcal{F}(S) \text{ if } Y \leq_a X \text{ then } \mathcal{M}, Y \models \varphi.$$

This says that if  $\varphi$  is true at all facets  $Y$  whose intersection with  $X$  contains an  $a$ -colored node, and such that the number of  $a$ -colored nodes in  $Y$  is bounded above by the number in  $X$ , then  $B_a \varphi$  is true at  $X$ .<sup>5</sup> The rough intuition here, as articulated in [11], is that facets containing more  $a$ -perspectives are less plausible (to agent  $a$ ) than those containing fewer, since  $a$  is in some sense less certain of their own state of mind in the former.

It’s easy to see that this definition validates  $K_a \varphi \rightarrow B_a \varphi$ . And it is certainly interesting from a formal perspective to relax the UCF assumption and leverage this extra freedom to define belief. However, the epistemic interpretation of facets with multiple  $a$ -perspectives remains somewhat murky. Unfortunately, if we re-impose the UCF condition, this of course implies that for all  $X \in \mathcal{F}(S)$ ,  $m_a(X) = 1$ , making  $\leq_a$  the total relation on  $\mathcal{F}(S)$ , hence  $\leq_a = \sim_a$ . In this case again we see that belief and knowledge coincide.

In contrast to these approaches, our framework provides a definition of belief in simplicial complexes satisfying *all* three of: (1) UCF, (2)  $K_a \varphi \rightarrow B_a \varphi$  is valid, and (3)  $B_a \varphi \rightarrow K_a \varphi$  is *not* valid. This has a few further upshots. The usual techniques for translating frame models into

<sup>5</sup>The paper also provides a stronger notion of belief, taking the belief relation to access worlds with a *minimum* number of perspectives, rather than all those with fewer than the starting world. The same result we show for the given definition also applies to this one.

simplicial models apply very cleanly (indeed, almost trivially) in our setting, yielding soundness and completeness with respect to the simple FULL axiom system (Theorem 2.5). The only wrinkle in the proof is the condition that the underlying relational frame be proper. This is circumvented using the result in [10], where it is shown that any improper relational model can be translated into an equivalent, proper one. This result, too, is easily lifted to the setting where there are both knowledge and belief relations.

As mentioned above, the existing literature has largely taken properness as a technical condition, introduced in order to facilitate translations from relational to simplicial models. [14, 28, 20, 19] However, as noted at the end of Section 2.3, in order to “make” a model proper we have to multiply worlds in a way that produces substantial redundancy. As a result, the structures become harder to parse—both the relational models and their simplicial correlates. We suspect, for example, that few will find the simplicial model in Figure 2.3 as easy to understand as the relational model in Figure 1.

It is natural to wonder, then, if there is any way to avoid imposing properness altogether. One promising approach utilizes the notion of *simplicial sets*.<sup>6</sup> Roughly speaking, simplicial sets generalize simplicial complexes by allowing us to count faces *with multiplicity*, similar to the generalization of a set to a multiset, or a graph to a multigraph. For instance, in a simplicial complex, the triangle  $\{a, b, c\}$  is uniquely determined by the three nodes it contains, whereas in a simplicial set, there could be multiple triangles on the same underlying set of nodes  $\{a, b, c\}$ . Visually, one may picture multiple triangular “membranes” connecting these nodes. Informally, we may define a simplicial set as a “multiset of multisets closed under subsets”.

Using this structure, one can capture the model in Figure 1 with a simplicial set  $S$  together with subsimplicial sets  $S_a$ ,  $S_b$ , and  $S_c$ , as depicted in Figure 4.

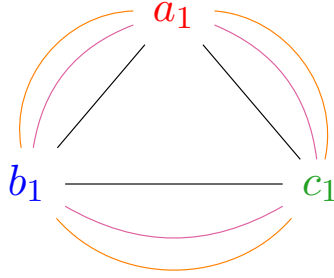


Figure 4: A translation of the model from Figure 1 into a simplicial simplicial set  $S$  together with subsimplicial sets  $S_a$ ,  $S_b$ , and  $S_c$  (purple facets belong to  $S_a$  and  $S_b$ , orange facets belong to  $S_a$  and  $S_c$ , while black facets belong to only  $S$ )

Intuitively, the node  $a_1$  corresponds to the (unique!) perspective that agent  $a$  has in Figure 1, namely the worlds  $w_1$  and  $w_2$ . Similarly,  $b_1$  corresponds to the view that only  $w_1$  is possible, and  $c_1$  corresponds to only seeing  $w_2$ . But we now have the freedom to associate not one but three facets with this set of perspectives. The black facet corresponds with  $w_0$ , the world accessible to none of the three agents. The purple facet belongs to  $S_a$  and  $S_b$ , but not  $S_c$ , corresponding to  $w_1$ , the world accessible to agents  $a$  and  $b$  but not  $c$ . And similarly the orange facet belongs to  $S_a$  and  $S_c$ , but not  $S_b$ , corresponding to  $w_2$ . Thus we see that simplicial sets give us the resources to represent non-proper models without artificially proliferating perspectives; this provides a promising direction for future research in simplicial semantics and particularly in the representation of belief therein.

<sup>6</sup>The idea of using simplicial sets in the context of simplicial models was introduced in [21].



## 2.6 Proofs

*Proof of Theorem 2.3.* Induction on formulas. Let  $P \in \mathfrak{P}$ . First suppose  $\mathcal{N}, w \models P$ ; then  $w \in v(P)$ , so  $f(w) \in f[v(P)] = L_{\mathcal{N}}(P)$ , which yields  $\mathcal{M}_{\mathcal{N}}, f(w) \models P$ , as desired. Conversely, suppose  $\mathcal{M}_{\mathcal{N}}, f(w) \models P$ ; then  $f(w) \in L_{\mathcal{N}}(P) = f[v(P)]$ . Because  $f$  is injective, we must then have  $w \in v(P)$ , which yields  $\mathcal{N}, w \models P$  as desired.

The case for  $\perp$  is trivial.

Now suppose inductively that for all  $w \in W$ ,  $\mathcal{N}, w \models \varphi$  iff  $\mathcal{M}_{\mathcal{N}}, f(w) \models \varphi$  and  $\mathcal{N}, w \models \psi$  iff  $\mathcal{M}_{\mathcal{N}}, f(w) \models \psi$ . We wish to show that this equivalence holds also for  $\varphi \rightarrow \psi$  and for  $K_a\varphi$ . For the former, observe that  $\mathcal{N}, w \models \varphi \rightarrow \psi$  if and only if  $\mathcal{N}, w \models \varphi$  implies  $\mathcal{N}, w \models \psi$ , which (by the inductive hypothesis) is true if and only if  $\mathcal{M}_{\mathcal{N}}, f(w) \models \varphi$  implies  $\mathcal{M}_{\mathcal{N}}, f(w) \models \psi$ . And of course this is true if and only if  $\mathcal{M}_{\mathcal{N}}, f(w) \models \varphi \rightarrow \psi$ , as desired.

At last we turn to the knowledge modalities. Suppose  $\mathcal{N}, w \models K_a\varphi$ . Fix arbitrary  $Y \in \mathcal{F}(S_{\mathcal{N}})$  such that  $\pi_a(Y) = \pi_a(f(w))$ . By definition of  $f$ ,  $\pi_a(Y) = \pi_a(\{([w]_a, a) : a \in Ag\}) = ([w]_a, a)$ . Because  $f$  is surjective, we can find  $w' \in W$  such that  $f(w') = Y$ . So  $([w']_a, a) = \pi_a(f(w')) = \pi_a(Y) = ([w]_a, a)$ , and therefore  $[w']_a = [w]_a$ , meaning  $w' \in [w]_a$ . We know from our initial supposition that for all  $w' \in [w]_a$ ,  $\mathcal{N}, w' \models \varphi$ . So, by the inductive hypothesis, we have  $\mathcal{M}_{\mathcal{N}}, f(w') \models \varphi$ , hence  $\mathcal{M}_{\mathcal{N}}, Y \models \varphi$ ; since  $Y$  was chosen arbitrarily, this yields  $\mathcal{M}_{\mathcal{N}}, f(w) \models K_a\varphi$ .

Conversely, suppose  $\mathcal{M}_{\mathcal{N}}, f(w) \models K_a\varphi$ . By definition, for all  $Y \in \mathcal{F}(S_{\mathcal{N}})$  such that  $\pi_a(Y) = \pi_a(f(w))$ ,  $\mathcal{M}_{\mathcal{N}}, Y \models \varphi$ . Let  $w' \in [w]_a$ . Then, since  $[w']_a = [w]_a$ , we must have  $\pi_a(f(w')) = \pi_a(f(w))$ , so  $\mathcal{M}_{\mathcal{N}}, f(w') \models \varphi$ . By the inductive hypothesis, then,  $\mathcal{N}, w' \models \varphi$ , and since  $w'$  was chosen arbitrarily, we conclude  $\mathcal{N}, w \models K_a\varphi$ .  $\square$

### 3 Simplicial Semantics for Belief Revision

#### Abstract

This chapter will give a definition of belief revision within simplicial semantics. Starting from the work presented in [26] as a baseline, this chapter modifies the semantics for belief given there to allow atomic formulae to be assigned to nodes, not facets. Conceptually and philosophically, this version of the semantics is better suited if one wishes to interpret the nodes of a simplicial model of epistemic logic as a “perspective” assigned to a particular agent. If nodes are perspectives, it then follows that worlds, or the facets of a simplicial model, are composed themselves of perspectives. This allows us to say that two worlds are more similar, or “nearer”, if they share more perspectives. With this conceptual notion of nearness in hand, two different formal presentations of revision are given. We conclude by exploring some conceptual pitfalls surrounding these definitions under iterated revision, and motivate a few potential solutions that involve giving the agents a “memory” of what has been announced so far.

#### 3.1 Introduction

Semantics for epistemic logic in simplicial complexes has been an ongoing area of research in the last few years. [20, 28, 14, 19, 17, 18, 21, 11, 26] Some of the earliest work in this area explored connections between simplicial semantics and Dynamic Epistemic Logic, particularly public announcements. [19] Recent work has attempted to use this setting to model belief. [11, 26, 14] What is relevant about belief, as compared to knowledge, is that belief is not generally taken to be factive. That is, the axiom schema **T**, often called factivity, is not sound. As a result, agents are able to have false beliefs. It may be the case that for some world  $w$ , and some formula  $\varphi$ , we have that  $B_a\varphi$  is true (read “agent  $a$  believes that  $\varphi$  is true”), but  $\varphi$  is false.

This chapter will explore the overlap between false beliefs and public announcements in simplicial semantics. In general, if  $\varphi$  is publicly announced, then any world  $w$  which agent  $a$  considers possible, they cease to consider possible. So, if agent  $a$  believes  $\varphi$ , and then  $\neg\varphi$  is announced, then agent  $a$  no longer considers any worlds possible. Their beliefs have become defunct. Belief revision is an alternative way of modeling how agents learn information in an announcement which avoids this issue. [24, 22, 1] The main idea in revision is that some worlds are “near” to each other. If agent  $a$  considers a world  $w$  possible which satisfies  $\varphi$ , and  $\neg\varphi$  is announced, the agent may “replace” or “revise”  $w$  to a “nearby” world  $w'$  that satisfies  $\neg\varphi$ .

The main issue with this notion of nearness is that it is arbitrary, in practice, to determine what worlds should be nearer or further from each other. We believe that simplicial semantics offers a solution here. In simplicial semantics, it is often assumed that worlds are composed of a simpler epistemic structure, namely “perspectives”. [17, 14] If this is true, we can say that two worlds are nearer to each other if they share more perspectives. this chapter defines a notion of revision in simplicial semantics that operates on precisely this notion. In Section 3.2, we outline the epistemology of “perspectives” that we will appeal to in this chapter. In Section 3.3, we briefly outline the semantics for belief that this chapter will appeal to, which is a modified form of the semantics in [26]. In Section 3.4, we give a detailed motivation of the concept of nearness at play in this chapter, and we formalize it in Section 3.5. We then proceed to give some examples, motivate a potential issue that arises if we assume announcements are indefeasible, and offer two potential solutions.

#### 3.2 The “Perspective” Perspective

The basic epistemic unit that we will start with we will call a “perspective” or “point of view,” very similar to the idea outlined in [17]. Given a particular agent, let’s call her Alice, the perspective of that agent consists abstractly of (some of) the information available to that

agent. For example, if the agent is sitting on a park bench, and sees a blue sky and ducks in the pond, all of this information is what we might consider part of the agent’s perspective. This is indeed an abstract notion, but crucial to us is that the information constituting perspectives can be broken down into two pieces. The first is the *hard* information, or the information that is in some sense indefeasible. In this scenario, where the agent on the park bench sees ducks, we consider the proposition representing this fact hard information. By contrast, suppose Alice has a friend, Barbara, who regularly goes to the park with her. Each time the pair has been to the park, they see ducks; however today, her friend is home sick. Alice still goes to the park and sees ducks, but Barbara of course does not. In this example, we will assume that both believe that there are ducks in the pond. However, for Alice, this is hard information. By contrast, Barbara believes there are ducks in the pond in a weaker sense. This is an example of *weak* information.

The line between hard and weak information is only intuitive. In fact, the distinction only matters as a modeling convention. In certain modeling contexts, it will be useful to represent some information as indefeasible, and other information as defeasible. That is all the distinction between *hard* and *weak* is meant to cache out. It rests solely on the intuition that an agent’s point of view is made up of facts that are immediately obvious to the agent, and other facts that are less so. As a technical and simplifying assumption, we will assume that the hard information is given by a possibly empty set of literals associated with that perspective.

Since we are taking perspectives as our foundational epistemic object, how can we use them to model uncertainty? One again imagines our agent at the pond with the ducks. Call her agent  $a$ . Her friend will be agent  $b$ . If  $D$  is the logical atom that means “The ducks are in the pond at the park,” then we can say that  $D$  is assigned to  $a$ ’s perspective, call it  $a_1$ .  $b$ ’s perspective does not, in this language, consist of any hard information, and so we associate  $b_1$  with the empty set. Since these perspectives are true simultaneously, we can group them together as the set  $\{a_1, b_1\}$ , as drawn in Figure 5

$$a_1(D) \text{ — } b_1$$

Figure 5: A model with two perspectives and one world/facet.

When we formally define simplicial semantics later, it will be clear that Figure 5 is a (very basic) simplicial complex. For readers unfamiliar with the exact workings of simplicial semantics, one should look at [14] and [26]. In order to read the above, for now, it suffices to recognize that Figure 5 is equivalent to a Kripke model, with one world where  $D$  is true, and two reflexive edges on that world, one for each agent. The single edge between the two perspectives in Figure 5 represents the set containing those two perspectives. This set with the two perspectives is the possible world, and each agent recognizes that world in their perspective, hence the reflexive edges.

Now let’s imagine a slightly more complicated example. Both friends are going to the park, but do not see each other yet. We will consider this from  $a$ ’s point of view. She sees the ducks in the pond, but is uncertain whether or not  $b$  sees the ducks yet. She cannot imagine that  $b$  sees there are not ducks in the pond, as that would be inconsistent with her own experience. But, she cannot tell whether or not  $b$ ’s perspective is one where they are, in fact, seeing ducks, or they have not seen the ducks yet. The latter is the same as  $b_1$  as above, but the former, which we call  $b_2$ , will have  $D$  associated with it, as seen in Figure 6.

$$b_2(D) \text{ — } a_1(D) \text{ — } b_1$$

Figure 6: A model with three perspectives and two worlds/facets.

Figure 6 is equivalent to a Kripke model with two worlds, one for each set, shown as edges.

The worlds are indistinguishable to agent  $a$ , and all reflexive edges for both agents are present. In the case of two agents, we will refer to sets containing two perspectives as *facets*. The facets in Figure 6 are red because they are associated with agent  $a$ . That is, these are facets, or worlds, that  $a$  considers possible. In general, in this chapter, we will assume that  $a$  is red,  $b$  is blue, and  $c$  is green.

### 3.3 Simplicial Semantics for Belief

this chapter will work in a slightly modified version of the semantics for belief, as presented in [26]. We will reproduce here the relevant formal details for those unfamiliar, as well as show how we need to modify the semantics for our purposes. The modification itself is similar. The semantics in [26] functions by assigning atomic propositions directly to the facets of the simplicial complexes. This follows the example of much of the previous literature. [20, 18, 21] By contrast, here we will assign atomic propositions to the nodes of the simplicial complexes. This, too, has a great deal of precedent in the literature. [28, 14, 19, 17] The reason for this change is that the notion of revision which we set out to motivate only makes sense if we can think of the individual nodes of a simplicial complex as *perspectives* or “points of view” belonging to a particular agent. That is, a node delineates what an agent is or is not “seeing”, or is aware of, in some immediate sense. We will return to this in Section 3.5.

Let  $\mathfrak{P}$  be a countable set of propositional atoms, and  $Ag$  a finite set of agents. Let  $N$  be a set of nodes,  $V : N \rightarrow Ag$  a function which assigns each node to an agent, called the *coloring* function, and  $L : N \rightarrow 3^{\mathfrak{P}}$  a function which assigns each node to a set of literals, which we call the *assignment*. The interpretation is that  $L(n)(P) = 1$  if and only if  $P$  is associated with  $n$ ,  $L(n)(P) = 0$  if and only if  $\neg P$  is associated with  $n$ , and  $L(n)(P) = 2$  if and only if neither is associated with  $n$ .  $L$  is the only difference between their presentation here as compared to [26]. As mentioned above,  $L$  assigns propositions to nodes.

Our first key idea is that we can use  $N$ ,  $V$ , and  $L$  to create a kind of **maximal** simplicial complex. Like much of the previous literature, we will assume our simplicial complexes are uniquely colored. Specifically, each facet of our complexes has a dimension of size  $|Ag|$ , and no two nodes are associated to the same agent. Put formally, if  $X \in S$  is such that for all  $Y \in S$ , if  $X \subseteq Y$  then  $X = Y$ , we have that  $|X| = |Ag|$ , and, if  $x, y \in X$ , then if  $V(x) = V(y)$ , we have that  $x = y$ . We call this condition UCF for “Uniquely Colored Facets”. The **Maximal Complex** is the set of subsets of  $N$  such that the associated logical content with that subset is consistent, and it satisfies the UCF condition. That is, it’s the subsets  $x \subseteq N$  such that  $|x| = |Ag|$ ,  $\bigcup_{u \in x} L(u)$  is consistent, and for all  $u, v \in x$ ,  $V(u) \neq V(v)$ . We refer to  $\mathfrak{M}(N, V, L)$  as the **Maximal Complex** of  $N$ ,  $V$ , and  $L$ . When the context is clear, we will refer to it simply as the maximal complex. The maximal complex can be defined set theoretically as follows:

$$\begin{aligned} \mathfrak{M}(N, V, L) := \{y \in 2^N \mid & \exists x \in 2^N (y \subseteq x \\ & \wedge \neg(\exists P \in \mathfrak{P} (\exists u, v \in x (L(u)(P) = 1 \wedge L(v)(P) = 0))) \\ & \wedge (|x| = |Ag|) \\ & \wedge (\forall u, v \in x (V(u) \neq V(v))))\} \end{aligned}$$

Note that  $\mathfrak{M}(N, V, L)$  is a simplicial complex. This is because, if  $Y \in \mathfrak{M}(N, V, L)$ , there is a set  $X$  such that  $Y \subseteq X$  and  $X$  satisfies certain properties. If  $Z \subseteq Y$ , then the same set  $X$  suffices to show that  $Z \in \mathfrak{M}(N, V, L)$ . All complexes we consider in this chapter will be UCF subcomplexes of the maximal complex.

If  $S$  is a simplicial complex, let  $\mathcal{F}(S)$  denote the facets<sup>7</sup> of  $S$ . Since all simplicial complexes we consider are UCF, we can define projection functions  $\pi_a : \mathcal{F}(S) \rightarrow N$  given by  $\pi(a)(X) = V^{-1}(a) \cap X$ . That is,  $\pi_a$  maps each facet to the unique  $a$ -colored node it contains.

We say a **simplicial belief model** is a tuple  $(N, V, L, S, \{S_a\}_{a \in Ag})$ , where  $N$ ,  $V$ , and  $L$  are as above,  $S$  is a UCF subcomplex of  $\mathfrak{M}(N, V, L)$ , and each  $S_a$  is a UCF subcomplex of  $S$ .

<sup>7</sup>Maximal faces under the subset ordering.

We will refer to each  $S_a$  as the *belief subcomplex* belonging to agent  $a$ .<sup>8</sup> Often, we will leave  $S$  unspecified. In this context, we assume that  $S = \mathfrak{M}(N, V, L)$ .<sup>9</sup>

In this setting, the language  $\mathcal{L}_B(Ag)$  recursively defined by

$$\varphi ::= P \mid \perp \mid \varphi \rightarrow \psi \mid B_a \varphi,$$

where  $P \in \mathfrak{P}$  and  $a \in Ag$ , can be interpreted in simplicial models as follows for any  $X \in \mathcal{F}(S)$ :

$$\begin{aligned} \mathcal{M}, X &\models P \text{ iff } \exists a \in Ag (L(\pi_a(X))(P) = 1) \\ \mathcal{M}, X &\not\models \perp \\ \mathcal{M}, X &\models \varphi \rightarrow \psi \text{ iff } \mathcal{M}, X \models \varphi \text{ implies } \mathcal{M}, X \models \psi \\ \mathcal{M}, X &\models B_a \varphi \text{ iff } \forall Y \in \mathcal{F}(S_a) \text{ if } \pi_a(Y) = \pi_a(X) \text{ then } \mathcal{M}, Y \models \varphi \end{aligned}$$

Similar to what is described in [17], we think of the nodes  $N$ , with coloring and assignment functions  $V$  and  $L$ , as *perspectives*. A perspective, as described in Section 3.2, is an informal notion meant to capture information that is “immediately” available to the agent. It often is useful to think of the perspective as capturing those logical atoms whose truth value the agent is directly observing. So, if  $P$  and  $\neg R$  are assigned to an  $a$ -colored node  $n$ , we think of  $n$  as representing the perspective where  $a$  is directly observing that  $P$  is true and  $R$  is false.

Often in the literature, atoms are assumed to be assigned to the agents themselves, and thus an atom represents a fact about the particular agent it is assigned to. Such a setting is referred to as the language of **Local Variables**. [14] In this setting, all atomic propositions belong to or are assigned to a particular agent, say agent  $a$ , and  $a$ -colored nodes fix the truth value of all  $a$ -colored propositions. More specifically, for each  $a \in Ag$ , fix a set of atoms  $\mathfrak{P}_a$ . We will refer to these as the atoms **local** to agent  $a$ . The total collection of atoms will be  $\bigcup_{a \in Ag} \mathfrak{P}_a$ . Fix  $P \in \mathfrak{P}_a$ . Then for each  $n \in N$ , we have that  $L(n)(P) \in \{0, 1\}$  and is only defined if  $V(n) = a$ . Finally, the semantic condition is modified so that  $\mathcal{M}, X \models P$  iff  $L(\pi_a(X))(P) = 1$ . In this way, the  $a$ -colored perspective of any facet fixes the truth value of all atoms in  $\mathfrak{P}_a$ .

There are many contexts in which presenting simplicial semantics this way is entirely appropriate. In particular, it’s clear how to model scenarios where each agent has a privately held bit value, and is attempting to signal its value to other agents. [28, 19] However, our presentation has the advantage of being slightly more general. Atomic formulae can be assigned to any agent’s perspective, and so a given proposition doesn’t have its truth value “fixed” by a particular agent at every world. This allows the methodology shown here to model a wider range of scenarios than the language of local variables. One such case is the classic example of the “Muddy Children Problem”. [16] We will present the case with three agents. Each agent, or child, can observe the muddiness of the other two children. So, agent  $a$  has in their perspective either  $M_b$  or  $\neg M_b$ , and either  $M_c$  or  $\neg M_c$ . This gives twelve total perspectives, four for each agent. If one is familiar with the usual Kripke structure for this scenario, it’s easy to check that at each world, every agent considers exactly two worlds possible. If we consider agent  $a$ , these are the world itself, and the world which differs only in the truth value of  $M_a$ . So, using the usual translation between Kripke models and simplicial models,<sup>10</sup> we have that each facet corresponding to a world is a triangle, and any two triangles share at most one perspective. Moreover, each perspective is present on exactly two triangles. The result is drawn in Figure 7.

Now we must assign atomic formulae to the individual nodes. Recall that, in the Muddy Children Problem, every child can see every other child. Hence, every child can observe the muddiness of every other child. Following our intuition for hard information, this would mean that, for every  $i \in Ag$ , and  $j, k \neq i$ , both the truth value of  $M_j$  and  $M_k$  are *hard* information

<sup>8</sup>The addition of multiple subcomplexes, one for each agent, is a modification to the standard simplicial semantics made to adapt those semantics for modeling belief. For more detail, see [26].

<sup>9</sup>It is possible to interpret a knowledge modality  $K_a$  on  $S$  in the standard way such modalities are given in the literature. [14] This would have the property that knowledge implies belief, i.e., for all  $a \in Ag$ ,  $K_a \varphi \rightarrow B_a \varphi$ , as in [26]. We do not discuss this here in order to keep the focus on revision.

<sup>10</sup>See, e.g., [26].

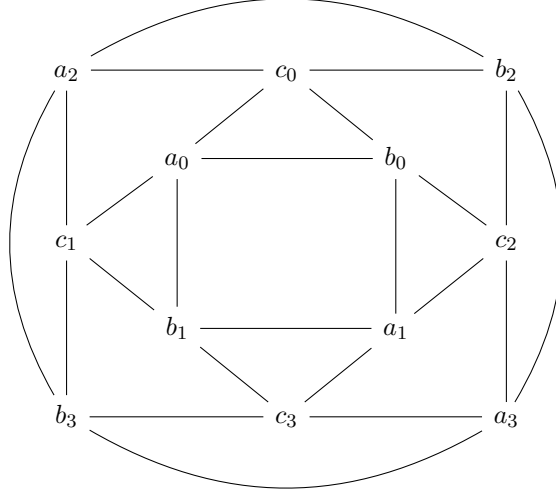


Figure 7: A simplicial complex capturing the following conditions: All facets are triangles, every node is in exactly two facets, and there are twelve nodes.

for agent  $i$ . Hence, at each  $i_n$ , either  $L(i_n)(M_j) = 1$  or  $L(i_n)(M_j) = 0$ , and the same for  $M_k$ . In particular, because  $i$  is uncertain about  $M_i$ ,  $L(i_n)(M_i) = 2$ . We draw this in Figure 8.

This configuration is surprising from the perspective of the language of local variables. [14] In this scenario, for every  $M_i$ , there must be an agent  $j \in Ag$  such that for all  $n \in \mathbb{N}$ ,  $L(j_n)(M_i) = 0$  or  $L(j_n)(M_i) = 1$ . A natural first guess to what this might be is that  $i = j$ . That is, assign the proposition which stands for “ $i$  is Muddy” to agent  $i$ . However, as we see in the argument above, this does not give the intended truth values for modal propositions. Indeed, such an assignment would make true  $B_i M_i \vee B_i \neg M_i$  for each  $i \in Ag$ , which contradicts that in the initial configuration,  $i$  must be *uncertain* about the truth value of  $M_i$ . While this isn’t borne out directly by the Muddy Children Problem, our system has the advantage that for a particular atom  $P$  and any agent  $i \in Ag$ , it can be the case that there is a model  $\mathcal{M}$  and a facet  $X$  such that  $\mathcal{M}, X \models \neg B_i P \vee \neg B_i \neg P$ . That is, for any agent, there can be a facet where that agent is uncertain about the truth value of  $P$ . This cannot be the case in the language of local variables, where agents are always certain about the truth values of their local variables. This is the sense in which our system is more general than local variables.

We can spell out this greater generality more specifically. In the language of local variables, the following axiom schema, locality, is sound. Let  $P \in \mathfrak{P}_a$ :

$$\mathbf{LOC} : P \rightarrow B_a P$$

If  $P$  is local to agent  $a$  and true, then  $a$  believes that  $P$ . By contrast, in our setting, the following schema is sound for any  $P \in \mathfrak{P}$ :

$$\mathbf{NU} : P \rightarrow \bigvee_{a \in Ag} B_a P$$

The name **NU** stands for “No Uncertainties”. The schema states that if an atom is true, then some agent believes it to be true. However, unlike in the language of local variables, which agent believes the atom to be true can vary from world to world.

It is clear that under the axioms of propositional logic,  $\mathbf{LOC} \vdash \mathbf{NU}$ . The converse is generally not true. Consider the model in Figure 9.

There are three facets,  $\{a_0, b_1\}$ ,  $\{a_1, b_0\}$ , and  $\{a_1, b_1\}$ . Call these  $w_0$ ,  $w_1$ , and  $w_2$  respectively. Facets  $w_0$  and  $w_1$  are the only two facets where  $P$  is true. However, at  $w_0$ ,  $B_a P$  is true but  $B_b P$

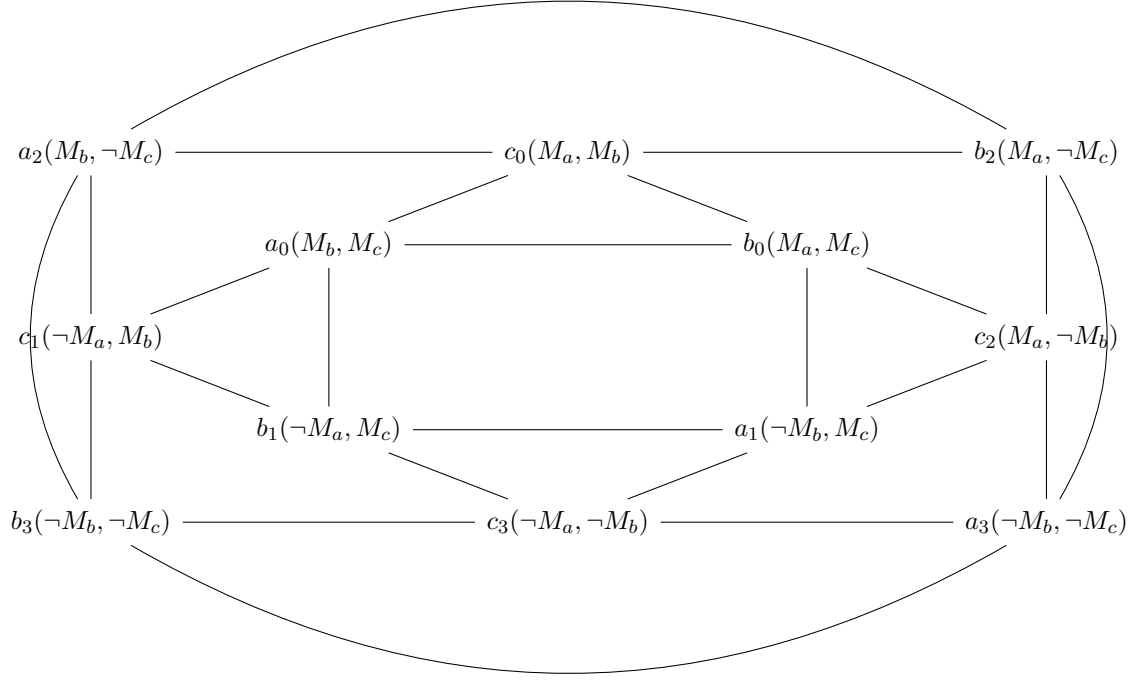


Figure 8: A simplicial model capturing the initial configuration of the Muddy Children problem with 3 agents. Note that  $S_a = S_b = S_c$ . In general, when a face is present for every agent, we will draw the face as black.

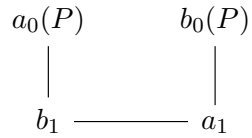


Figure 9: A simplicial belief model with two agents and three worlds/facets.

is false. Conversely, at  $w_1$ ,  $B_a P$  is false and  $B_b P$  is true. So, **LOC** is not valid in this model.

We have the following two theorems.

**Theorem 3.1** (Soundness and Completeness). *The axiom schema **K45+NU**, plus the axioms of propositional logic and the inference rules necessitation and modus ponens are sound and complete with respect to simplicial belief models.*

*Proof.* Appendix 3.7 □

A reasonable question to ask is why the seriality schema, **D**, is not sound. This is because it is possible for an  $a$ -colored perspective, call it  $n$ , to be such that there is no  $F \in \mathcal{F}(S_a)$  such that  $n \in F$ . We will call such perspectives **isolated**. It is easy enough to assume that no perspectives are isolated, in which case **D** is indeed sound. However, our revision concept will take as input models with no isolated perspectives, and create as output models with isolated perspectives. We will explore this in detail in Section 3.5.

### 3.4 The Revision Concept

Belief revision is certainly not a new idea in the realm of formal philosophy. [22, 24, 1, 3] The basic idea is quite simple. When an agent learns  $\varphi$ , rather than simply eliminate all those possible worlds which she formally considered possible, but now are ruled out because they satisfied  $\neg\varphi$ , she instead may replace some of those eliminated worlds with “nearby” worlds that satisfy  $\varphi$ . For example, suppose agent  $a$  considers two worlds possible,  $w_1$  and  $w_2$ . Suppose there are two worlds she does not consider possible,  $w_3$  and  $w_4$ . Let  $P$  be true at  $w_1$ ,  $w_3$ , and  $w_4$ , and  $Q$  be true at  $w_2$  and  $w_3$ . Now suppose  $a$  learns that  $P$  is true. She now no longer considers  $w_2$  possible. However, under revision, instead of merely eliminating  $w_2$ , she may replace it with a “nearby” world that satisfies  $P$ . Both  $w_3$  and  $w_4$  are candidates for replacement, then. However, one might further argue that  $w_3$  is “nearer” to  $w_2$  than  $w_4$ , as both  $w_2$  and  $w_3$  make true  $Q$ . In this case, after learning  $P$ , agent  $a$  would switch from considering the set  $\{w_1, w_2\}$  possible to considering  $\{w_1, w_3\}$  possible. Call this Scenario 1. If, instead, one considered  $w_3$  and  $w_4$  equally near to  $w_2$ , then after learning  $P$  she would consider  $\{w_1, w_3, w_4\}$  possible. Call this Scenario 2.

Many variations have been given on this basic premise. In the initial presentation by David Lewis, one can specify a “nearness” function that takes any world/formula pair to the nearest set of worlds satisfying the given formula.[24] More specifically,  $f : W \times \mathbf{Form} \rightarrow 2^W$  is a nearness function, where **Form** is the set of formulas.<sup>11</sup> Under all forms of revision, if an agent starts out believing in the set of worlds  $X$ , then after learning  $\varphi$ , they believe the set  $\bigcup_{w \in X} f(w, \varphi)$ . In Scenario 1 above,  $f(w_2, P) = \{w_3\}$  and  $f(w_1, P) = \{w_1\}$ .<sup>12</sup> So, the agent revises her beliefs from  $X = \{w_1, w_2\}$  to  $\bigcup_{w \in X} f(w, P) = \{w_1, w_3\}$ . In Scenario 2,  $f(w_2, P) = \{w_3, w_4\}$  and  $f(w_1, P) = \{w_1\}$ . So, the agent revises her beliefs from  $X = \{w_1, w_2\}$  to  $\bigcup_{w \in X} f(w, P) = \{w_1, w_3, w_4\}$ . Importantly, in this initial presentation, Lewis makes no restrictions on the nearness function.

Grove, by contrast, does make some restrictions on nearness functions. [22] For Grove, nearness is given by a system of nested spheres on the set of worlds  $W$ <sup>13</sup>. More specifically,  $\mathbf{S} \subseteq 2^W$  is a system of nested spheres if and only if:<sup>14</sup>

1.  $\mathbf{S}$  is totally ordered under subset.
2. There is a nonempty set  $X \in \mathbf{S}$  such that  $X$  is minimal under the subset ordering.
3.  $W \in \mathbf{S}$

<sup>11</sup>This is not the notation that Lewis uses but it will be clearer for our purposes.

<sup>12</sup>It is typically assumed that if a world  $w$  satisfies a formula  $\varphi$ ,  $f(w, \varphi) = w$ .

<sup>13</sup>Grove assumes that the set of worlds is the set of all maximal consistent sets of formulas. We will present his theory in a more general context here.

<sup>14</sup> $|\varphi|$  is the set of worlds satisfying  $\varphi$ .



4. For any formula  $\varphi$ , there is a set  $V \in \mathbf{S}$  such that  $|\varphi| \cap V \neq \emptyset$  and for all  $U \in \mathbf{S}$  such that  $|\varphi| \cap U \neq \emptyset$ ,  $V \subseteq U$ . That is, for any formula  $\varphi$ , there is a subset minimal sphere  $V$  intersecting that formula's satisfaction set.<sup>15</sup>

The reader should note that the minimal set under the ordering,  $X$ , is conceptually important.  $X$  is the set of worlds the agent starts out believing, before she learns any new information. From this, Grove defines the notion of the “closest” worlds in  $W$  to  $X$  in which a formula  $\varphi$  holds. Let  $\min_\varphi(\mathbf{S})$  be the smallest “sphere” in  $\mathbf{S}$  which contains worlds satisfying  $\varphi$ . That is,  $\min_\varphi(\mathbf{S})$  is the unique  $V \in \mathbf{S}$  such that  $|\varphi| \cap V \neq \emptyset$  and for all  $U \in \mathbf{S}$ , then if  $|\varphi| \cap U \neq \emptyset$ , then  $V \subseteq U$ . With this in hand, we say that the closest worlds in  $W$  to  $X$  satisfying  $\varphi$  are given by

$$f'(X, \varphi) = |\varphi| \cap \min_\varphi(\mathbf{S})$$

That is,  $f'(X, \varphi)$  is the set  $V$  as specified in the 4th property of a system of nested spheres.  $f'$  is not a nearness function as we have defined it, however. Ultimately, Grove appeals to the following nearness function:

$$f(w, \varphi) = \begin{cases} X \cap |\varphi| & \text{if } X = f'(X, \varphi) \\ f'(X, \varphi) & \text{if } X \neq f'(X, \varphi) \end{cases}$$

Note that  $X = f'(X, \varphi)$  if and only if  $X \cap |\varphi| \neq \emptyset$ . So, the specified nearness function says that if the agent believes  $\varphi$  is possible, the nearest world to  $w$  is always those worlds in  $X$  satisfying  $\varphi$ , namely  $X \cap \varphi$ . Otherwise, the agent shifts to those worlds satisfying  $\varphi$  in the smallest sphere that contains such worlds, i.e.,  $f'(X, \varphi)$ . This ensures that the agent does not *revise* unless she rules out *all* of the worlds she initially considers possible. By “revise”, we mean specifically the process of considering new worlds possible which she did not consider possible before. This is distinct from merely “updating”, which could be revision, or simply removing those worlds contradicting what was just learned. So, in the Grove model, the agent “updates” each time she learns  $\varphi$ , but she only “revises” when  $X \cap |\varphi| = \emptyset$ . By contrast, in the more general Lewis setting, the agent may revise any time she learns a new proposition.

Let's look at an example of the Grove model in action. Recall our story from before. Agent  $a$  considers two worlds possible,  $w_1$  and  $w_2$ . There are two worlds she does not consider possible,  $w_3$  and  $w_4$ .  $P$  is true at  $w_1$ ,  $w_3$ , and  $w_4$ , and  $Q$  is true at  $w_2$  and  $w_3$ . Since  $X = w_1, w_2$ , the initial set of possible worlds, is such that  $X \cap |P| \neq \emptyset$ ,  $f(w_1, P) = f(w_2, P) = X \cap |P| = \{w_1\}$ , giving that after learning  $P$  the agent believes in the singleton set of worlds  $\{w_1\}$ . This captures the intuition that the agent should not revise her beliefs unless what she learns *contradicts* her beliefs.<sup>16</sup> This presents us with two different intuitions for revision definitions. On the one hand, an agent can revise every time she learns a new piece of information, seeking to replace any world which she has ruled out. On the other hand, an agent only revises if she has ruled out every world she initially considers possible. Ultimately we will provide a notion of revision which can be modified to be compatible with either intuition.

Both of these examples nicely illustrate what we take to be the biggest issue with belief revision as a model of learning. That is, in practice, it is very difficult to know which worlds should be considered nearer to any other. Lewis, of course, makes no restrictions on nearness whatsoever. While Grove does make restrictions on nearness, they do not tell a modeler which worlds should go into which spheres in a practical sense. If one were to attempt to apply Grove's model of revision in practice, no clues seem to be given as to which worlds are nearer to any other. Some of this is by design. For example, when using revision to give a semantics for counterfactuals or other natural language constructs, one can select worlds to be near to each other precisely to get the intended speaker intuition about a sentence to fall out. [27] However, when one does not have such speaker intuition, or any other clear notion about which worlds should be nearer to each other, it remains difficult to practically use revision.

<sup>15</sup>As before, this is not given in the original notation, but is clearer for our purposes.

<sup>16</sup>This is the third positive axiom (+3) of the “AGM” postulates, as presented in [1]. Of course, Grove shows that nested sphere models satisfy all of the 8 positive AGM postulates. [22]

The ultimate goal of this chapter will be to motivate a particular concept of “nearness” which is very easily modeled in simplicial semantics. We should not overstate our aims here, however. While we will motivate what we hope to be an intuitive notion of nearness, we do not believe this is the only, or even best, notion of nearness which one could appeal to. Nor do we believe that this notion of nearness will apply in every situation one can imagine. It will suffice for us that this notion of nearness applies in a wide range of contexts. All we are hoping to demonstrate to the reader is a particular intuition which seems reasonable enough that it is worth following up on in a formal setting.

Consider the following case: You are agent  $a$ . You have three friends, call them agents  $b$ ,  $c$ , and  $d$ . Therefore the facets in this case are tetrahedrons with four perspectives total, one for each agent. There are three facts, call them  $B$ ,  $C$ , and  $D$ . The fact  $X$ , should be read as “ $x$  is at the park”, hence  $B$  means that “ $b$  is at the park” and so on.

Initially, you consider a single world/facet possible, consisting of your perspective, and a perspective for each of your friends. We will label your perspective  $a'$ . The perspectives of each of your friends, which we will label  $x_1$ , consist of the fact that  $X$  is true, and no other facts. So, for example, the  $b$ -perspective  $b_1$  is such that  $L(b_1)(B) = 1$ , and  $L(b_1)(C) = L(b_1)(D) = 2$ . Ambiently, there are additional perspectives for your friends, call them  $x_0$  for each friend  $x$ . These are such that  $X$  is false, and no other facts are true or false. Or, more specifically, for the  $b$ -perspective  $b_0$ ,  $L(b_0)(B) = 0$ , and  $L(b_0)(C) = L(b_0)(D) = 2$ . As we said before, initially, you only consider one world/facet possible, and this is the tetrahedron  $\{a', b_1, c_1, d_1\}$ . Hence, you believe that each of your friends is at the park.

Now you learn the following fact: Either  $b$  and  $c$  are both not at the park,  $d$  is not at the park, or all three are not at the park. This is the proposition  $(\neg B \wedge \neg C) \vee \neg D$ . Since this contradicts your beliefs, you no longer consider any worlds possible, and hence you must revise your beliefs. How should you do so? Let’s consider some possible candidates for revision. Intuitively, these worlds should preserve your perspective, since we interpret facts at your perspective as indefeasible as you learn things. These are the candidate replacement facets for the contradicted facet  $X$ :

- $X_1 := \{a', b_1, c_1, d_0\}$
- $X_2 := \{a', b_0, c_0, d_1\}$
- $X_3 := \{a', b_0, c_0, d_0\}$

Which of these is the best replacement? We want to argue that  $X_1$  is a better candidate than  $X_2$ , and  $X_2$  is a better candidate than  $X_3$ , corresponding to the fact that  $|X \cap X_1| > |X \cap X_2| > |X \cap X_3|$ .  $X_1$  and  $X$  share “more in common” with each other than  $X$  and  $X_2$ . Indeed, if worlds are made of perspectives, as is the conceit of simplicial semantics, then  $X$  and  $X_1$  sharing more perspectives than  $X$  and  $X_2$  becomes a simple metric by which we can say  $X$  and  $X_1$  are “more similar” or “nearer”. In practice, this consists of wishing to preserve as many of your fellow agents’ perspectives, or points of view, as possible. So, in this example,  $X_1$  is nearer than  $X_2$  because  $X_1$  preserves both  $b$ ’s and  $c$ ’s perspectives, while  $X_2$  preserves only  $d$ ’s perspective.  $X_3$ , of course, preserves nothing.

This example is fairly extreme, however. This is a case of revision where you are revising your beliefs in light of all worlds you consider possible being ruled out. As mentioned above, we want to allow for less forced cases of revision, where you may revise even if you still consider some worlds possible. In the new setup, you initially consider two worlds possible,  $X$  and  $X_3$ . Note that this means you are initially uncertain about all three of  $B$ ,  $C$ , and  $D$ . You then learn that  $d$  is at the park, that is, you learn that  $D$ . This is of course consistent with  $X$ , but rules out  $X_3$ . So, what are the candidates for replacing  $X_3$ ? As before, they must contain  $a'$ , and they should make true  $D$ . This gives us the following candidates:

- $X_2 := \{a', b_0, c_0, d_1\}$
- $X_4 := \{a', b_1, c_0, d_1\}$
- $X_5 := \{a', b_0, c_1, d_1\}$

- $X := \{a', b_1, c_1, d_1\}$

Note that  $|X_3 \cap X_2| > |X_3 \cap X_4| = |X_3 \cap X_5| > |X_3 \cap X|$ . If we use our intuition from previously, that preserving more perspectives gives the best notion of nearness, then  $X_2$  is our best candidate replacement world. This also has a few other niceties. We learn  $D$ , which in this setup means we learn information about  $d$ 's perspectives. In this setup, there is no overlap between the perspectives of your friends. Each perspective corresponds uniquely to a specified literal. So, choosing  $X_2$  as the replacement facet for  $X_3$  means following the intuition to minimally alter your opinion of  $b$ 's perspective and  $c$ 's perspective when learning about  $d$ 's perspective. In this case, you preserve your uncertainty about the facts  $B$  and  $C$  when you learn  $D$ . There is another intuition, however, which suggests that in this case, you should actually pick  $X$  as the replacement for  $X_3$ , despite the fact that it shares the *fewest* perspectives with  $X_3$  out of all the possible candidates. This is because you already consider  $X$  possible. In terms of uncertainty, this choice captures the fact that your initial uncertainty about  $B$ ,  $C$ , and  $D$  are tied together. Really, you believe that all three facts are true, or all three are false. So, if you learn that one fact is true, based on your initial setup, you infer that all three are true. This, of course, corresponds with the usual Bayesian update on new information. Under this intuition, revision would be reserved *only* for the case where *every* world you consider possible is ruled out. We will provide formal definitions that capture both of the above intuitions.

However, all of these examples are restricted to the case where there is a one-to-one correspondence between perspectives and literals.<sup>17</sup> Hence, we could equally motivate all of the above examples by arguing we are saying worlds are nearer the more literals they validate in common, rather than the more perspectives they share. We should check that our “preserving perspectives” notion makes sense in an example where this is not true. As before, your perspective is  $a'$ . Following our example from the introduction, we will imagine a case where our friends are at the park, and they are observing the status of a pond at said park. Let  $D$  stand for “there are ducks in the pond” and  $F$  stand for “there are frogs in the pond”. We set up the perspectives as follows. For any  $x \in \{b, c, d\}$ , the perspective  $x_{i,j}$  is such that  $L(x_{i,j})(D) = i$  and  $L(x_{i,j})(F) = j$ . So,  $b_{1,0}$  is the  $b$ -perspective where  $b$  is seeing that there are ducks in the pond, and there are no frogs in the pond. By contrast,  $c_{1,2}$  is the  $c$ -perspective where  $c$  is seeing that there are ducks in the pond, but they are not seeing either that there are or are not frogs in the pond.

Initially, you consider two worlds possible. These are  $X := \{a', b_{1,1}, c_{1,1}, d_{1,1}\}$ , and  $Y := \{a', b_{1,0}, c_{1,2}, d_{1,2}\}$ . In the first world, all of your friends see ducks and frogs. In the second world, all of your friends see ducks, while  $b$  sees that there are no frogs, and  $c$  and  $d$  neither see frogs nor see that there are no frogs. Then you learn that  $F$ , which contradicts  $Y$ . One can easily check that the nearest candidate replacement world is given by  $Y' := \{a', b_{1,1}, c_{1,2}, d_{1,2}\}$ . At this world,  $D$  and  $F$  are both true. But what about  $Y'' := \{a', b_{1,2}, c_{1,1}, d_{1,2}\}$ ? This world satisfies the same literals as  $Y'$ , making true both  $D$  and  $F$ . However,  $1 = |Y \cap Y'| > |Y \cap Y''| = 2$ . This is because  $Y$  differs from  $Y'$  only in  $b$ 's perspective, but  $Y''$  differs from  $Y$  in both  $b$  and  $c$ 's perspective. This motivates the notion of nearness best.  $Y$  was a world making true  $D$  and  $\neg F$ . However,  $b$ 's perspective was the reason that  $F$  was false at  $Y$ . So, when we learn that  $F$  is actually true, when finding nearby candidate worlds, we should focus on changing  $b$ 's perspective when building our replacement for  $Y$ .

This “building” intuition is useful to elaborate on. When imagining candidate replacement worlds, the following story is useful. Revision has, at its basis, the idea that we replace worlds with nearby ones on a world-by-world basis. That is, it doesn't (necessarily) matter that  $w$  is related to  $w'$  when determining nearness. So, when  $Y$  is ruled out by learning  $F$ , we can focus on finding a replacement for  $Y$  in isolation. When we look at *why*  $Y$  is ruled out by  $F$ , we find, in this case, that the culprit is  $b$ 's perspective. So, when searching for a replacement, we start by swapping out other  $b$ -perspectives in until we get something that works. This is the sense in which we are “building” the replacement(s) for  $Y$ . As it turns out, in this example, changing  $b$ 's perspective alone is sufficient. If it weren't, we could, intuitively, *then* try swapping out  $c$

<sup>17</sup>As such, all of the above examples may be given in the language of local variables. [14]

perspectives and  $d$  perspectives until we found something that worked. But, since swapping the  $b$  perspective is a forced choice, we try that alone first. Only if that fails do we try swapping more perspectives. Hence, sharing more perspectives gives us our notion of nearness.

A final note is that the notion of a formula being true at a particular agent's perspective is not formalized directly in our language. For example,  $Y''$  and  $Y'$  satisfy exactly the same non-modal formulas. However, doing so would seem to be simple. One could add a modal operator  $A_x$  for all  $x \in Ag$  which applies only to literals, and for a model  $\mathcal{M}$  and facet  $X$ , we could say that  $\mathcal{M}, X \models A_x P$  iff  $L(\pi_x(X))(P) = 1$ , and  $\mathcal{M}, X \models A_x \neg P$  iff  $L(\pi_x(X))(P) = 0$ . It would follow that  $\mathcal{M}, Y' \models A_b P$ , but  $\mathcal{M}, Y'' \not\models A_b P$ .

It seems that this could be a fruitful connection between simplicial semantics and something like awareness logic.[15] One can read  $A_x(P)$  as stating that “ $x$  is aware of  $P$ ”, and as a consequence think of the literals true at a perspective as constituting what the agent is “aware” of at that perspective. Because what is and is not assigned to perspectives, and hence what agents are and are not aware of, plays such a major role in our conception of revision, this would seem to be a fruitful direction for future work. However, it is beyond the scope of this chapter.

### 3.5 Simplicial Semantics for Belief Revision

In order to formalize the intuition set out in Section 3.4, we will turn to an example. Let  $Ag = \{a, b, c\}$ , and suppose each agent has a privately held bit value, denoted with the proposition  $P_i$  for  $i \in Ag$ .  $P_i$  denotes that  $i$ 's bit value is 1, and  $\neg P_i$  denotes that  $i$ 's bit value is 0. Because the value is privately held,  $i$  is aware of the truth value of  $P_i$  (it is in  $i$ 's perspective) but no agent other than  $i$  is aware of the truth value of  $P_i$ . This gives us six perspectives, two for each agent. We model this formally as  $N = \{a_0, a_1, b_0, b_1, c_0, c_1\}$ ,  $V(i_j) = i$  and  $L(i_j)(P_i) = j$  for all  $i \in Ag$  and  $j \in \{0, 1\}$ . These are depicted in Figure 10.

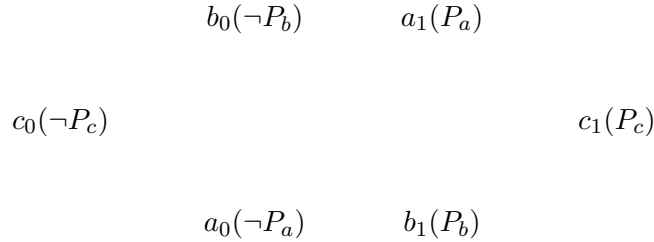


Figure 10: 6 perspectives modelling 3 agents each with a privately held bit value.

Now, assume that for all  $i \in Ag$ ,  $\mathcal{F}(S_i) = \{\{a_1, b_1, c_1\}, \{a_1, b_1, c_0\}, \{a_0, b_0, c_0\}\}$ . We will call these facets  $w_0 = \{a_0, b_0, c_0\}$ ,  $w_1 = \{a_1, b_1, c_1\}$ , and  $w_2 = \{a_1, b_1, c_0\}$  for brevity. The choice of worlds in this example is arbitrary except insofar as they will be illustrative when considering our revision mechanism. This is drawn in Figure 11.

Now we suppose that  $P_a$  is publicly announced. How should we imagine each agent learning this piece of information? It is useful to think perspective by perspective. First, imagine you are agent  $a$ . Figure 11 represents your epistemic frame prior to learning any information. If you are in perspective  $a_1$ , you consider two worlds possible:  $w_1$  and  $w_2$ . Both of these worlds make true  $P_a$ , so you do not cease to consider either possible after announcement. However, suppose you are in perspective  $a_0$ . What has just been publicly announced,  $P_a$ , contradicts what is true at the only world you consider possible, namely  $w_0$ . So, you no longer consider this world possible. Moreover, revision is not possible. Candidate revision worlds should at minimum preserve your perspective. Since at  $a_0$ ,  $\neg P_a$  is in your perspective, any world containing  $a_0$  will contradict  $P_a$ . This update is drawn in Figure 12. Note that agent  $a$ 's update is the usual public announcement update. Worlds which contradict the announcement are removed with no revision.

Things are trickier for agent  $b$ . Again, it is useful to consider things perspective by perspective. Suppose you are in perspective  $b_1$ . Then you consider two worlds possible, those

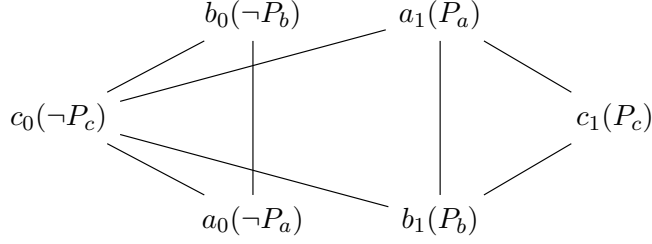


Figure 11: The model of 3 agents with privately held bit values prior to announcement. The three triangles correspond to the three worlds.

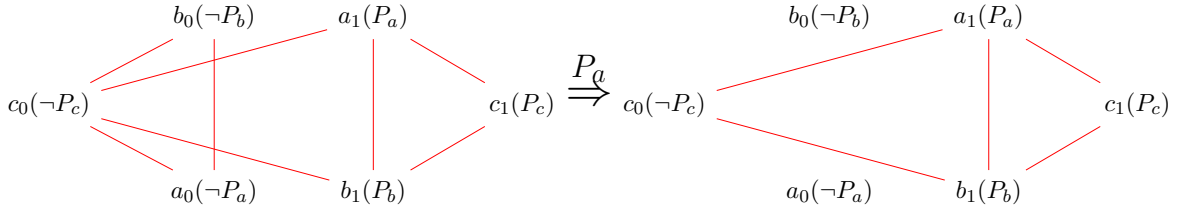


Figure 12:  $S_a$  before and after  $P_a$  is publicly announced

being  $w_1$  and  $w_2$ . Since both of these worlds satisfy  $P_a$ , neither is removed or changed after the announcement. Now suppose you are in perspective  $b_0$ . Before the announcement, you consider only world  $w_0$  possible. This is then contradicted by the announcement, and so must be removed. However, unlike agent  $a$ ,  $b$  has the possibility to revise. As mentioned before, candidates for revision should preserve  $b$ 's point of view. Moreover, they need to satisfy what was just announced,  $P_a$ . The third condition is the one motivated in Section 3.4. Namely, replacement worlds should preserve as many perspectives from the replaced world as possible. In this case, the world to be replaced is  $w_0 = \{a_0, b_0, c_0\}$ . By the first condition, candidate replacement worlds need to contain  $b_0$ . By the second condition, they need to satisfy  $P_a$ . In our example, that means candidate replacement worlds for  $w_0$  after  $P_a$  is announced must contain  $a_1$ . This gives us two potential candidates:  $w_3 = \{a_1, b_0, c_0\}$  and  $w_4 = \{a_1, b_0, c_1\}$ . Note that  $|w_3 \cap w_0| > |w_4 \cap w_0|$ . Hence,  $w_3$  and  $w_0$  share more perspectives, namely two, than  $w_4$  and  $w_0$ , which only share one perspective. So, from our informal intuition,  $w_3$  is the best candidate replacement for  $w_0$  after  $P_a$  is announced. This is shown in Figure 13.

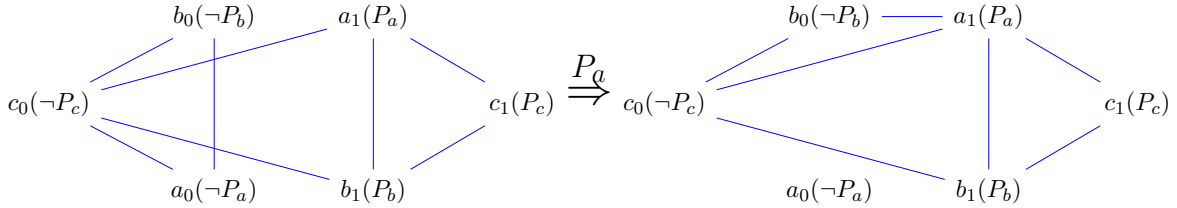


Figure 13:  $S_b$  before and after  $P_a$  is publicly announced

The last agent to explore is agent  $c$ . Suppose you are in perspective  $c_1$ . Before the announcement, you consider one world possible, namely  $w_1$ . This is not contradicted by the announcement, so there is no change. Now suppose you are in perspective  $c_0$ . You consider two worlds possible, namely  $w_0$  and  $w_2$ . Since  $w_0$  satisfied  $\neg P_a$ , this is ruled out by the announce-

ment of  $P_a$ . Moreover, the same revision calculation carried out for agent  $b$  reveals that the best candidate for  $w_0$  is  $w_3$ . This is drawn in Figure 14.

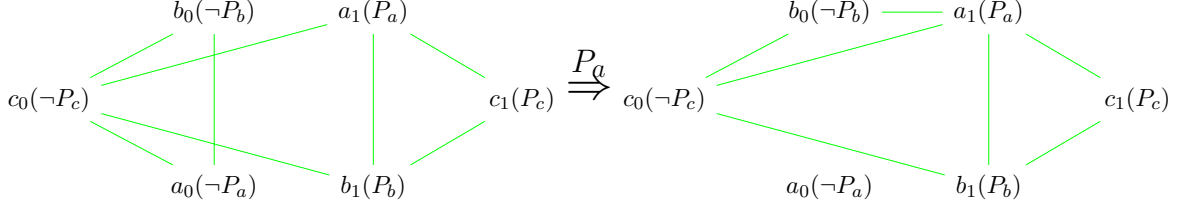


Figure 14:  $S_c$  before and after  $P_a$  is publicly announced, assuming revision happens to every eliminated world.

However, this is not the only intuition one could have about how  $c$  should revise. When  $b$  revised around perspective  $b_0$ , it is notable that all of the worlds containing  $b_0$  had been eliminated by the announcement. By contrast, both there are worlds consistent with the announcement containing both  $c_1$ , and in particular,  $c_0$ , namely  $w_2$ . Because  $c$  already considers  $w_2$  possible, it is reasonable to argue that  $w_2$  is “nearer” to  $w_0$ , even though  $|w_3 \cap w_0| > |w_2 \cap w_0|$ . This would track the intuition that revision should only happen if all worlds the agent considers possible are eliminated, as per Grove’s nested sphere model, discussed in Section 3.4. [22] In this case, the revision looks as it appears in Figure 15.

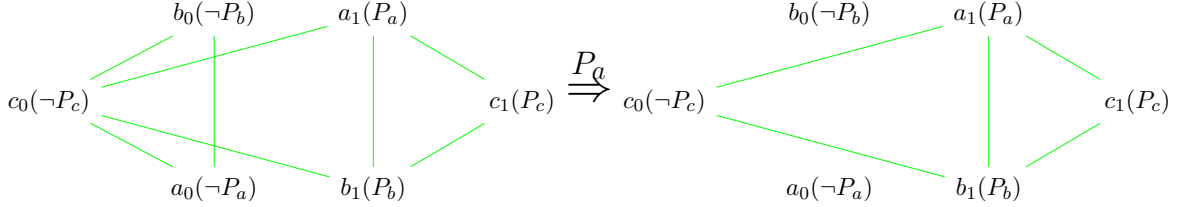


Figure 15:  $S_c$  before and after  $P_a$  is publicly announced, assuming revision happens only when a perspective is no longer contained in any world.

Fix a formula  $\varphi$  which contains no modalities in its recursive construction. That is,  $\varphi$  is a proposition. Following the intuitions above, given a model  $\mathcal{M}$ , if a facet  $X \in \mathcal{F}(S_a)$  is ruled out by a public announcement of  $\varphi$ , candidate replacement facets  $Y$  needs to satisfy the following three conditions:

1.  $X$  and  $Y$  need to share the same  $a$ -perspective, or more formally,  $\pi_a(X) = \pi_a(Y)$
2.  $Y$  need to satisfy  $\varphi$ , or more formally,  $\mathcal{M}, Y \models \varphi$
3. If  $Z$  satisfies conditions 1 and 2,  $Y$  needs to share at least as many perspectives with  $X$  as  $Z$ , or more formally,  $|Y \cap X| \geq |Y \cap Z|$ .

We can formalize these three conditions into the following revision rule:

**Definition 3.1.**  $\mathcal{M}[\varphi]$

Consider a UCF simplicial model  $\mathcal{M} = \langle N, V, L, \{S_a\}_{a \in A} \rangle$ , and  $\varphi$  a formula with no modalities in its recursive construction. Let  $N_\varphi := N$ ,  $V_\varphi := V$ , and  $L_\varphi := L$ . Further, let  $\mathbf{F}_\varphi$  be the subset of  $\mathcal{F}(\mathfrak{M}((N_\varphi, V_\varphi, L_\varphi)))$  which satisfy  $\varphi$ . We need to define a function  $\mathcal{R}_a : \mathcal{F}(S_a) \times \mathcal{L}_B(\text{Ag}) \rightarrow 2^{\mathbf{F}_\varphi}$ .

$$\mathcal{R}_a(F, \varphi) := \{G \in \mathbf{F}_\varphi \mid (\pi_a(F) = \pi_a(G)) \wedge \forall X \in \mathbf{F}_\varphi ((\pi_a(X) = \pi_a(G)) \rightarrow (|F \cap G| \geq |X \cap G|))\}$$

Note that  $\mathcal{R}_a : \mathcal{F}(S_a) \times \mathcal{L}_B(Ag) \rightarrow 2^{\mathbf{F}_\varphi}$  is a nearness function.<sup>18</sup> Further, note that, if  $G \in \mathbf{F}_\varphi$ , then  $\mathcal{R}_a(G, \varphi) = \{G\}$ . Let each  $S_{a,\varphi}$  be the simplicial complex whose facets are elements of the set  $\bigcup_{F \in \mathcal{F}(S_a)} \mathcal{R}_a(F, \varphi)$ . Then  $\mathcal{M}[\varphi] := \langle N_\varphi, V_\varphi, L_\varphi, \{S_{a,\varphi}\}_{a \in A} \rangle$ .

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If  $S$  is not maximal, simply take each  $S_{a,\varphi}$  be the simplicial complex whose facets are elements of the set  $\mathcal{F}(S) \cap \bigcup_{F \in \mathcal{F}(S_a)} \mathcal{R}_a(F, \varphi)$ . It is easy to check that the above definition gives a UCF sub-simplicial complex of the maximal complex of  $N_\varphi$ ,  $V_\varphi$ , and  $L_\varphi$  (or a UCF subcomplex of  $S$  if  $S$  is non-maximal). Therefore, the above definition gives us a model  $\mathcal{M}[\varphi]$  which represents the result of a model  $\mathcal{M}$  after each agent revises their beliefs in light of a public announcement of  $\varphi$ .

A potential fourth condition is that, if there is already a facet  $Y \in \mathcal{F}(S_a)$  such that  $\pi_a(X) = \pi_a(Y)$  and  $\mathcal{M}, Y \models \varphi$ , then  $Y$  is the nearest facet to  $X$ , regardless of intersection size. This potential fourth condition is motivated by the intuition that revision should only happen if a perspective becomes isolated, i.e., an  $a$ -perspective is no longer contained in a facet of  $S_a$ . We formalize this as follows:

**Definition 3.2.**  $\mathcal{M}[\varphi]_G$ <sup>20</sup>

Consider a UCF simplicial model  $\mathcal{M} = \langle N, V, L, \{S_a\}_{a \in A} \rangle$ , and  $\varphi$  a formula with no modalities in its recursive construction. Let  $N_\varphi := N$ ,  $V_\varphi := V$ , and  $L_\varphi := L$ . We need to define a function  $\mathcal{R}_{a,G} : \mathcal{F}(S_a) \times \mathcal{L}_B(Ag) \rightarrow 2^{\mathbf{F}_\varphi}$ . Fix the set  $\mathbf{T}_a(F, \varphi) := \{X \in \mathbf{F}_\varphi \cap \mathcal{F}(S_a) \mid \pi_a(X) = \pi_a(Y)\}$ . Then

$$\mathcal{R}_{a,G}(F, \varphi) := \begin{cases} \mathbf{T}_a(F, \varphi) & \text{if } \mathbf{T}_a(F, \varphi) \neq \emptyset \\ \mathcal{R}_a(F, \varphi) & \text{if } \mathbf{T}_a(F, \varphi) = \emptyset \end{cases}$$

Note that  $\mathcal{R}_{a,G} : \mathcal{F}(S_a) \times \mathcal{L}_B(Ag) \rightarrow 2^{\mathbf{F}_\varphi}$  is a nearness function. Further, note that, if  $X \in \mathbf{F}_\varphi$ , then  $X \in \mathcal{R}_{a,G}(X, \varphi)$ . Let each  $S_{a,\varphi,G}$  be the simplicial complex whose facets are elements of the set  $\bigcup_{F \in \mathcal{F}(S_a)} \mathcal{R}_{a,G}(F, \varphi)$ . Then  $\mathcal{M}[\varphi]_G := \langle N_\varphi, V_\varphi, L_\varphi, \{S_{a,\varphi,G}\}_{a \in A} \rangle$

As before, it is easy to check that the above definition gives a UCF sub-simplicial complex of the maximal complex of  $N_\varphi$ ,  $V_\varphi$ , and  $L_\varphi$ . Therefore, the above definition gives us a model  $\mathcal{M}[\varphi]_G$  which represents the result of a model  $\mathcal{M}$  after each agent revises their beliefs in light of a public announcement of  $\varphi$ , where revision is restricted to the case where perspectives become isolated, represented by the case where  $\mathbf{T}_a(F, \varphi) = \emptyset$ . While we present both concepts here, for the remainder of the text, we will use  $\mathcal{M}[\varphi]$  unless otherwise stated.

### 3.5.1 Examples

The purpose of these examples is to become familiar with the updating procedure, and to motivate a particular issue that arises under certain interpretations of announcement. Figure 16 captures the idea of “everything going right.” This is the kind of story we want to be able to capture.

Suppose that the “real” world/facet is given by  $\{a_1, b_1, c_1\}$ . Then, at this facet, it is true that each agent has a true bit value of 1, but mistakenly believes that the other agents have a bit value of 0. Each public announcement of the true bit values brings the agents closer to consensus. There is no interesting higher order modal information here, as the different colored simplicial complexes do not interact with each other. Note that at each stage, there is exactly one red, one blue, and one green facet. Therefore, there can be no two worlds, or in our case facets, which are the same color and share a perspective of that color. Hence, there is no uncertainty about worlds/facets, and thus no higher order information.

<sup>18</sup>The total set of worlds  $S$  is  $\mathfrak{M}((N_\varphi, V_\varphi, L_\varphi) = \mathfrak{M}(N, V, L)$ , as  $\mathcal{F}(S_a) \subseteq \mathfrak{M}(N, V, L)$  and  $\mathbf{F}_\varphi \subseteq \mathfrak{M}(N_\varphi, V_\varphi, L_\varphi)$ .

<sup>19</sup>This definition of revision, and the one following, also seems perfectly well motivated in settings where one uses the language of Local Variables. [14] So, if one prefers that setting, this definition should be conceptually compatible.

<sup>20</sup>The subscript stands for “Grove”.

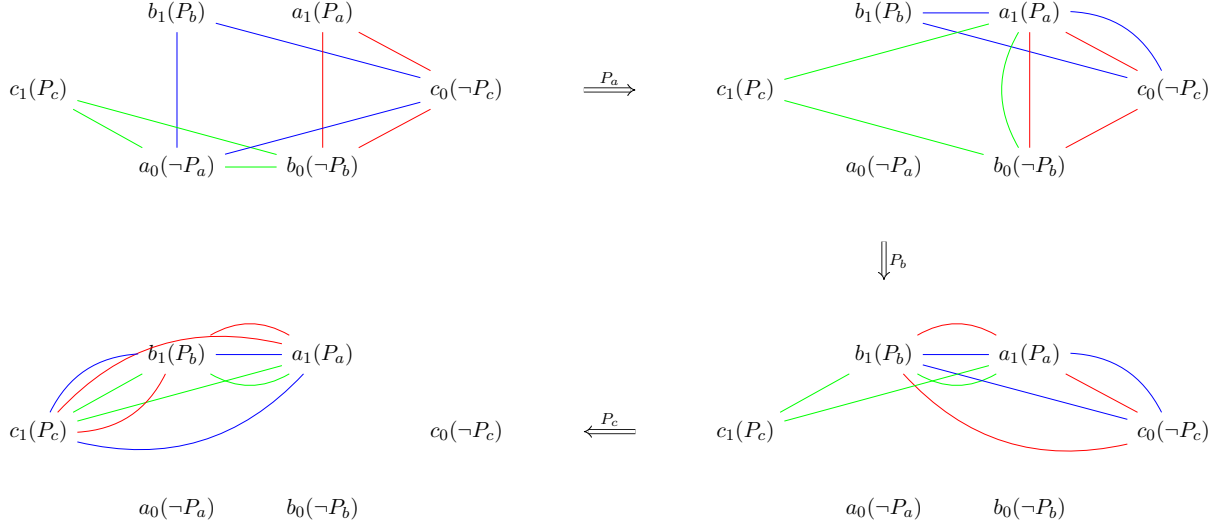


Figure 16: Four models in a sequence of three announcements. The initial configuration depicts three agents, each of whose private bit value is 1, but has the mistaken belief that the other two agents have private bit value of zero. The announcements sequentially correct this misconception.

Figure 17 also captures “things going well,” but importantly motivates details about how we should think about “candidate alternative” for perspectives.

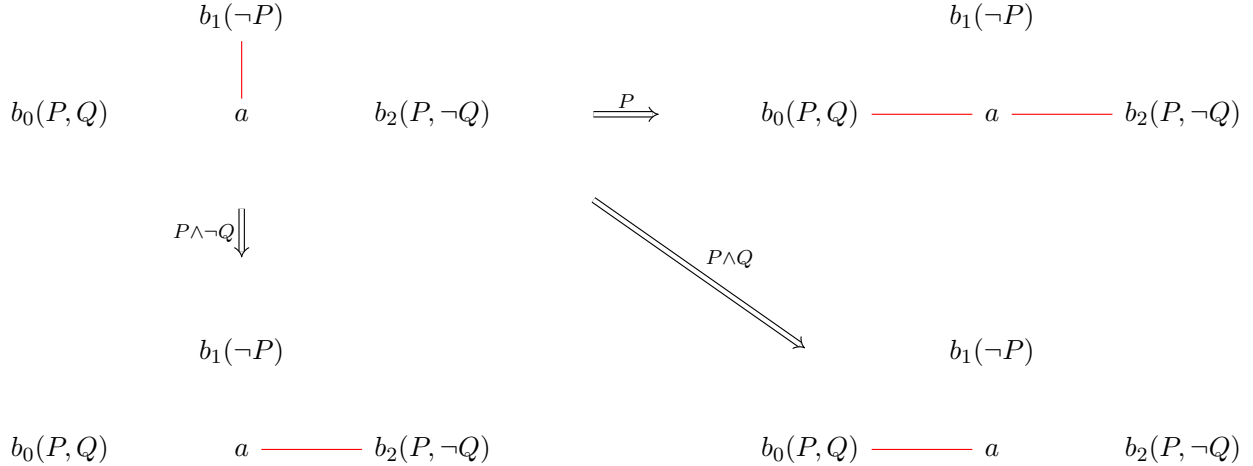


Figure 17: Four models and three distinct updates based on three distinct announcements.

The complexes shown in Figure 17 are all  $S_a$ . There are two key takeaways from this example. Firstly, what is announced significantly affects the update. When  $P \wedge Q$  is announced,  $a$  switches to the available perspective where both of these are true. When  $P \wedge \neg Q$  is announced, a different perspective. Moreover, when only  $P$  is announced, intuitively,  $a$  should become uncertain, as there is no way for them to differentiate between  $b_0$  and  $b_2$  on the available information.

Figure 18 motivates a flaw in this version of the formalism. Agents are, at present, very forgetful of what they’ve learned. If we take announcements to have a high degree of veracity, this seems like an issue that is worth overcoming.

The complexes shown in Figure 18 are all  $a$ ’s complexes. In the example,  $a$  in the first



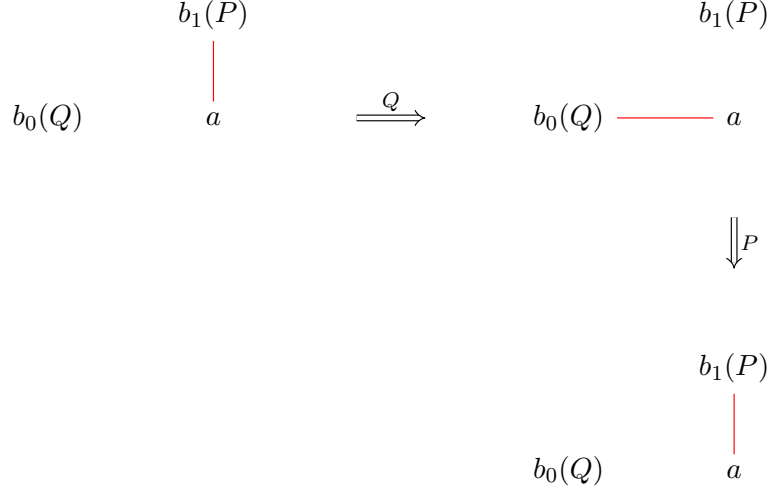


Figure 18: Three models with two agents, showing an initial model with sequential updates after a public announcement of  $Q$  then of  $P$ . Agent  $a$  begins by believing  $P$ .

complex believes  $P \wedge \neg Q$ . Then  $Q$  is announced, and  $a$  believes  $\neg P \wedge Q$ . Then when  $P$  is announced, the agent believes  $P \wedge \neg Q$  again. In this sense, the agent is extremely forgetful of what has just been announced. The agent goes right back to believing  $\neg Q$ , even though they had just heard that  $Q$  was announced.

One might think that the example in Figure 18 is a bit forced. Yes, the agents are forgetful; the example is extremely constrained. Given the facets available to agent  $a$ , the two announcements are in fact inconsistent with each other. As we see in Figure 19, however, adding in more potential facets does not solve the issue.

Figure 19 is similar to Figure 18, except that there is a potential consistent facet for every combination of truth values of  $P$  and  $Q$ . Note that, before any announcements,  $a$  considers  $\neg P \wedge Q$  possible. However, after the announcements of  $P$  and  $Q$ , they should know  $P \wedge Q$ . Yet they consider  $\neg P \wedge Q$  possible again! The situation is not ideal, to say the least.

### 3.5.2 Memory

We will show a simple solution for this “forgetfulness” by using the  $S$  complex as a kind of “memory”.  $S$  is the set from which all facets are selected already. In sequential updates, if subsequent  $S$  keep a record that of everything that has been announced, then the candidate facets for revision, too, will keep such a record. We can define this formally (the new parts of the definition are highlighted in red):

**Definition 3.3.**  $\mathcal{M}[\varphi]_m$

Consider a UCF simplicial model  $\mathcal{M} = \langle N, V, L, \mathbf{S}, \{S_a\}_{a \in A} \rangle$ , and  $\varphi$  a formula with no modalities in its recursive construction.. Let  $N_\varphi := N$ ,  $V_\varphi := V$ ,  $L_\varphi := L$ . Further, let  $\mathbf{F}_\varphi$  be the subset of  $\mathcal{F}(\mathfrak{M}((N_\varphi, V_\varphi, L_\varphi)))$  which satisfy  $\varphi$ . **Also let  $S_\varphi$  be the simplicial complex whose facets are given by  $\mathcal{F}(\mathbf{S}) \cap \mathbf{F}_\varphi$ .** We need to define a function  $\mathcal{R}_a : \mathcal{F}(S_a) \times \mathcal{L}_B(Ag) \rightarrow 2^{\mathbf{F}_\varphi}$ .

$$\mathcal{R}_a(F, \varphi) := \{G \in \mathbf{F}(S_\varphi) \mid (\pi_a(F) = \pi_a(G)) \wedge \forall X \in \mathbf{F}_\varphi ((\pi_a(X) = \pi_a(G)) \rightarrow (|F \cap G| \geq |X \cap G|))\}$$

Note that  $\mathcal{R}_a : \mathcal{F}(S_a) \times \mathcal{L}_B(Ag) \rightarrow 2^{\mathbf{F}_\varphi}$  is a nearness function. Further, note that, if  $G \in \mathbf{F}(S_\varphi)$ , then  $\mathcal{R}_a(G, \varphi) = \{G\}$ . Let each  $S_{a,\varphi}$  be the simplicial complex whose facets are elements of the set  $\bigcup_{F \in \mathcal{F}(S_a)} \mathcal{R}_a(F, \varphi)$ . Then  $\mathcal{M}[\varphi]_m := \langle N_\varphi, V_\varphi, L_\varphi, \mathbf{S}_\varphi, \{S_{a,\varphi}\}_{a \in A} \rangle$ .

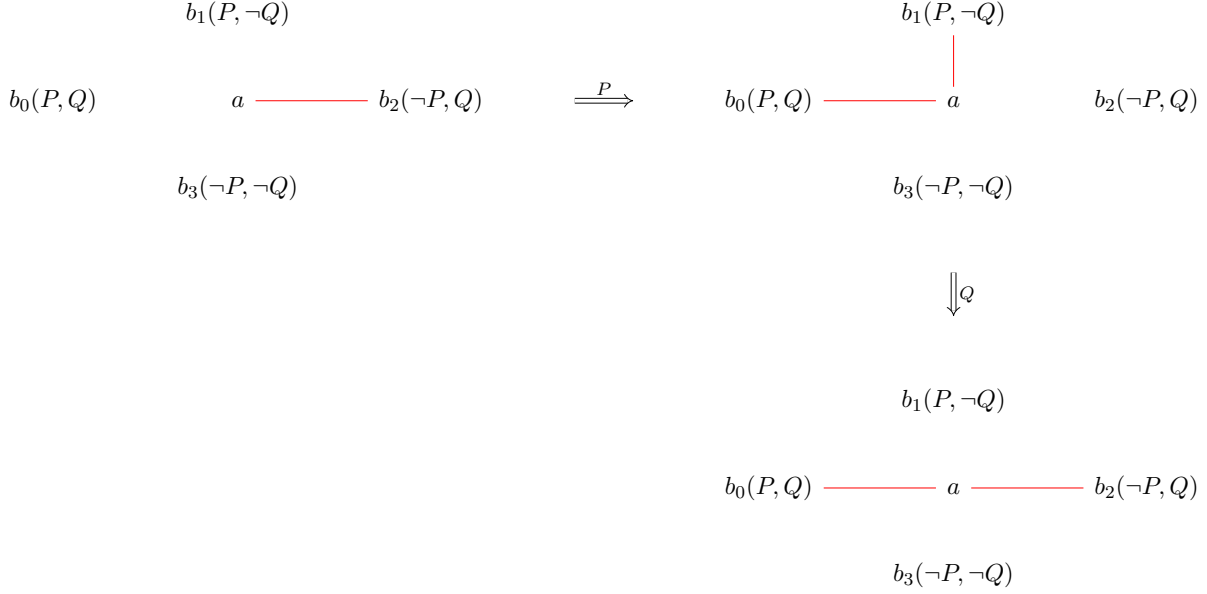


Figure 19: Three models with two agents, showing an initial model with four facets in  $W = \mathcal{M}(N, V, L)$ , corresponding to all possible arrangements of the atoms  $P$  and  $Q$ . As in Figure 18, there are two sequential updates. First the agent updates by  $P$  and then  $Q$ . Agent  $A$  begins by believing  $\neg P \wedge Q$ .

Let us observe how this new definition solves our previous example. Consider Figure 20.

Let us track how  $S$  alters with each update in Figure 20. In the initial configuration,  $\mathcal{F}(S) = \mathcal{M}(N, V, L) = \{\{a, b_0\}, \{a, b_1\}, \{a, b_2\}, \{a, b_3\}\}$ . After  $P$  is announced,  $\mathcal{F}(S_P) = \mathcal{F}(S) \cap \mathbf{F}_P = \{\{a, b_0\}, \{a, b_1\}\}$ . This does not affect the first update, except insofar as the worlds  $\{a, b_2\}$  and  $\{a, b_3\}$  are no longer available for subsequent revisions. This fact, however,  $\mathcal{F}(S_{P,Q}) = \mathcal{F}(S_P) \cap \mathbf{F}_Q = \{\{a, b_0\}\}$ . This means the only available world to revise to is  $\{a, b_0\}$ , and indeed, this is what we see in Figure 20. The issue where the agent has gone back to considering the world  $\{a, b_2\}$  possible, despite it being ruled out in the first update, has been taken care of.

Another potential solution to the issue of memory is to instead “write” what has been announced directly into perspectives. In the conception of memory given above, announcements are taken to be infeasible. But this new conception would treat announcements as “hard” information available to the agents. Working this idea out formally does require altering definitions somewhat. In particular, we will need to allow for “inconsistent” perspectives, that is, perspectives that may contain both  $P$  and  $\neg P$  for some atom  $P$ . To account for this, Let  $L : N \rightarrow 4^{\mathfrak{P}}$ , where  $L(n)(P) = 4$  is interpreted as  $P$  and  $\neg P$  both being assigned to  $n$ . The maximal complex is restricted so that no faces containing a node for which some atom is assigned to 4 are included, in order to preserve the consistency of every face.

This notion of update is in greater accordance with the kinds of updates explored in the literature connecting simplicial complexes and distributed computing. [19, 23] There, announcements change, and generally increase, what is true at a perspective.<sup>21</sup> For our purposes, for this notion of announcement to make sense, we must assume that only atoms are announced.<sup>22</sup> We define this as follows:

**Definition 3.4.**  $\mathcal{M}[P]_h$

<sup>21</sup>In the case where announcements are not public, the number of perspectives increases at each time step. Forthcoming work will seek to model this via action models.

<sup>22</sup>We leave the possibility of generalizing this idea for future work.

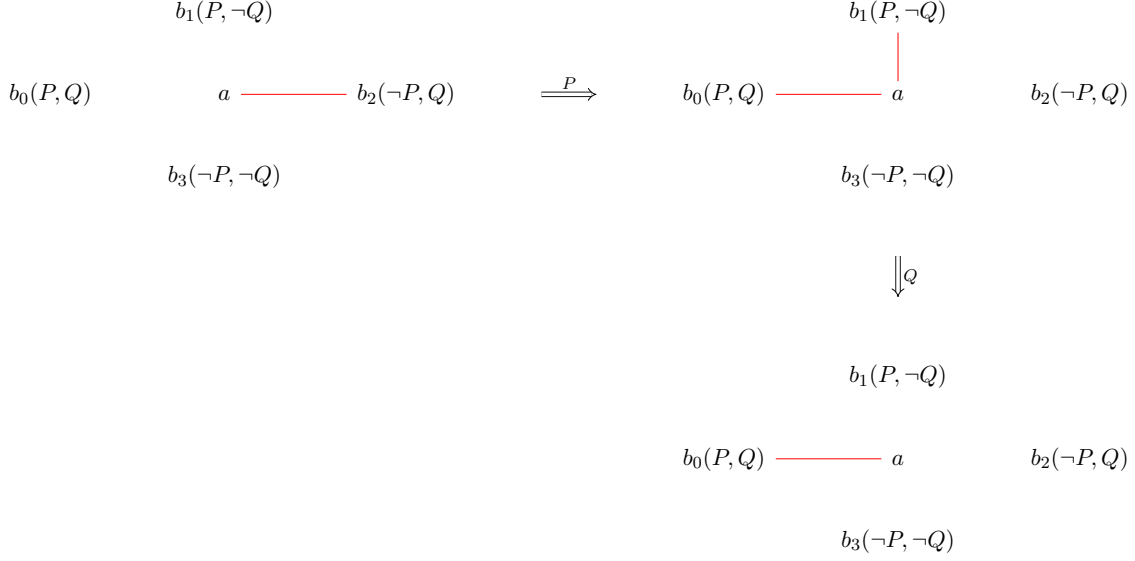


Figure 20:  $S_c$  before and after  $P_a$  is publicly announced, assuming revision happens only when a perspective is no longer contained in any world.

Consider a UCF simplicial model  $\mathcal{M} = \langle N, V, L, \{S_a\}_{a \in A} \rangle$ , and  $\varphi$  a formula with no modalities in its recursive construction. Let  $N_P := N$ ,  $V_P := V$ , and

$$L_P(n)(P) := \begin{cases} 1 & \text{if } L(n)(P) \in \{1, 3\} \\ 4 & \text{if } L(n)(P) \in \{2, 4\} \end{cases}$$

Further, let  $\mathbf{F}_P$  be the subset of  $\mathcal{F}(\mathfrak{M}((N_P, V_P, L_P)))$  which satisfy  $P$ . We need to define a function  $\mathcal{R}_a : \mathcal{F}(S_a) \times \mathcal{L}_B(Ag) \rightarrow 2^{\mathbf{F}_P}$ .

$$\mathcal{R}_a(F, P) := \{G \in \mathbf{F}_P \mid (\pi_a(F) = \pi_a(G)) \wedge \forall X \in \mathbf{F}_P ((\pi_a(X) = \pi_a(G)) \rightarrow (|F \cap G| \geq |X \cap G|))\}$$

Note that  $\mathcal{R}_a : \mathcal{F}(S_a) \times \mathcal{L}_B(Ag) \rightarrow 2^{\mathbf{F}_P}$  is a nearness function. Further, note that, if  $G \in \mathbf{F}_P$ , then  $\mathcal{R}_a(G, P) = \{G\}$ . Let each  $S_{a,P}$  be the simplicial complex whose facets are elements of the set  $\bigcup_{F \in \mathcal{F}(S_a)} \mathcal{R}_a(F, P)$ . Then  $\mathcal{M}[P] := \langle N_P, V_P, L_P, \{S_{a,P}\}_{a \in A} \rangle$ .

Note that Figure 21 is the same as Figure 20. These two definitions do not always give the same output, however. Consider Figure 22. On the left is the update given by our procedure, and on the right is the update given by introducing the announced formula to every perspective. This suggests a subtle difference between the nature of hard information at a perspective, and the sense in which announcements are hard information, or indefeasible. As we see in Figure 22, an indefeasible announcement of  $P$  rules out a facet where  $P$  is false simply because no agent has hard information of  $P$ . By contrast, if we interpret announcements instead as giving agents hard information in their perspectives, the facet remains.<sup>23</sup>

It is worth emphasizing that this notion of “memory” is only well motivated if announcements are taken to be, in some sense, indefeasible. If announcements can be contradicted, Figure 19 might be a coherent picture. This is because this notion of “memory” would ensure that if an

<sup>23</sup> Above, we acknowledged that one can interpret a knowledge modality on  $S$ . It seems that, since when we use  $S$  as memory, we force it to update with the usual, factive account, this is entirely conceptually consistent with interpreting a knowledge modality on  $S$ . Again, however, we do not explore that here, in order to keep the focus on revision.

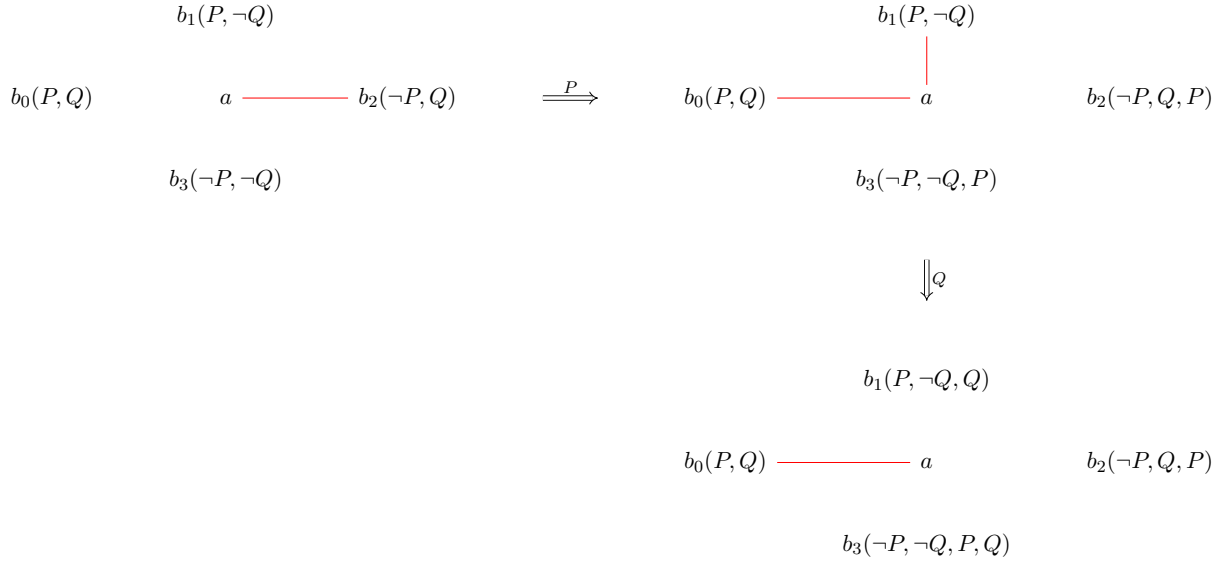


Figure 21: Figure 20 using Definition 3.4 instead of 3.3.

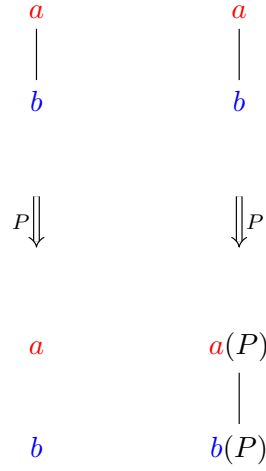


Figure 22: Two sequences of two models. The first, on the left, shows an initial configuration with two perspectives composing one world. After  $P$  is announced, using Definition 3.3, that world is erased. On the right, the same initial configuration is given and the same announcement is made, but using Definition 3.4, the world remains.

agent learns something that entails  $\varphi$ , and then later learns something which entails  $\neg\varphi$ , there are no worlds for the agent to revise to. Future work will want to consider precisely such kinds of contradictory announcements, and so will not make use of this notion of memory.

### 3.6 Conclusion

This chapter sought to give a model of belief revision in simplicial semantics. In particular, the hope was that by appealing to the fact that in simplicial semantics, worlds are composed of a different structure, namely perspectives, we could say that two worlds were “nearer” if they shared more perspectives. Two different formalisms were given on the basis of this intuition, the first of which allowed revision on every world ruled out by an announcement, and the second of which restricted revision to situations where perspectives become “isolated”, that is, an  $a$ -colored perspective is no longer contained in a facet of  $S_a$ . After exploring some examples, we saw that, if we took announcements to be infeasible, repeated revisions led to some strange behavior, where agents could go back-and-forth on whether or not they considered a particular world possible, simply believing whatever was announced to them most recently. To solve this issue, we introduced the notion of “memory”, and motivated two different ways we could force agents to “remember” what had been announced, avoiding the previous issue.

There are quite a few directions for future work. A simple project would involve connecting the use of  $S$  as “memory” and the possibility of interpreting a knowledge modality on  $S$ , as in [26]. Whether or not these two formal applications of  $S$  are conceptually consistent with each other will take further analysis. Another possibility for future work is exploring how different notions of trust affect this model of revision. Indeed, if some agents treat others as “more trustworthy” than others, presumably their perspectives should “outrank” other agents’ perspectives in determining nearness. Spelling this out formally, and seeing all the possible ways one might do this, would be a compelling direction to take this notion of revision.

It would be nice to explore more connections between this semantics and other presentations of belief revision. In particular, future work should explore connections between our semantics and the “AGM” presentation of revision. [1, 22] For example, we should check if and how we can use our semantics to model multi-agent belief revision. [3] Another possibility would be to use the same perspective-based structure of worlds in simplicial semantics to model belief merging instead of belief revision. We can therefore phrase the question of merging thusly: Given a model, and two perspectives in the model, what is the “best” set of worlds one could create using the two sets of worlds containing each of the two perspectives, respectively. A simple answer is obviously the intersection. Given two perspectives, call them  $x_1$  and  $x_2$ , let  $S_i$  be the set of facets in a model  $\mathcal{M}$  containing  $x_i$ . Then merging the perspectives  $x_1$  and  $x_2$  could be given by the set of worlds  $S_1 \cap S_2$ . But we could obviously say something more nuanced. Perhaps  $X \in S_1$  but  $X \notin S_2$ . However, there is a world  $X'$  that differs from  $X$  in only one perspective, and  $X' \in S_1 \cap S_2$ . Perhaps the merge set should include worlds like  $X'$ . Similar to our discussion of announcements, one can also modify this so that the update is  $S_1 \cap S_2$ , unless this intersection is empty, in which case one selects some set of nearby worlds to  $S_1 \cup S_2$ .

The most fleshed out project for future work involves using simplicial semantics to model Gossip Protocols, and to incorporate revision into those protocols. [8] The idea would be to use action models to model protocols as a first step. This already has had some exploration in the literature, including the simplicial semantics literature. [19, 14, 2, 25, 29, 13] Once a definition of action model is sufficiently worked out for our purposes, the second step would be to modify this definition of action model to update via revision. This, too, already has some history of work. [6, 7] With all of these pieces in play, the goal would be to then describe protocols where the update procedure is interpreted as revision. A possible future step after this would be to simulate said protocols, with the potential goal of comparing protocols with and without revision in order to assess the efficacy of revision as a learning mechanism in certain contexts.

### 3.7 Proofs

This section largely sets out to prove a completeness theorem. For this theorem, along the way we show a categorical equivalence between a category of Kripke models and a category of simplicial models. Only one direction of this equivalence is needed for the proof of completeness. However, showing the full categorical equivalence gives a sense of what kind of “information” is preserved when translating back and forth between the two settings. It’s also consistent with how such proofs are done in the literature. [14, 19]

*Proof of Theorem 3.1.* The proof follows many of the standard techniques in the literature. [14]

**Definition 3.5** ( $\mathfrak{S}$ ). Let  $S_1$  and  $S_2$  be *UCF* simplicial models. We say that  $f : S_1 \rightarrow S_2$  is a morphism iff, treating  $f$  as a function on  $\mathcal{F}(S_1)$ , we have that for any facets  $X, Y \in S_1$  and  $X \in S_{1,b}$ ,  $Y \in S_{1,a}$  and  $\pi_a(X) = \pi_a(Y)$  iff  $f(Y) \in S_{2,a}$  and  $\pi_a(f(X)) = \pi_a(f(Y))$ , and furthermore,  $\bigcup_{a \in A} L(\pi_a(X)) = \bigcup_{a \in A} L(\pi_a(f(X)))$ . We say that the category whose objects are *UCF* simplicial models with these morphisms is  $\mathfrak{S}$ .

**Definition 3.6** ( $\mathfrak{K}$ ). Let  $K_1$  and  $K_2$  be transitive and Euclidean proper Kripke models such that for any world  $w$  and atom  $P$  true at that world, there exists an agent  $a \in A$  such that if  $wR_a u$  and  $w \in \ell(P)$ , then  $u \in \ell(P)$ . It’s easy to see that this makes **NU** sound. We call such Kripke models *Ideal*. We say that  $g : K_1 \rightarrow K_2$  is a morphism iff, treating  $g$  as a function on worlds,  $wR_{a,1} u$  iff  $g(w)R_{a,2} g(u)$ , and  $w \in \ell_1(P)$  iff  $g(w) \in \ell_2(P)$ . We say that the category whose objects are ideal Kripke models with these morphisms is  $\mathfrak{K}$ .

**Lemma 3.2.**  $\mathfrak{S}$  and  $\mathfrak{K}$  are equivalent categories.

*Proof.* First we need to define two functors. For the first functor, define  $F : \mathfrak{S} \rightarrow \mathfrak{K}$  such that for a *UCF* model  $S$ ,  $F(S)$  is a Kripke model such that for each maximal facet  $X$  in some  $S_a$ , there is a unique world  $w_X$  in  $F(S)$ , and  $w_X R_a w_Y$  iff  $\pi_a(X) = \pi_a(Y)$  and  $Y \in \mathcal{F}(S_a)$ . By design, worlds and facets are in bijection. Furthermore,  $w_X \in \ell(P)$  iff  $P \in \bigcup_{x \in X} L(x)$ . We need to show that this is an ideal Kripke model. It’s obvious that  $R_a$  is transitive and Euclidean for each  $a \in A$ . Now consider any atom  $P$  and world  $w_X$ . Suppose that  $w_X \in \ell(P)$ . It follows that  $P \in \bigcup_{x \in X} L(x)$ . Fix  $x \in X$  such that  $P \in L(x)$ . Since  $S$  is *UCF*, fix  $a$  such that  $V(x) = a$ . Suppose further that  $w_X R_a w_Y$ . Then  $x = \pi_a(X) = \pi_a(Y)$ , and so  $P \in L(\pi_a(Y)) \subseteq \bigcup_{x \in Y} L(x)$ . It follows that  $w_Y \in \ell(P)$ , as desired. Finally, if  $w_X R_a w_Y$  for all  $a \in A$ , then  $\pi_a(X) = \pi_a(Y)$  for all  $a \in A$ . This means that  $X = Y$ , and so  $w_X = w_Y$ , which tells us that  $F(S)$  is a proper Kripke model. This shows that we have an ideal Kripke model.

Now we must show that  $F$  as given above is a functor. Let  $f : S_1 \rightarrow S_2$  be a  $\mathfrak{S}$  morphism. Then  $F(f) : F(S_1) \rightarrow F(S_2)$  is defined as follows:  $F(f)(w_X) = w_{f(X)}$ . We need to show that this is a  $\mathfrak{K}$  morphism. Indeed, by design,  $w_{f(X)}$  is a world in  $F(S_2)$ . Let  $R_1$  be the relation in  $F(S_1)$  and  $R_2$  be the relation in  $F(S_2)$ . Suppose  $w_X R_1^a w_Y$ . This is true if and only if  $\pi_a(X) = \pi_a(Y)$  and  $Y \in \mathcal{F}(S_{1,a})$ . Call this shared vertex  $v$ . This follows if and only if  $f(v) \in f(X) \cap f(Y)$ , and since  $f(Y)$  is in  $S_{2,a}$ , this is true if and only if  $w_{f(X)} R_2^a w_{f(Y)}$ , as desired. Suppose  $w_X \in \ell_1(P)$ . This is true iff  $P \in \bigcup_{x \in X} L(x)$ . Because  $\bigcup_{x \in X} L(x) = \bigcup_{x \in f(X)} L(x)$ , this is true if and only if  $P \in \bigcup_{x \in f(X)} L(x)$ . This is true iff  $w_{f(X)} \in \ell_2(P)$ , as desired. This shows that  $F$  is well defined.

It’s obvious that  $F$  preserves the identity. We must show that it preserves composition to establish its functoriality. Let  $f_1 : S_1 \rightarrow S_2$  and  $f_2 : S_2 \rightarrow S_3$  be morphisms. Then the following holds:

$$F(f_2 \circ f_1)(w_X) = w_{f_2 \circ f_1(X)} = w_{f_2(f_1(X))} = F(f_2)(w_{f_1(X)}) = F(f_2)(F(f_1)(w_X)) = (F(f_2) \circ F(f_1))(w_X)$$

From this it follows that  $F(f_2 \circ f_1) = F(f_2) \circ F(f_1)$ , as desired. So, we have given our first functor, and shown that it is a functor.

Now we must define the second functor. Define  $G : \mathfrak{K} \rightarrow \mathfrak{S}$  such that for an ideal model  $K$ ,  $G(K)$  is a simplicial model where the facets of  $S$  are of the form  $X_w$  for each world  $w$  in  $K$ , and the facets of each  $S_a$  are of the form  $X_w$  for each world  $w$  in  $K$  such that  $wR_a w$ . Because  $K$

is ideal, worlds and facets are in bijection. We assume each facet has a unique vertex for each agent. The only further stipulation is that  $\pi_a(X_w) = \pi_a(X_u)$  and  $X_u \in \mathcal{F}(S_a)$  if and only if  $wR_a u$ .<sup>24</sup> For each atom  $P$  true at  $w$ , we say that  $P \in L(\pi_a(X_w))$  iff  $P$  is true at all  $u$  such that  $wR_a u$ . By the fact that  $K$  is ideal, we know such an  $a$  exists. It is easy to see that this is a *UCF* model. Indeed, the disjoint collection of facets  $X_w$  is vacuously *UCF*, and each association of two vertices preserves the *UCF* property.

Now we must show that  $G$  as given above is a functor. Let  $g : K_1 \rightarrow K_2$  be a  $\mathfrak{K}$  morphism. Then  $G(g) : G(K_1) \rightarrow G(K_2)$  is defined as follows:  $G(g)(X_w) = X_{g(w)}$ . We need to show this is a  $\mathfrak{S}$  morphism. It suffices to show that  $G$  preserves facets and the logical information. Since  $g$  preserves reflexive edges,  $G(g)$  therefore preserves facets in each  $S_{1,a}$  in  $G(K_1)$ . Let  $X$  be a facet in  $S_1$  in  $G(K_1)$ . Fix  $w$  in the worlds of  $K_1$  such that  $X = X_w$ . Then  $G(g)(X_w) = X_{g(w)}$  which is by definition a facet in  $G(K_2)$ , since  $g(w)$  is a world in  $K_2$ . Now suppose  $P \in L(\pi_a(X_w))$ . This is true iff  $u \in \ell(P)$  for all  $u$  such that  $wR_a u$ . This is true iff  $g(u) \in \ell(P)$  for all  $u$  such that  $wR_a g(u)$ . This is true iff  $P \in L(\pi_a(g(w)))$ . This is true iff  $P \in L(G(g)(\pi_a(X_w)))$ , as desired. This shows that  $G$  is well defined.

It's obvious that  $G$  preserves the identity. We must show that it preserves composition to establish its functoriality. Let  $g_1 : K_1 \rightarrow K_2$  and  $g_2 : K_2 \rightarrow K_3$  be morphisms. Then the following holds:

$$G(g_2 \circ g_1)(X_w) = X_{g_2 \circ g_1(w)} = X_{g_2(g_1(w))} = G(g_2)(X_{g_1(w)}) = G(g_2)(G(g_1)(X_w)) = (G(g_2) \circ G(g_1))(X_w)$$

From this it follows that  $G(g_2 \circ g_1) = G(g_2) \circ G(g_1)$ , as desired.

What remains to be shown is that for any Kripke model  $K$ ,  $FG(K)$  is isomorphic to  $K$  and for any Simplicial model  $S$ ,  $GF(S)$  is isomorphic to  $S$ . For the first case, note that the worlds in  $FG(K)$  are of the form  $w_{X_w}$  where  $w$  is a world in  $K$ . The above demonstrated this identification is a bijection. Moreover, for any worlds  $w$  and  $u$  in  $K$ ,  $wR_a u$  in  $K$  iff  $X_u$  is in  $S_a$  and  $\pi_a(X_w) = \pi_a(X_u)$ . This is true iff  $w_{X_w} R_a u_{X_u}$ , as desired. Moreover,  $w \in \ell(P)$ , iff there is some  $a \in Ag$  such that  $P \in L(\pi_a(X_w))$ , which is true iff  $w_{X_w} \in \ell(P)$ . This shows that  $K$  and  $FG(K)$  are isomorphic.

Suppose  $X$  and  $Y$  are facets in  $S$ ,  $\pi_a(X) = \pi_a(Y)$  and  $Y \in S_a$ . This is true iff  $w_X R_a w_Y$ , which, because  $Y \in S_a$ , and so  $w_Y R_a w_Y$ , is true iff  $\pi_a(X_{w_X}) = \pi_a(Y_{w_Y})$  and  $Y_{w_Y} \in \mathcal{F}(S_a)$  for  $GF(S)$ . Suppose instead that  $\bigcup_{a \in A} L(\pi_a(X)) = \bigcup_{a \in A} L(\pi_a(Y))$ . This is true iff  $w_X, w_Y \in \ell(P)$ . Because  $F(S)$  is ideal, for each  $P$ , there is an agent  $a \in Ag$  such that if  $w \in \ell(P)$ , then for all  $u$  such that  $wR_a u$ ,  $u \in \ell(P)$ . By definition of  $G$ , this means that  $\bigcup_{a \in A} L(\pi_a(X_{w_X})) = \bigcup_{a \in A} L(\pi_a(Y_{w_Y}))$ . This shows that  $S$  and  $GF(S)$  are isomorphic.  $\square$

Now, we need to show that  $F$  and  $G$  are logic preserving functors. This is a simple structural induction.

Suppose  $P$  is an atomic formula,  $K$  is a Kripke model, and  $S$  is a simplicial model. Suppose further that  $K, w \models P$  and  $S, X \models P$ . Then,  $w \in \ell(P)$  and there is  $a \in Ag$  such that  $P \in L(\pi_a(X))$ . So,  $w_X \in \ell(P)$ , and  $P \in L(\pi_a(X_w))$ , showing that  $F$  and  $G$  preserve atomic formulas.

Assume the obvious inductive hypothesis for formulas up to depth  $n$ . The only interesting case is the modal one. Let  $\varphi$  be a formula of depth  $n$ . Suppose that  $K, w \models B_a \varphi$  and  $S, X \models B_a \varphi$ . We will handle the Kripke case first. This means that, for all  $u$  such that  $wR_a u$ ,  $K, u \models \varphi$ . By the IH, and the definition of  $G$ , for all facets  $Y \in S_a$  such that  $\pi_a(Y) = \pi_a(X_u)$ ,  $G(K), X_u \models \varphi$ . The result follows. Because  $S, X \models B_a \varphi$ , then for all  $Y \in S_a$  such that  $\pi_a(X) = \pi_a(Y)$ , then  $S, Y \models \varphi$ . By the IH, and the definition of  $F$ ,  $F(S), w_Y \models \varphi$ . The result follows.

The next step is to construct a canonical **K45+NU** frame model, using the usual “unboxing” method. That is, construct a model  $K_C$  whose set of worlds  $W_C$  is the set of all maximal consistent **K45+NU** sets of formulas,  $\ell(P) = \{w \in W_C \mid P \in w\}$ , and  $wR_a u$  if and only if all formulas  $\varphi$ , if  $B_a \varphi \in w$ , then  $\varphi \in u$ . It is easy to show that this model is such that  $K_C, w \models \varphi$  if

<sup>24</sup>By the Euclidean axiom, we have that, if  $wR_a u$ , then  $uR_a u$ , so  $X_u \in \mathcal{F}(S_a)$  is actually redundant as a condition.

and only if  $\varphi \in w$ . In general, this model is not proper, but we can apply the translation from [10] to generate a bisimilar model which is. Then, applying  $G$  to this proper model gives us a canonical simplicial model. Completeness follows. □



## 4 Simplicial Actions

### Abstract

This chapter sets out to explore the implementation of action models in the setting of simplicial semantics for modal logic. Such an idea is not entirely new to the literature, showing up in both [19] and [14]. However, we will explore action models in a more general setting. In particular, we will allow for action models for simplicial models for *belief*, as in [26]. This will let us incorporate the notion of belief revision, as developed in Chapter 3, into these action models. Moreover, we explicitly connect action models in the simplicial setting to distributed protocols as defined in [23]. We conclude with some speculation on how we might interpret distributed protocols with revision.

### 4.1 Introduction

Seeing as the impetus for Simplicial Semantics is the use of combinatorial topology in distributed computing, most of that literature acknowledges and is indebted to this connection already. [20, 28, 14, 19, 17, 18, 23] In general, the procedures for using action models to describe distributed protocols are understood but new, showing up in [19], [14], [2], [25], [29], and [13]. Moreover, the papers [14], [26], and [11] are already attempts to give a semantics for belief, or non-factive (defeasible) knowledge. Our paper will seek to build on this work by implementing simplicial actions in the setting of belief models. In doing so, we will be able to describe actions where the agents update via belief revision as described in Chapter 3. Moreover, we will explicitly connect these action models to distributed protocols as defined in [23].

The paper will proceed as follows. In Section 4.2, we will outline the basic definitions for simplicial semantics used throughout the paper. In Section 4.3, we explore the basics of connecting distributed protocols to combinatorial topology. In Section 4.4, we explore how action models in the non-simplicial setting have been used to model protocols. In Section 4.5, we define action models for both simplicial models of knowledge and belief, and also define the latter where agents update via belief revision. We also give examples of some existing protocols and show how to model them using simplicial actions.

### 4.2 Simplicial Semantics

This section will give the particulars for how we will define simplicial semantics throughout the paper. For a more thorough overview, we recommend reading [14], [26], or Chapter 3.

The literature on simplicial semantics divides roughly into two approaches to defining valuations for propositional atoms. Loosely speaking, the first approach assigns truth values directly to the facets [20, 18, 21], while the second “vertex-based” approach assigns them (in a partial way) to the vertices and then “lifts” them to facets. [28, 14, 19, 17] This chapter will focus on the latter approach.

Let  $\mathfrak{P}$  be a countable set of propositional atoms, and  $Ag$  a finite set of agents. Let  $N$  be a set of **nodes**, called the **coloring** function, and  $L : N \rightarrow 3^{\mathfrak{P}}$  a function which assigns each node to a set of literals, which we call the **assignment**. The interpretation is that  $L(n)(P) = 1$  if and only if  $P$  is associated with  $n$ ,  $L(n)(P) = 0$  if and only if  $\neg P$  is associated with  $n$ , and  $L(n)(P) = 2$  if and only if neither is associated with  $n$ .  $L$  is the only difference between their presentation here as compared to [26]. As mentioned above,  $L$  assigns propositions to nodes.

Our first key idea is that we can use  $N$ ,  $V$ , and  $L$  to create a kind of **maximal** simplicial complex. Like much of the previous literature, we will assume our simplicial complexes are uniquely colored. Specifically, each facet of our complexes has a dimension of size  $|Ag|$ , and no two nodes are associated to the same agent. Put formally, if  $X \in S$  is such that for all  $Y \in S$ , if  $X \subseteq Y$  then  $X = Y$ , we have that  $|X| = |Ag|$ , and, if  $x, y \in X$ , then if  $V(x) = V(y)$ , we have that  $x = y$ . We call this condition UCF for “Uniquely Colored Facets”. The **Maximal Complex** is the set of subsets of  $N$  such that the associated logical content with that subset

is consistent, and it satisfies the UCF condition. That is, it's the subsets  $x \subseteq N$  such that  $|x| = |A|$ ,  $\bigcup_{u \in x} L(u)$  is consistent, and for all  $u, v \in x$ ,  $V(u) \neq V(v)$ . We refer to  $\mathfrak{M}(N, V, L)$  as the **Maximal Complex** of  $N$ ,  $V$ , and  $L$ . When the context is clear, we will refer to it simply as the maximal complex. The maximal complex can be defined set theoretically as follows:

$$\begin{aligned} \mathfrak{M}(N, V, L) := \{y \in 2^N \mid \exists x \in 2^N (y \subseteq x \\ \wedge \neg(\exists P \in \mathfrak{P}(\exists u, v \in x(L(u)(P) = 1 \wedge L(v)(P) = 0))) \\ \wedge (|x| = |A|) \\ \wedge (\forall u, v \in x(V(u) \neq V(v))))\} \end{aligned}$$

Note that  $\mathfrak{M}(N, V, L)$  is a simplicial complex. This is because, if  $Y \in \mathfrak{M}(N, V, L)$ , there is a set  $X$  such that  $Y \subseteq X$  and  $X$  satisfies certain properties. If  $Z \subseteq Y$ , then the same set  $X$  suffices to show that  $Z \in \mathfrak{M}(N, V, L)$ . All complexes we consider in this chapter will be UCF subcomplexes of the maximal complex.

If  $S$  is a simplicial complex, let  $\mathcal{F}(S)$  denote the facets<sup>25</sup> of  $S$ . Since all simplicial complexes we consider are UCF, we can define projection functions  $\pi_a : \mathcal{F}(S) \rightarrow N$  given by  $\pi(a)(X) = V^{-1}(a) \cap X$ . That is,  $\pi_a$  maps each facet to the unique  $a$ -colored node it contains.

We can now define a **Simplicial Model For Knowledge**:

**Definition 4.1** (Simplicial Model for Knowledge). A **Simplicial Model for Knowledge**  $\mathcal{M}$  is a tuple  $(N, V, L, S)$  where  $N$  is a nonempty set of nodes,  $V$  is a coloring function,  $L$  is an assignment function, and  $S$  is a UCF subcomplex of  $\mathfrak{M}(N, V, L)$ .

In this setting, the language  $\mathcal{L}_K(Ag)$  recursively defined by

$$\varphi ::= P \mid \perp \mid \varphi \rightarrow \psi \mid K_a \varphi,$$

where  $P \in \mathfrak{P}$  and  $a \in Ag$ , can be interpreted in simplicial models for knowledge as follows for any  $X \in \mathcal{F}(S)$ :

$$\begin{aligned} \mathcal{M}, X \models P &\text{ iff } \exists a \in Ag(L(\pi_a(X))(P) = 1) \\ \mathcal{M}, X &\not\models \perp \\ \mathcal{M}, X \models \varphi \rightarrow \psi &\text{ iff } \mathcal{M}, X \models \varphi \text{ implies } \mathcal{M}, X \models \psi \\ \mathcal{M}, X \models K_a \varphi &\text{ iff } \forall Y \in \mathcal{F}(S) \text{ if } \pi_a(Y) = \pi_a(X) \text{ then } \mathcal{M}, Y \models \varphi \end{aligned}$$

In order to incorporate belief revision, as we will do in Section 4.5.2, it will be necessary to talk about models for belief rather than models for knowledge. The following definition is borrowed from Chapter 3:

**Definition 4.2** (Simplicial Model for Belief). A **Simplicial Model for Belief**  $\mathcal{M}$  is a tuple  $(N, V, L, S, \{S_a\}_{a \in Ag})$  where  $N$  is a nonempty set of nodes,  $V$  is a coloring function,  $L$  is an assignment function,  $S$  is a UCF subcomplex of  $\mathfrak{M}(N, V, L)$ , and for each  $a \in Ag$ ,  $S_a$  is a UCF subcomplex of  $S$ . Often, we will leave  $S$  unspecified. In this case, we assume that  $S = \mathfrak{M}(N, V, L)$ .

In this setting, the language  $\mathcal{L}_B(Ag)$  recursively defined by

$$\varphi ::= P \mid \perp \mid \varphi \rightarrow \psi \mid B_a \varphi,$$

where  $P \in \mathfrak{P}$  and  $a \in Ag$ , can be interpreted in simplicial models for knowledge as follows for any  $X \in \mathcal{F}(S)$ :

$$\begin{aligned} \mathcal{M}, X \models P &\text{ iff } \exists a \in Ag(L(\pi_a(X))(P) = 1) \\ \mathcal{M}, X &\not\models \perp \\ \mathcal{M}, X \models \varphi \rightarrow \psi &\text{ iff } \mathcal{M}, X \models \varphi \text{ implies } \mathcal{M}, X \models \psi \\ \mathcal{M}, X \models B_a \varphi &\text{ iff } \forall Y \in \mathcal{F}(S_a) \text{ if } \pi_a(Y) = \pi_a(X) \text{ then } \mathcal{M}, Y \models \varphi \end{aligned}$$

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<sup>25</sup>Maximal faces under the subset ordering.

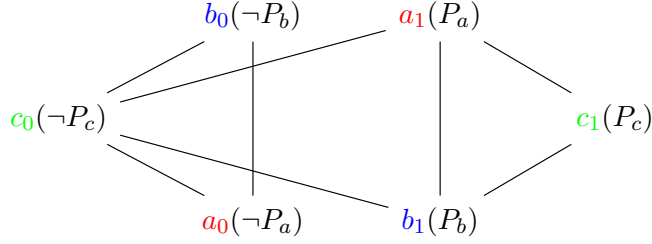


Figure 23: A simplicial model for belief with three agents and three worlds. In this example,  $S_a = S_b = S_c$ , and so we draw the complex as black rather than with distinct colors.

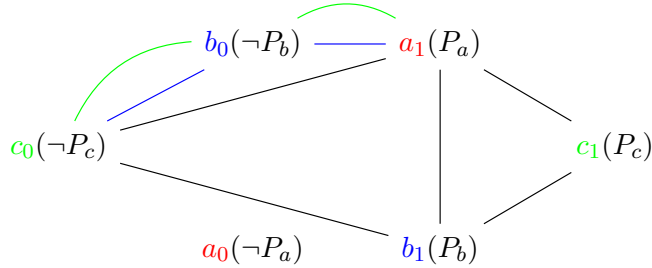


Figure 24: A simplicial model for belief with three agents and three worlds. In this example,  $S_a \neq S_b$ ,  $S_a \neq S_c$ , and  $S_b = S_c$ .

Following common convention, if  $\mathcal{M}$  is a simplicial model for either knowledge or belief, the claim  $x \in \mathcal{M}$  is short hand for  $x \in N$  where  $N$  is the set of nodes in  $\mathcal{M}$ .

Since belief complexes and how we draw them are relatively novel, we will look at a few examples. Consider Figure 23. In this example, there are three agents,  $a$ ,  $b$ , and  $c$ . Hence, facets will be of size 3 and are drawn as triangles. There are six nodes,  $a_i$ ,  $b_i$ , and  $c_i$  for  $i \in 2$ , and in each  $S_i$  there are three facets,  $\{a_1, b_1, c_1\}$ ,  $\{a_1, b_1, c_0\}$ , and  $\{a_0, b_0, c_0\}$ . There are three atomic propositions,  $P_i$  for  $i \in 2$ , and  $V(i_j)(P_k) = j$  if  $k = i$  and  $V(i_j)(P_k) = 2$  otherwise.<sup>26</sup>

We should consider an example where the  $S_i$  are distinct. Consider Figure 24. This example is drawn on the same set of nodes with the same valuation, but  $S_a \neq S_b$  and  $S_a \neq S_c$ . Facets which are present in  $S_a$  are drawn in red, those in  $S_b$  are drawn in blue, and those in  $S_c$  are drawn in green. If a facet is present in all three complexes, we draw it in black, as before. Hence, the two facets  $\{a_1, b_1, c_1\}$  and  $\{a_1, b_1, c_0\}$  are present in all three  $S_i$ . However, the facet  $\{a_1, b_0, c_0\}$  is present only in  $S_b$  and  $S_c$ .

### 4.3 Simplicial Complexes and Protocols

There is, at this point, a fairly well established discipline of applying combinatorial topology in modeling distributed protocols. [20, 28, 14, 19, 17, 18, 23] For those unfamiliar, we will briefly outline the broad methodology as sketched in the textbook “Distributed Computing through Combinatorial Topology”. [23] A “protocol” is a description of a sequence of signals between multiple processes, or agents. A good example is the “Alternating Message Passing Protocol” on pages 30-31 of the textbook. In this protocol, there are two agents, call them  $a$  and  $b$ . Each agent has a privately held bit value, and is aware of their own bit value. However, each agent is uncertain of the other agent’s bit value. We will denote these bit values as  $P_a$  and  $P_b$ ,

<sup>26</sup>It follows that this example can be given in the language of “local variables”. [14] In fact, many of the examples and definitions in this chapter are very easily lifted to that setting, as we are simply working in a slightly more general setting. See Chapter refSimpBelRev for more.

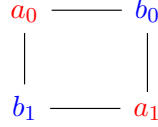


Figure 25: The Initial Configuration for the “Alternating Message Passing Protocol”.

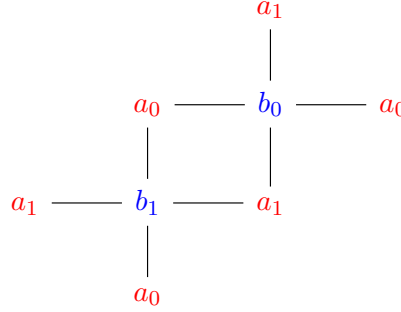


Figure 26: The second step for the “Alternating Message Passing Protocol”.

respectively. So,  $a$  is certain of the truth value of  $P_a$ , but uncertain of  $P_b$ , and vice versa.

One agent is assumed to be the first sender, and we will assume this is  $b$ . In the protocol, the agents will send back and forth. So, once  $a$  receives a message,  $a$  becomes the next sender. However, the catch is that there’s always a chance a message fails to be received. In this case, all sending stops.

One of the core ideas underlying applying simplicial complexes to these protocols is that we can describe each step of the protocol as a series of perspectives, and hence, each step as a simplicial model.<sup>27</sup> Consider the initial setup of the alternating message passing protocol. There are four perspectives, two for each agent. Let  $a_1$  be the perspective where  $P_a$  is true, and  $a_0$  the perspective where it is false. Similarly with  $b_1$  and  $b_0$ . This initial configuration is drawn on page 30 of “Distributed Computing through Combinatorial Topology”, and we replicate that below in Figure 25.

The main idea then is that perspectives will change and duplicate in each stage. Consider  $a_0$ . There are three possibilities for how the first step can go for this perspective. Either  $P_b$  is received,  $\neg P_b$  is received, or nothing is received. So, in the second step,  $a_0$  has to split into three perspectives. A similar change of reasoning will show that  $a_1$  also splits into three perspectives.

Now let’s consider  $b_0$ . Really, only one thing can happen in the first stage, and that’s that you send  $\neg P_b$ . Same for  $b_1$ . So, this leaves us with 3 copies of  $a_0$ , 3 copies of  $a_1$ , a single copy of  $b_0$ , and a single copy of  $b_1$  in the second stage of the protocol. Again, this is drawn on page 30 of “Distributed Computing through Combinatorial Topology”, and we replicate that drawing in Figure 26.

The same sort of reasoning, given the description of the signal pattern for the protocol, can give a very predictable way in which perspectives are duplicated from stage to stage of the protocol. The crucial idea is that we can describe a kind of morphism which maps protocols to their duplicates, or more generally, faces of the complex to faces composed of perspectives which are duplicates of those composing the initial face. This determines the “pattern” by which perspectives are duplicated. Ultimately, the textbook is interested in this sense of protocol. A protocol specifies how one complex transforms into another (how one stage proceeds to the next stage) by specifying a morphism which shows which faces map to which.

<sup>27</sup>The textbook never actually interprets epistemic formulae in its models, but appeals to the intuition that two faces sharing the same perspective means that perspective is uncertain between those faces at many points throughout.

Let us spell this intuitive story out formally. The following is equivalent to Definition 8.4.1 in the textbook “Distributed Computing through Combinatorial Topology”: [23]

**Definition 4.3** (Simplicial Protocol). A **Simplicial Protocol** is a triple  $(\mathcal{I}, \mathcal{P}, \Xi)$  where  $\mathcal{I}$  is a simplicial model for knowledge, called the “**Input Complex**”,  $\mathcal{P}$  is a simplicial model for knowledge called the “**Protocol Complex**” (better thought of in our context as the output complex) and  $\Xi : \mathcal{I} \rightarrow 2^{\mathcal{P}}$  is called the “**Execution Map**”, which takes faces in  $\mathcal{I}$  to sets of faces in  $\mathcal{P}$  such that if  $\sigma \subseteq \tau$ ,  $\Xi(\sigma) \subseteq \Xi(\tau)$ , and  $\Xi(\sigma \cap \sigma') = \Xi(\sigma) \cap \Xi(\sigma')$ . Moreover, for any face  $\sigma$  in  $\mathcal{I}$ ,  $\{a \in Ag \mid \exists x \in \sigma(V_{\mathcal{I}}(x) = a)\} = \{a \in Ag \mid \exists X \in \Xi(\sigma), x \in X(V_{\mathcal{P}}(x) = a)\}$ . When this is true we say that  $\Xi$  is **Chromatic**. And, additionally,  $\mathcal{P} = \bigcup_{\sigma \in \mathcal{I}} \Xi(\sigma)$ .

In general,  $\Xi$  specifies which input perspectives go to which output perspectives. The restrictions on  $\Xi$  bear further analysis. Consider in particular the intersection property:  $\Xi(\sigma \cap \sigma') = \Xi(\sigma) \cap \Xi(\sigma')$ . The textbook justifies it by saying no agent “can “tell” whether the execution started with inputs from  $\sigma$  or from  $\sigma'$ ”. [23] Here, “tell” refers to the usual understanding of knowledge in simplicial complexes, where the accessibility of two facets is given by their nonempty intersection.<sup>28</sup> Suppose that a node  $x$  is in the image of  $\sigma$  and  $\sigma'$ . That is,  $x \in \Xi(\sigma) \cap \Xi(\sigma')$ . Then, it follows from our intersection rule that  $x \in \Xi(\sigma \cap \sigma')$ . By our intersection semantics in Section 4.2, these are precisely the nodes which are in the image of those that cannot distinguish between  $\sigma$  and  $\sigma'$ . So, nodes which cannot tell whether they are in the image of  $\sigma$  or  $\sigma'$  are precisely those whose preimage cannot tell whether they are in  $\sigma$  or  $\sigma'$ .

#### 4.4 Action Models and Protocols

In this section, we will first briefly outline action models in their usual setting of Kripke or frame semantics for those unfamiliar with them. We will make use of definitions for action models over Kripke frames akin to those in [5]. Then, we will discuss the existing work connecting action models to distributed protocols, before developing our own definition of actions similar to [14] and [19] in Section 4.5, though modified for our new particular simplicial semantics.

Consider the following informal description of a signal:

“ $a$  sends a message to  $b$  whose content is  $P_a$ . However,  $c$  is unaware that any message was sent at all, and furthermore, there is a chance the message will not be received, in which case  $b$  will think no message was sent.”

What we would like to do is find a way of representing this signal such that, given an input model  $\mathcal{M}$ , define the model  $\mathcal{M}'$  which represents this input after the above signal is sent. As it turns out, to model this signal as an action model, we need to represent it using something very similar to a Kripke frame. For those unfamiliar with the basics of Kripke semantics, we recommend [16] or [9]. For us, a **Kripke Frame** is a tuple  $\langle W, \{\mathcal{R}_a\}_{a \in Ag} \rangle$ , and a **Kripke model** is a tuple  $\langle W, \{\mathcal{R}_a\}_{a \in Ag}, \ell \rangle$  which augments a frame with a valuation function  $\ell : \mathfrak{P} \rightarrow 2^W$ . Since we want to build something akin to a Kripke frame, the first step is to describe the worlds which capture the above signal. As it stands, there are three. The first world, which we shall call  $k$ , is the world where a message is sent, but not received. The second world, which we shall call  $m$ , is the world where a message is sent, but is received. Naturally,  $a$  cannot tell these worlds apart. However,  $b$  can. At  $m$ ,  $b$  considers  $m$  possible. However, at  $k$ ,  $b$  should not consider  $k$  possible. After all, if the message is not received,  $b$  thinks that no message was sent. This tells us that we need a third world,  $l$ , which is the world where nothing happens. At world  $k$ ,  $b$  falsely believes the world is  $l$ . Similarly, at both  $k$  and  $m$ ,  $c$  falsely believes the world is  $l$ . The only curious touch is that, at world  $l$ ,  $a$  should think the world is  $l$ . After all, this is the world where no message was sent - if  $a$  doesn’t send a message at all, they think they are in world  $l$ . This description gives us the Kripke frame in Figure 27

As before, red corresponds to  $\mathcal{R}_a$ , blue to  $\mathcal{R}_b$ , and green to  $\mathcal{R}_c$ . Black edges are those edges present in all three relations. However, the above frame is insufficient to describe the action.

<sup>28</sup>This is one place where the textbook gestures at intuitions afforded by the simplicial semantics.

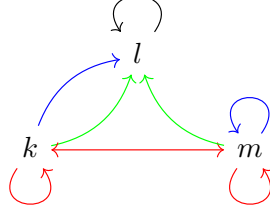


Figure 27: The Kripke Frame corresponding to the signal 4.4.

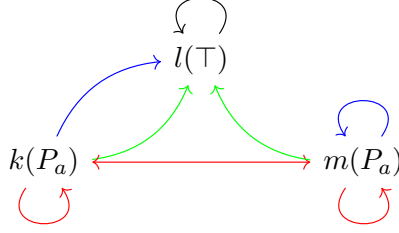


Figure 28: The Kripke Model corresponding to the signal 4.4.

After all, these worlds are not compatible with every possible input. Imagine an input world  $w$  where  $\neg P_a$  was true. Presumably, since  $P_a$  is a fact that  $a$  is aware of, i.e., a secret pertaining to  $a$  or a bit value which  $a$  is aware of, it is not possible that  $a$  sends out  $P_a$  at  $w$  (we are assuming that  $a$  is not lying and the signal always goes through correctly if it goes through). Hence,  $w$  is incompatible with  $k$  and  $m$ . Actions use the notion of “preconditions” to bear this out. Basically, in order for world  $k$  to obtain, it must be the case, and hence, it is a precondition, that  $P_a$  is true. We will mark preconditions with parentheses, like we do for truth conditions in typical models, and let context differentiate. However, preconditions, at least for actions over Kripke frames, are distinct from truth conditions, as a world can have a precondition which is not an atomic formula. Writing in the preconditions gives us the model depicted in Figure 28

$l$  is compatible with any input, however,  $k$  and  $m$  require that  $P_a$  to be true in order for them to obtain. Nicely, this is precisely the action given on page 22 of [4], just with the precondition  $K_a\varphi$  swapped out for  $P_a$ : these preconditions are equivalent so long as we only ever assign  $P_a$  to  $a$ -colored perspectives in the input, and hence,  $P_a$  will be true if and only if  $B_a P_a$  is true.

Now that we understand what actions look like, let’s imagine what updating an input with an action model might yield. Consider the input model as drawn in Figure 29. This input is easy enough to interpret - it is the Kripke equivalent of Figure 23, using the translation given in Chapter 3. Following our heuristic description of how to update some input with an action, we know that worlds in the update are ordered pairs, whose first term is from the input, and whose second term is from the action. However, not all ordered pairs are allowed - if at a world  $w$ , the precondition for the action world  $n$  is false, then the pair  $(w, n)$  cannot be allowed in the action update. Worlds in the update model are typically given as ordered pairs, with a world from the input in the first place and a world from the output in the second place. The allowable pairs

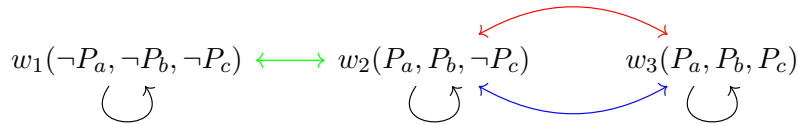


Figure 29: The Kripke Model we interpret as the input, or starting configuration.

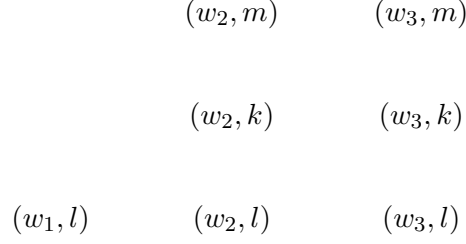


Figure 30: The worlds in the output where Figure 29 is the input and Figure 28 is the action.

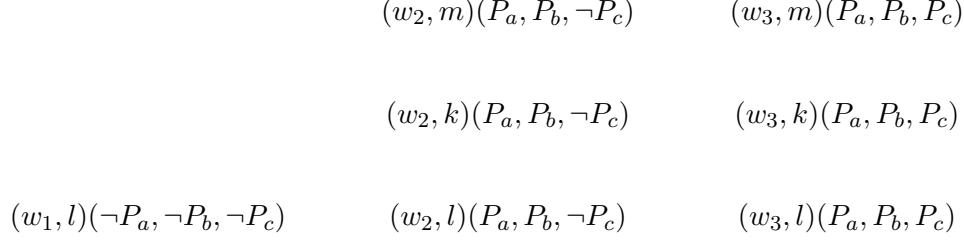


Figure 31: The worlds with truth values in the output where Figure 29 is the input and Figure 28 is the action.

are those where the first world satisfies the preconditions of the second world. For our example, using figures 28 and 29, this gives us the collection of worlds drawn in Figure 30.

What should the truth conditions at these worlds be? Well, if  $(w_2, k)$  is the world  $w_2$ , but where the message is sent but not received, then this should take on the same truth values as  $w_2$ . Indeed, actions don't change what's true at a world. Rather, they create copies of worlds which can be differently interpreted. This gives us the model as drawn in Figure 31.

All that is left to determine is each  $\mathcal{R}_i$  for  $i \in Ag$ . When should  $(w, n)\mathcal{R}_i(w', n')$ ? Well,  $(w, n)$  is world  $w$  where  $n$  has occurred. So, if  $w$  and  $w'$  are differentiable, so should  $(w, n)$  and  $(w', n')$ . The action does not change what is true at worlds, and so does not make worlds which were differentiable suddenly undifferentiable. However, if  $n$  and  $n'$  are differentiable, then  $(w, n)$  and  $(w', n')$  should be differentiable, even if  $w$  and  $w'$  are undifferentiable. After all, this is the main point - these were the same world, or at least undifferentiable worlds, but something different has occurred in each. This gives us a very natural definition:  $(w, n)\mathcal{R}_i(w', n')$  if and only if  $w\mathcal{R}_i w'$  and  $n\mathcal{R}_i n'$ . This gives us the output model as drawn in Figure 32.

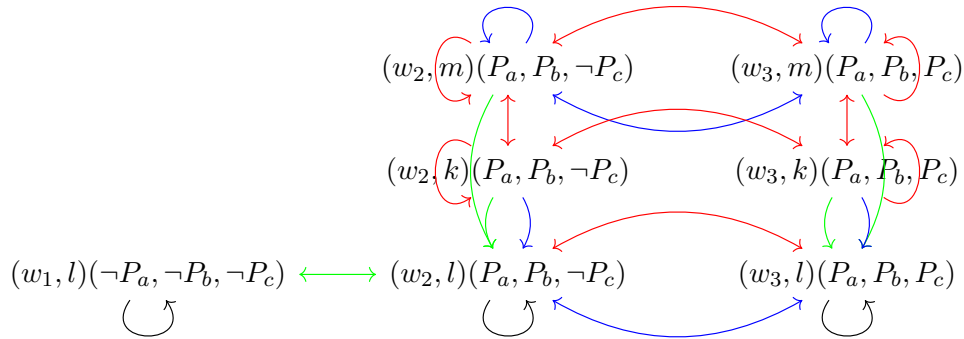


Figure 32: The Kripke model output where Figure 29 is the input and Figure 28 is the action.

Using our heuristic understanding, it's not difficult to imagine what each world in the above model means. At any world  $(w_j, n)$ ,  $c$  should only consider possible those worlds  $(w_h, l)$  such that  $w_j \mathcal{R}_c w_h$ , because  $c$  always falsely believes no signal was sent. Similarly, at world  $(w_2, k)$ , all four of  $(w_2, k)$ ,  $(w_2, m)$ ,  $(w_3, k)$ , and  $(w_3, m)$  are indistinguishable for  $a$ , as  $a$  cannot tell apart  $w_2$  and  $w_3$ , nor can  $a$  tell apart  $k$  from  $m$ . In general, one can think of an action update as creating (restricted) “copies” of the input model, and these “copies” are indistinguishable in terms of the relations in the action. Hence, the  $l$  “copy” of the input model contains the only worlds which  $c$  considers possible, etc.

This gives us the following formal definition of an action, equivalent to the one in [5]:

**Definition 4.4.** Formally, an action model  $A$  is a tuple  $\langle W_A, \{\mathcal{R}_{A,a}\}_{a \in Ag}, pre \rangle$  such that  $pre$  is a function from worlds  $w \in W$  to sets of formulas.<sup>29</sup> Given a Kripke model  $\mathcal{M} := \langle W, \{\mathcal{R}_a\}_{a \in Ag}, \ell \rangle$ , and an action model  $A$ , we define the action update model  $\mathcal{M}[A] := \langle W[A], \{\mathcal{R}_a[A]\}_{a \in Ag}, \ell[A] \rangle$  as follows:<sup>30</sup>

$$\begin{aligned} W[A] &:= \{(x, y) \in W \times W_A \mid \mathcal{M}, x \models pre(y)\} \\ \mathcal{R}_a[A](x, y) &:= \{(x', y') \in W[A] \mid x' \in \mathcal{R}_a(x) \wedge y' \in \mathcal{R}_{A,a}(y)\} \\ \ell[A](P) &:= \{(x, y) \in W[A] \mid x \in \ell(P)\} \end{aligned}$$

We conclude this section by outlining the existing applications of action models to distributed protocols. In general, the procedures for using action models to describe protocols are understood but new, showing up in [19], [14], [2], [25], [29], and [13]. In Section 4.5.1, we will translate two of the examples of protocols in [23] into action models, broadly following the trends of these papers. The key idea is that we can think of each “step” in the protocol as the application of an action. This is why we used the intuition of a “signal” when giving our account of actions. Not all actions can or should be interpreted as signals between agents (they are very general objects), but those actions that can be interpreted as signals can be interpreted as steps in a protocol. Suppose we are in a setting where agents have a privately held bit value. If the first step of a protocol in this setting is that all agents broadcast their bit value, with a chance that this broadcast is not received by some or all of the other agents, we can model this step as an action in the following sense: if  $\mathcal{M}$  represents the protocol before this step, and  $A$  is the action corresponding with this signal,  $\mathcal{M}[A]$  is the protocol after this step. Hence,  $\mathcal{M}[A]$  will be the input for the next step, which itself will be represented by a possibly distinct action, and so on.

Let's again consider the “Alternating Message Passing Model”. [23] All Kripke models we will consider in this example are equivalence relations. We again consider the case with two agents, call them  $a$  and  $b$ . Each agent has a privately held bit value, which we will denote using the proposition  $P_a$  and  $P_b$ , respectively. Hence, in the input, there are four worlds, corresponding with all possible configurations of the bit values. This is drawn in Figure 33. It is easy to determine which of these worlds the agents cannot tell apart. Agent  $a$  is uncertain about the truth value of  $P_b$ , but certain about the truth value  $P_a$ , and vice versa for agent  $b$ . So, agent  $a$  cannot tell apart  $w_0$  and  $w_2$ , and cannot tell apart  $w_1$  and  $w_3$ . Hence,  $w_0 \mathcal{R}_a w_2$  and  $w_1 \mathcal{R}_a w_3$ . We will denote  $\mathcal{R}_a$  with red edges and  $\mathcal{R}_b$  with blue edges. This is drawn in Figure 34. Call this model  $I$ .

Recall how the “Alternating Message Passing Protocol” proceeds. In this protocol, one agent is randomly selected as the first sender. For our purposes, this will be agent  $a$ . In the first step, agent  $a$  sends her bit value to agent  $b$ . However, there is a chance that this message is not received. This gives us four worlds. There are two worlds where  $P_a$  is sent, and two worlds where  $\neg P_a$  is sent. In one of the worlds where  $P_a$  is sent, it is received by agent  $b$ , and in another, it is not. Similarly for the worlds where  $\neg P_a$  is sent. Let  $1_r$  denote the world where  $P_a$  is sent and received,  $1_{\sim r}$  denote the world where  $P_a$  is sent and not received,  $0_r$  the world where  $\neg P_a$  is sent and received, and  $0_{\sim r}$  the world where  $\neg P_a$  is sent but not received. Hence,

<sup>29</sup> $W_A$  is a set of worlds, and each  $\mathcal{R}_{A,a}$  is a binary relation on those worlds.

<sup>30</sup> $W[A]$  is a subset of  $W \times W_A$ . Each  $\mathcal{R}_a[A]$  is a binary relation on  $W[A]$ , and  $\ell[A] : \mathfrak{P} \rightarrow 2^{W[A]}$ , just as in any Kripke model.



$$\begin{array}{cc}
w_0(P_a, P_b) & w_1(\neg P_a, P_b) \\
\\
w_2(P_a, \neg P_b) & w_3(\neg P_a, \neg P_b)
\end{array}$$

Figure 33: The four worlds in the input for the “Alternating Message Passing Protocol”.

$$\begin{array}{ccc}
w_0(P_a, P_b) & \longleftrightarrow & w_1(\neg P_a, P_b) \\
\updownarrow & & \updownarrow \\
w_2(P_a, \neg P_b) & \longleftrightarrow & w_3(\neg P_a, \neg P_b)
\end{array}$$

Figure 34: The first stage of the “Alternating Message Passing Protocol” with two agents.

$1_r \mathcal{R}_a 1_{\sim r}$  and  $0_r \mathcal{R}_a 0_{\sim r}$ . If  $b$  is in  $1_r$  they know they have received the message  $P_a$ , and hence only consider  $1_r$  possible. Similarly for  $0_r$ . However,  $b$  cannot tell apart  $1_{\sim r}$  and  $0_{\sim r}$ , as these are the worlds where no message is received. Note that  $a$  will only send their actual bit value. Hence,  $a$  will only send  $P_a$  if  $P_a$  is true. So, the precondition for  $1_r$  and  $1_{\sim r}$  is  $P_a$ , and the precondition for  $0_r$  and  $0_{\sim r}$  is  $\neg P_a$ . We draw this in Figure 35. Call this action  $A_a$ . Hence, we can compute  $I[A_a]$ . This is drawn in Figure 36. It’s easy to check that  $I[A_a]$  corresponds with the “One Step” drawing on page 30 in [23], using a translation between simplicial models and Kripke models similar to those given in [14].<sup>31</sup>

After  $b$  receives a message from  $a$ , she sends a response which consists of her own bit value. As before, there is a chance this message is not received. This action is the exact same as  $A_a$ , just with the agents swapped. We call this action  $A_b$  and draw it in Figure 37. Unfortunately,  $(I[I_a])[I_b]$  does not correspond to the “Two Steps” drawing in [23]. The reason is because even worlds where  $a$ ’s message failed to be received may become worlds where  $b$  still sends a message as we have defined things. In principle, however, accounting for this would be easy. One simply must define “postconditions” where what is true at a world is “overwritten”, as is done for the action models in [14]. This would let us apply a tag to worlds like  $(w_0, 1_{\sim r})$  which reads “not received”, and we can take as a precondition for all worlds in  $I_b$  that this tag be false. However, rather than explore this more carefully here, we will leave a formal exploration of postconditions for Section 4.5. In Section 4.5.1, we will use postconditions to fully capture the Alternating Message Passing protocol.

Our next goal will be to apply the notion of action models to simplicial semantics, similar to [20] and [14], though slightly modified to apply to our particular simplicial semantics. After this, we will modify this definition of action to account for the notion of belief revision given in Chapter 3. Modifying action models where agents learn via belief revision is also an explored area, with work such as [6] and [7]. However, none of these models will use the specific model of revision we will motivate in Section 4.5. That is because this work on using revision in protocols does not use the formal apparatus for describing protocols in simplicial complexes as seen in [23],

<sup>31</sup>Despite being called a “graph”, the drawings in Section 2 of [23] should be thought of as simplicial models for knowledge with two agents.

$$1_r(P_a) \longleftrightarrow 1_{\sim r}(P_a) \longleftrightarrow 0_{\sim r}(\neg P_a) \longleftrightarrow 0_r(\neg P_a)$$

Figure 35: The action representing the first step of the “Alternating Message Passing Protocol”: Agent  $a$  sends her true bit value to agent  $b$  with a chance the message fails to be received.

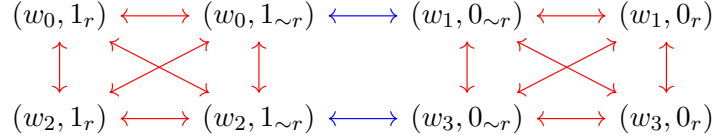


Figure 36: The output of applying the action shown in Figure 35 to the input shown in Figure 34.

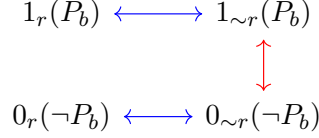


Figure 37: The action representing the first step of the signal describes as follows: agent  $b$  sends her true bit value to agent  $a$  with a chance the message fails to be received.

and thus cannot take advantage of the model of revision presented in Chapter 3.

## 4.5 Simplicial Actions

Before we define action models in the simplicial setting, we need to give three important definitions. First,  $\pi_0$  and  $\pi_1$  are the projections onto the first and second coordinates of an ordered pair, respectively. That is, for any ordered pair  $(x, y)$ ,  $\pi_0((x, y)) = x$  and  $\pi_1((x, y)) = y$ . For the third definition, let  $N$  be a set of nodes,  $V$  a coloring function, and  $L$  an assignment. Suppose  $F$  is a face in  $\mathfrak{M}(N, V, L)$ . We say that  $F$  is **consistent with respect to  $L$**  when for all  $x, y \in F$ , there is no  $P \in \mathfrak{P}$  such that  $L(x)(P) = 1$  and  $L(y)(P) = 0$ .

We can now give the definitions for simplicial actions formally. In general, we will use definitions for actions similar to those in [20], [14], and [19]. In particular, as in [14], actions will have both pre and post conditions. Importantly, we will assume that both preconditions and postconditions are atomic formulae in this chapter. It is essential that they be non-modal formulae, unlike in the general case of action models, so that we can interpret revision, and restricting to the atomic case is sufficient for our examples. In drawings, we will denote preconditions with parentheses, and postconditions with brackets.

**Definition 4.5** (Simplicial Action). A **Simplicial Action**  $A$  is a tuple  $\langle N_A, V_A, L_A, S_A, Post \rangle$ , where  $Post : N_a \rightarrow 3^{\mathfrak{P}}$ .<sup>32</sup>

Given a simplicial model for knowledge  $\mathcal{M} = \langle N, V, L, S \rangle$ , and a simplicial action  $A = \langle N_A, V_A, L_A, S_A, Post \rangle$ , we define the update model  $\mathcal{M}[A] := \langle N[A], V[A], L[A], S[A] \rangle$ <sup>33</sup> as follows:

<sup>32</sup>  $N_A$  is a set of nodes,  $V_A : N_A \rightarrow Ag$ ,  $L_A : N_A \rightarrow 3^{\mathfrak{P}}$ , and  $S_A$  is a UCF subcomplex of  $\mathfrak{M}(N_A, V_A, L_A)$ , just as in a regular simplicial model.

<sup>33</sup>  $N[A]$  is a subset of  $N \times N_A$ ,  $V[A] : N[A] \rightarrow Ag$ ,  $L[A] : N[A] \rightarrow 3^{\mathfrak{P}}$ , and  $S[A]$  is a UCF subcomplex of  $\mathfrak{M}(N[A], V[A], L[A])$ , just as in a regular simplicial model.

$$\begin{aligned}
N[A] &:= \{(x, y) \in N \times N_A \mid V(x) = V_A(y) \\
&\quad \wedge \forall P \in \mathfrak{P}((L(x)(P) = 1 \rightarrow L_A(y)(P) \neq 0) \wedge (L(x)(P) = 0 \rightarrow L_A(y)(P) \neq 1))\} \\
V[A]((x, y)) &:= V(x) \\
L[A]((x, y))(P) &:= \begin{cases} Post(y)(P) & \text{if } Post(y)(P) \neq 2 \\ L_A(y)(P) & \text{if } L(x)(P) = 2 \\ L(x) & \text{otherwise} \end{cases} \\
\mathcal{F}(S[A]) &:= \left\{ F \in 2^{N[A]} \mid \begin{array}{l} \pi_0(F) \in \mathcal{F}(S) \\ \wedge \pi_1(F) \in \mathcal{F}(S_A) \\ \wedge F \text{ is consistent w.r.t. } L[A] \end{array} \right\}
\end{aligned}$$

Note that if  $F \in 2^{N[A]}$ ,  $\pi_0(F) = \{x \mid \exists y \in S_A((x, y)) \in F\}$ , and similarly for  $\pi_1$ . Let us show that for any simplicial model for knowledge  $\mathcal{M}$  and simplicial action  $A$ ,  $\mathcal{M}[A]$  satisfies the UCF property.

**Theorem 4.1.** *Let  $\mathcal{M}$  be a simplicial model for knowledge and  $A$  a simplicial action model. Then the output model  $\mathcal{M}[A]$  also satisfies the UCF property.*

*Proof.* Section 4.8 □

Our definition of  $N[A]$  is not the only one which could be motivated. In fact, all three of the definitions below are defensible.

- $\forall P \in \mathfrak{P}((L(x)(P) = 1 \rightarrow L_A(y)(P) = 1) \wedge (L(x)(P) = 0 \rightarrow L_A(y)(P) = 0))$
- $\forall P \in \mathfrak{P}((L_A(y)(P) = 1 \rightarrow L(x)(P) = 1) \wedge (L_A(y)(P) = 0 \rightarrow L_A(x)(P) = 0))$
- $\forall P \in \mathfrak{P}((L(x)(P) = 1 \rightarrow L_A(y)(P) \neq 0) \wedge (L(x)(P) = 0 \rightarrow L_A(y)(P) \neq 1))$

We will refer to these as modes 0-2. Indeed, mode 2 is the one introduced in our definition above. For our purposes in this chapter, it will suffice.

We now show the sense in which action models are protocols. Let  $I$  be a simplicial model for knowledge and  $A$  an action model. Intuitively, our protocol complex should be given by  $I[A]$ . What we need to define is an execution map  $\Xi : I \rightarrow 2^{I[A]}$ . As it turns out, the correct map is given by

$$\Xi(F) := \{X \in I[A] \mid \pi_0(X) \subseteq F\}$$

Where  $X \in I[A]$  is shorthand for “ $X$  is a face in the simplicial complex of  $I[A]$ ”. This is analogous to the  $\pi_{\mathcal{I}}$  function from [19]. This gives us the following:

**Theorem 4.2.** *Let  $I$  be a simplicial model for knowledge and  $A$  an action model. The tuple  $(I, I[A], \Xi)$  is a Simplicial Protocol.*

*Proof.* Section 4.8 □

### 4.5.1 Examples

As we have already mentioned, there is existing work which establishes the connection between action models and protocols. [19, 14, 2, 25, 29, 13] In particular, [14] and [19] even give some examples. We will do much the same, using examples from the text “Distributed Computing through Combinatorial Topology” as our baseline. [23] These examples are nice because they are already given in the framework of simplicial complexes. All we need to do is prove that it is possible to represent these existing protocols using action models.

We can now fully capture the Alternating Message Passing Protocol on page 30 of [23]. In this protocol, there are two agents, call them  $a$  and  $b$ . As usual, these will be the red and blue

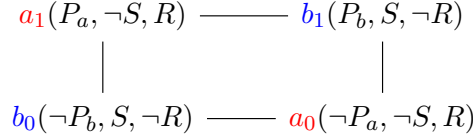


Figure 38: The Initial Configuration for the “Alternating Message Passing Protocol”.

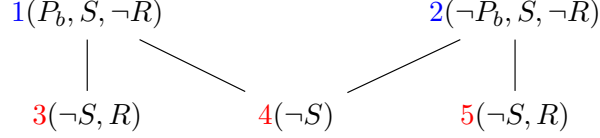


Figure 39: The fragment of the action for the “Alternating Message Passing Protocol” corresponding with  $b$  sharing their bit value, sans postconditions.

agents respectively. The initial arrangement is that each agent has a personal bit value, and they will be sharing that bit value with each other. As above, we will call these bit values  $P_a$  for  $a$ 's bit value and  $P_b$  for  $b$ 's bit value. One agent is designated the first sender, and we shall pick  $b$ . The input complex will have four perspectives, two for each agent corresponding to the possible bit values, and total uncertainty. Since  $b$  is the first sender, the two  $b$  perspectives will be marked with the literals  $S$  and  $\neg R$ , and since  $a$  is the first received, they will be marked with the literals  $\neg S$  and  $R$ . This gives us the input as drawn in Figure 38. Note how this is the same as the first stage drawing in [23], which we replicated in Figure 25.

Now we must describe the protocol. The main idea of the alternating message protocol is that agents are sending messages back and forth, and there is always a chance a message does not go through. If an agent doesn't receive a message, they don't send. Let's imagine what happens when  $b$  sends the first message. There are two  $b$  perspectives, corresponding to where  $b$  sends  $P_b$  and where  $b$  sends  $\neg P_b$ . These should have preconditions  $S$ ,  $\neg R$ , and  $P_b$  or  $\neg P_b$  as appropriate. We will call these 1 and 2. By contrast, there are three  $a$  perspectives. These correspond with  $a$  learning  $P_b$ ,  $a$  learning  $\neg P_b$ , and  $a$  learning nothing. We will call these 3, 5, and 4 respectively. The preconditions on 3 and 5 are  $\neg S$  and  $R$ , while the precondition on 4 is merely  $\neg S$ . This last point will be important for repeated applications of the protocol - any  $a$  perspective which is not a sender can receive nothing, possibly after having received nothing for many turns. This partially defined action model is drawn in Figure 39.

We now must draw the postconditions. Consider perspective 1. Having just sent, they turn into a receiver. So the postconditions here should be  $\neg S$  and  $R$ . Same for 2. Similarly, receivers become senders, so the postconditions on 3 and 5 should be  $S$  and  $\neg R$ . Lastly, the postcondition on 4 should be  $\neg R$ . If  $a$  receives nothing, they continue to not be a sender, but now are also not a receiver. This action model is drawn in Figure 40.

The last step in the protocol is to create the mirror image of this for when  $a$  is the sender and attach it to the existing drawing. This is seen in Figure 41.

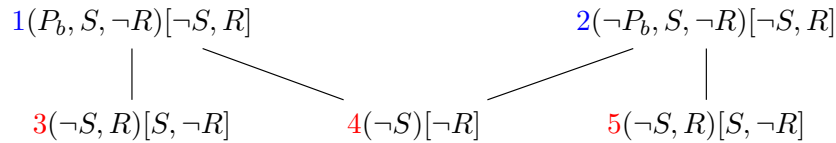


Figure 40: The fragment of the action for the “Alternating Message Passing Protocol” corresponding with  $b$  sharing her bit value.

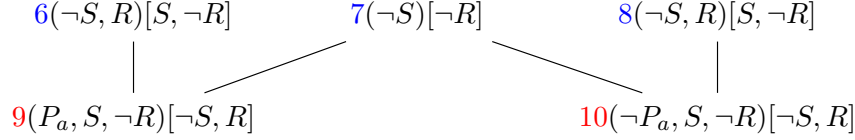


Figure 41: The fragment of the action for the “Alternating Message Passing Protocol” corresponding with  $a$  sharing her bit value.

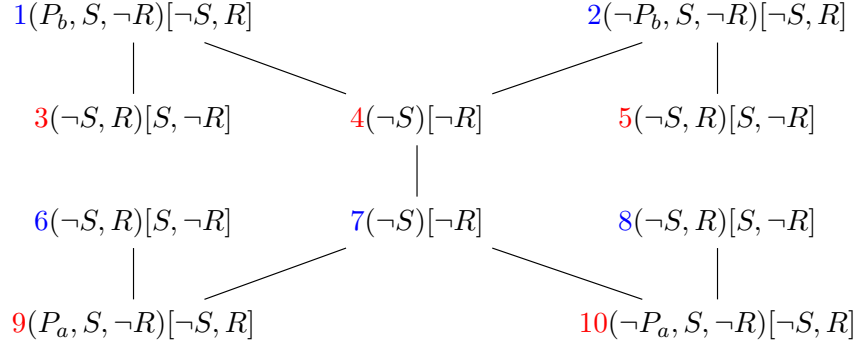


Figure 42: The action for the “Alternating Message Passing Protocol”.

The last thing to acknowledge is that an edge must be drawn between 4 and 7 - faces between two edges where nothing is received continue to be faces after any step of the protocol, as these are the worlds where nothing happens. This gives us the complete action model, which is drawn in Figure 42.

Now let's compute the output of applying this action model to the specified input. The output vertices are  $(a_1, 4)$ ,  $(a_1, 3)$ ,  $(a_1, 5)$ ,  $(b_1, 1)$ ,  $(b_0, 2)$ ,  $(a_0, 4)$ ,  $(a_0, 3)$ ,  $(a_0, 5)$ . This gives the output as drawn in Figure 43. Note how this is the same as the drawing of the second step for the protocol as given in [23], which we replicated in Figure 26. There are new perspectives created for when  $a$  learns the truth value of  $y$ , but also the original  $a$  perspectives are present, now corresponding to a failed signal. Let's apply the action again. The output vertices are  $((a_1, 4), 4)$ ,  $((a_0, 4), 4)$ ,  $((b_1, 1), 7)$ ,  $((b_1, 1), 6)$ ,  $((b_1, 1), 8)$ ,  $((b_0, 2), 7)$ ,  $((b_0, 2), 6)$ ,  $((b_0, 2), 8)$ ,  $((a_0, 3), 10)$ ,  $((a_1, 3), 9)$ ,  $((a_0, 5), 10)$ ,  $((a_1, 5), 9)$ . This again gives the result seen on page 30 of [23].

An important observation - when creating the action, we have to think through copies of perspectives. Even though creating a “base” copy of the  $b_1$  world in the input (i.e., pairing it with 7) would seem to make sense, it doesn't. We need to transform this perspective from a sender to a non-sender. This is the role 1 and 2 play in the action. The base (4 and 7) duplicate everything up to the sender perspective. Then, 1 duplicates the sender, while 3 duplicates the receiver (already duplicated by 4, hence an additional copy).

Let us prove by induction that this example works no matter how many times we apply the protocol.

*Proof.* The above examples are our base case. Suppose  $\text{Output}(n)$  is a square of four nodes, with two strings off each blue node, alternating in color, of length  $n$  (so four strings total). Everything is  $\neg S, \neg R$ , except the last two nodes on each string, which are  $\neg S, R$  just before the tip and  $S, \neg R$  at the tip. We will show that applying the action increases the length of these strings by 1. Suppose, without loss of generality, that the tips of the strings are red. Then 4 and 7 copy everything up until the tip of each string. Each tip is copied by either 9 or 10, and the nodes just before the tip are additionally copied by 6 and 8. Because the tip of each node is either copied by 9 or 10, the additional copies from 6 uniquely attach to their respective

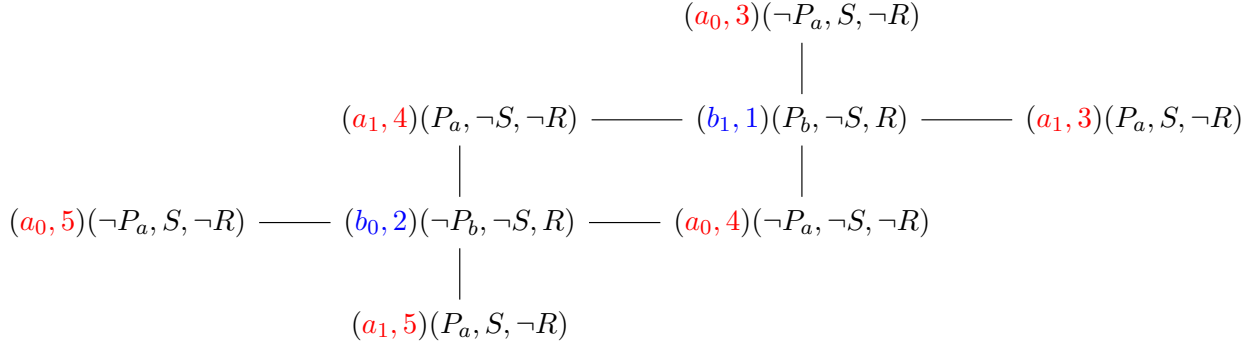


Figure 43: The output of applying the action in Figure 42 to the input in Figure 38.

$$\textcolor{red}{a}(\neg P_a, S) \text{ — } \textcolor{blue}{b}(P_b, S)$$

Figure 44: Restricted input for the “Layered Message Passing Protocol” where  $a$  has bit value 0 and  $b$  has bit value 1.

9-tipped string, and the 8 copies their respective 10-tipped string, as desired.  $\square$

We will now turn towards our next example, the “Layered Message Passing Protocol”. Here, the two agents simultaneously write their bit value to a ledger at each time interval (so the protocol is synchronous). However, at each interval, there is a chance that at most one agent will fail. Upon observing a failure, the agents will stop. We will consider the restricted situation where  $a$  has bit value 0 and  $b$  has bit value 1 first. Each of course starts as a sender, giving us the input as drawn in Figure 44.

The protocol itself is trickier. The action has four fairly obvious perspectives, these being where  $a$  receives the message,  $a$  does not, and the same for  $b$ . But there are two other perspectives, one for each agent, which are less obvious because they are not relevant in the first step of the protocol. That is the perspective of  $a$  where at some point in the past, they failed to receive a message, and therefore on this particular step of the protocol did not send any message. All told, this gives the action model drawn in Figure 45. For example, perspective 2 is one where  $P_b$  sent, but  $b$  failed to receive. 4 is where  $P_b$  sent and received, and 6 is where  $b$  did not send. Same for the  $a$  perspectives. The most nonideal thing about this setup is the need for the  $\neg P_a$  pre and post-condition in 4, and the similar setup in 3. These will be necessary to prevent unwanted duplications, corresponding to receiving  $\neg P_a$  and then  $P_a$ , in the setup where the bit values are not fixed in the input, as we shall see. But as presented, they do seem overly restricted. Other workarounds would be ideal.

If we apply the action model to the input a single time, we get the drawing in Figure 46, which is as given on page 31 of [23]. If we apply it again, we get the output as drawn in Figure

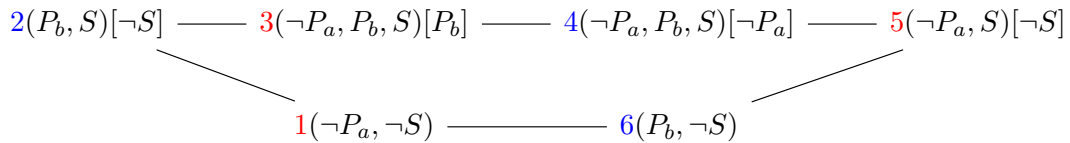


Figure 45: The action for the restricted case of the “Layered Message Passing Protocol” where  $a$  has bit value 0 and  $b$  has bit value 1.

$$(a, 5)(\neg P_a, \neg S) \text{ — } (b, 4)(\neg P_a, P_b, S) \text{ — } (a, 3)(\neg P_a, P_b, S) \text{ — } (b, 2)(P_b, \neg S)$$

Figure 46: The output of applying the action in Figure 45 to the input in Figure 44, representing a single step of the restricted case of the “Layered Message Passing Protocol”.

$$\begin{array}{ccccccc} ((b, 4), 2)(\neg P_a, P_b, \neg S) & \text{—} & ((a, 3), 3)(\neg P_a, P_b, S) & \text{—} & ((b, 4), 4)(\neg P_a, P_b, S) & \text{—} & ((a, 3), 5)(P_b, \neg P_a, \neg S) \\ & \searrow & & & & \swarrow & \\ & ((a, 5), 1)(\neg P_a, \neg S) & & & & ((b, 2), 6)(P_b, \neg S) & \end{array}$$

Figure 47: The output of applying the action in Figure 45 to the input in Figure 46, representing the second step of the restricted case of the “Layered Message Passing Protocol”.

47, which is again as desired.

Now we want to apply this intuition to a more complicated setting, where the bit values are not predetermined in the input. This input is drawn in Figure 48.

Delightfully, the action model is easy to describe given the work we’ve already done. We simply need four copies of the action model previously, corresponding with the four combinations of pairs of bit values. We then stitch these together and we get Figure 49.

Applying the action model once gives us the model in Figure 50, exactly as we would want per page 31 of [23]. Applying it a second time gives us the model in Figure 51. The values of variables are left off for ease of reading. It is here that it is important we include the non-ideal additional pre- and post- conditions. Otherwise, we could have perspectives in the above like  $(a_0, (2, 4), (4, 2))$ , which would be where  $a_0$  first learns  $P_b$  and then  $\neg P_b$ . Nevertheless, we can show that the protocol applied  $n$  times is a square with sidelength  $2n - 1$ :

*Proof.* The above examples are our base case. Suppose  $\text{Output}(n)$  is a square of sidelength  $2(n + 1)$ . The center two nodes on each edge are labeled  $S$ , all other nodes  $\neg S$ , and every node but the corners contains a unique  $a_i, b_j$  pair. Applying the action duplicates every node but the centers of the edges of the input. These are duplicated by the center four nodes of the action, namely  $(2, 3)$ ,  $(3, 3)$ ,  $(2, 2)$ , and  $(3, 2)$ . The two center nodes of each edge are first copied by their respective  $a_i, b_j$  pair. These copies will be the center two nodes of the string of four nodes that will replace each center pair. In this case, the  $a_0 b_1$  edge are copied by  $(2, 4)$ ,  $(3, 4)$ . Then the  $a$  is copied additionally by the  $a_i$  corner, and the  $b$  node is copied additionally by the  $b_j$  corner. These are the outer two nodes of that string of four nodes. No other copies are possible, as these nodes specify both the  $a_i$  and  $b_j$  value, and are  $S$  nodes. So, as desired, we have a single copy of every node except the center of each edge, which gets two copies each, as desired.  $\square$

#### 4.5.2 Incorporating Revision

There are two definitions. There’s updating by a belief action, and revising by a belief action.

$$\begin{array}{ccc} b_1(P_b, S) & \text{—} & a_0(\neg P_a, S) \\ | & & | \\ a_1(P_a, S) & \text{—} & b_0(\neg P_b, S) \end{array}$$

Figure 48: The full input for the “Layered Message Passing Protocol”.

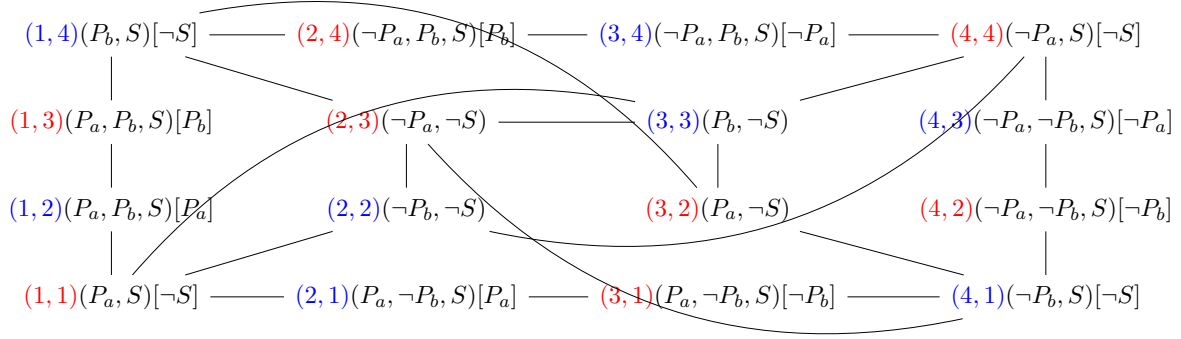


Figure 49: The full action for the “Layered Message Passing Protocol”.

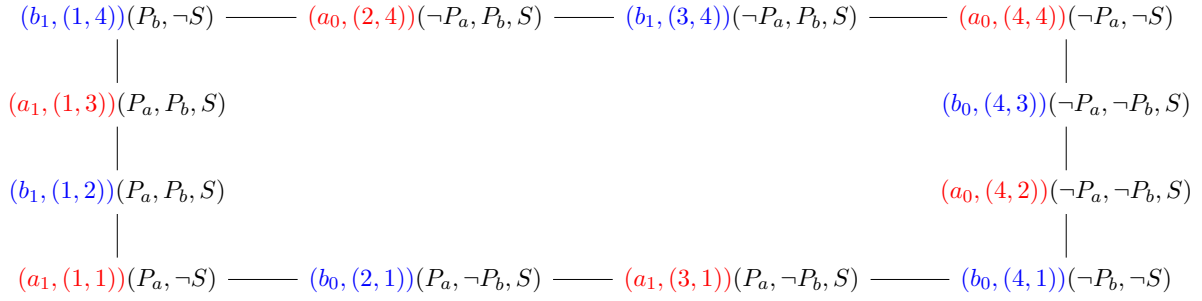


Figure 50: The output of applying the action in Figure 49 to the input in Figure 48, representing the a single step of the full “Layered Message Passing Protocol”.

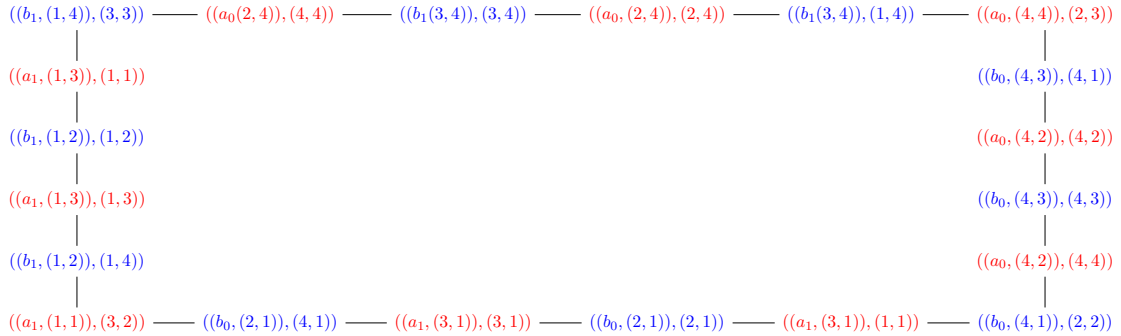


Figure 51: The output of applying the action in Figure 49 to the input in Figure 50, representing the second step of the full “Layered Message Passing Protocol”.



**Definition 4.6** (Simplicial Action for Belief). A **Simplicial Action for Belief**  $A$  is a tuple  $\langle N_A, V_A, L_A, \{S_{a,A}\}_{a \in Ag}, Post \rangle$ , where  $Post : N_A \rightarrow 3^{\mathfrak{P}}$ .<sup>34</sup>

Given a simplicial model for belief  $\mathcal{M} = \langle N, V, L, S \rangle$ , And a simplicial action for belief  $A = \langle N_A, V_A, L_A, \{S_{a,A}\}_{a \in Ag}, Post \rangle$ , we define the update model  $\mathcal{M}[A]_B := \langle N[A]_B, V[A]_B, L[A]_B, \{S_a[A]_B\}_{a \in Ag} \rangle$ <sup>35</sup> as follows:

$$\begin{aligned} N[A]_B &:= \{(x, y) \in N \times N_A \mid V(x) = V_A(y) \\ &\quad \wedge \forall P \in \mathfrak{P}((L(x)(P) = 1 \rightarrow L_A(y)(P) \neq 0) \wedge (L(x)(P) = 0 \rightarrow L_A(y)(P) \neq 1))\} \\ V[A]_B((x, y)) &:= V(x) \\ L[A]_B((x, y))(P) &:= \begin{cases} Post(y)(P) & \text{if } Post(y)(P) \neq 2 \\ L_A(y)(P) & \text{if } L(x)(P) = 2 \\ L(x) & \text{otherwise} \end{cases} \\ \mathcal{F}(S_a[A]_B) &:= \left\{ F \in 2^{N[A]_B} \mid \begin{array}{l} \pi_0(F) \in \mathcal{F}(S_a) \\ \wedge \pi_1(F) \in \mathcal{F}(S_{a,A}) \\ \wedge F \text{ is consistent w.r.t. } L[A]_B \end{array} \right\} \end{aligned}$$

We should check that the update model in this setting is UCF.

**Theorem 4.3.** *Let  $\mathcal{M}$  be a simplicial model for belief and  $A$  a simplicial action model for belief. Then the output model  $\mathcal{M}[A]_B$  also satisfies the UCF property.*

*Proof.* Proof is almost identical to knowledge case. Showing each  $S_a[A]_B$  is UCF is the same as showing  $S[A]$  is UCF. More detail in Section 4.8.  $\square$

We would like to say that this notion of an action is a protocol. However, the notion of protocol we used previously, the one given in [23], is insufficient for our purposes. Intersection preservation is meant to ensure that no agent “knows” whether they came from face  $X$  or  $Y$ . Since we have shifted over to belief, that is no longer available to us. We need the corresponding version of the intersection property which uses the definition of truth for modalities in the belief setting. In particular, we will need an execution map for each agent. But, it will only be relevant that the  $a$ -colored execution map preserve the intersections of  $a$ -colored vertices.<sup>36</sup> This motivates the following definition of a **Belief Protocol**:

**Definition 4.7** (Belief Protocol). A **Belief Protocol** is a triple  $(\mathcal{I}, \mathcal{P}, \{\Xi_a\}_{a \in Ag})$  where  $\mathcal{I}$  is a simplicial model for belief, called the “**Input Complex**”, and  $\mathcal{P}$  is a simplicial model for belief called the “**Protocol Complex**” (better thought of in our context as the output complex). Let  $\mathcal{I}_a$  denote the  $a$ -complex for  $\mathcal{I}$  and similarly for  $\mathcal{P}_a$ . For each  $a \in Ag$  a function  $\Xi_a : \mathcal{I} \rightarrow 2^{\mathcal{P}}$  is called the “**Execution Map**”, which takes faces in  $\mathcal{I}_a$  to sets of faces in  $\mathcal{P}_a$  such that if  $\sigma \subseteq \tau$ ,  $\Xi_a(\sigma) \subseteq \Xi_a(\tau)$ , and  $\Xi_a(\pi_a(\sigma \cap \sigma')) = \pi_a(\Xi_a(\sigma)) \cap \pi_a(\Xi_a(\sigma'))$ . Moreover, for any face  $\sigma$  in  $\mathcal{I}_a$ ,  $\{a \in Ag \mid \exists x \in \sigma(V_{\mathcal{I}}(x) = a)\} = \{a \in Ag \mid \exists X \in \Xi_a(\sigma), x \in X(V_{\mathcal{P}}(x) = a)\}$ . When this is true we say that  $\Xi_a$  is **Chromatic**. And, additionally, for each  $a \in Ag$ ,  $\mathcal{P}_a = \bigcup_{\sigma \in \mathcal{I}} \Xi_a(\sigma)$ .

We now show the sense in which action models for belief are protocols. Let  $I$  be a simplicial model for belief whose  $a$  complex is denoted  $I_a$ , and  $A$  a simplicial action for belief. As before, intuitively, our protocol complex should be given by  $I[A]_B$ . What we need to define is for each  $a \in Ag$  an execution map  $\Xi_a : I_a \rightarrow 2^{I_a[A]_B}$ . As it turns out, the correct map is given by the analogous map to the knowledge case:

$$\Xi_a(F) := \{X \in I_a[A]_B \mid \pi_0(X) \subseteq F\}$$

<sup>34</sup> $N_A$  is a set of nodes,  $V_A : N_A \rightarrow Ag$ ,  $L_A : N_A \rightarrow 3^{\mathfrak{P}}$ , and each  $S_{a,A}$  is a UCF subcomplex of  $\mathfrak{M}(N_A, V_A, L_A)$ , just as in a regular simplicial model.

<sup>35</sup> $N[A]_B$  is a subset of  $N \times N_A$ ,  $V[A]_B : N[A]_B \rightarrow Ag$ ,  $L[A]_B : N[A]_B \rightarrow 3^{\mathfrak{P}}$ , and  $S_a[A]_B$  is a UCF subcomplex of  $\mathfrak{M}(N[A]_B, V[A]_B, L[A]_B)$ , just as in a regular simplicial model.

<sup>36</sup>This is especially relevant in the revision setting.

Where  $X \in I_a[A]_B$  is shorthand for “ $X$  is a face in the simplicial complex of  $I_a[A]_B$ .” This gives us the following:

**Theorem 4.4.** *Let  $I$  be a simplicial model for belief and  $A$  an action model. The tuple  $(I, I[A]_B, \{\Xi_a\}_{a \in Ag})$  is a Belief Protocol.*

*Proof.* Section 4.8 □

Morally, this is true because of the following. Chromaticity follows roughly because  $V[A]_B((x, y)) = V(x)$ . This then entails the intersection property roughly as follows. If  $F$  and  $G$  share an  $a$ -node, call it  $x$ ,  $\Xi_a$  will send that  $a$ -node to a collection of  $a$ -nodes in  $I[A]_B$ . In particular,  $\Xi_a(\{x\})$  is the set of all singletons  $\{(x, y)\}$  in  $I_a[A]_B$ . Now consider  $\Xi_a(F)$ . By definition, this is the set of all faces  $X$  in  $I_a[A]_B$  such that  $\pi_0(X) \subseteq F$ . Since  $\pi_0(X) = \{z \mid (z, y) \in X\}$ , if  $X \in \Xi_a(F)$ , then  $\pi_a(\pi_0(X)) = \{x\}$ . Hence,  $\Xi_a(F)$  contains the set of all faces  $X$  in  $I_a[A]_B$  whose  $a$ -node is a tuple  $(x, y)$ . Hence,  $\pi_a(\Xi_a(F))$  is the set of all singletons  $\{(x, y)\}$  in  $I_a[A]_B$ . The same reasoning applies to  $G$  as to  $F$ . So, we can conclude that

$$\Xi_a(\pi_a(F \cap G)) = \pi_a(\Xi_a(F)) = \pi_a(\Xi_a(G))$$

as desired.

Now we need to modify this definition of an action to use the notion of revision from Chapter 3. The key idea is to modify the facets from  $\mathcal{F}(S_a[A]_B)$ .

**Definition 4.8** (Simplicial Action for Belief with Revision). Given a simplicial model for belief  $\mathcal{M} = \langle N, V, L, S \rangle$ , and a simplicial action for belief  $A = \langle N_A, V_A, L_A, \{S_{a,A}\}_{a \in Ag}, Post \rangle$ , we define the update model **with revision**  $\mathcal{M}[A]_{BR} := \langle N[A]_{BR}, V[A]_{BR}, L[A]_{BR}, \{S_a[A]_{BR}\}_{a \in Ag} \rangle$ <sup>37</sup> as follows:

$$\begin{aligned} N[A]_{BR} &:= \{(x, y) \in N \times N_A \mid V(x) = V_A(y) \\ &\quad \wedge \forall P \in \mathfrak{P}((L(x)(P) = 1 \rightarrow L_A(y)(P) \neq 0) \wedge (L(x)(P) = 0 \rightarrow L_A(y)(P) \neq 1))\} \\ V[A]_{BR}((x, y)) &:= V(x) \\ L[A]_{BR}((x, y))(P) &:= \begin{cases} Post(y)(P) & \text{if } Post(y)(P) \neq 2 \\ L_A(y)(P) & \text{if } L(x)(P) = 2 \\ L(x) & \text{otherwise} \end{cases} \end{aligned}$$

Defining  $S_a[A]_{BR}$  will take some more work. Fix  $a \in Ag$  and let  $X$  be a UCF facet in  $2^{N \times N_A}$  such that  $\pi_0(X) \in \mathcal{F}(S_a)$  and  $\pi_1(X) \in \mathcal{F}(S_{a,A})$ . Call the set of such  $X$   $\mathcal{F}^a(N \times N_A)$ . Let  $X$  be a UCF facet in  $2^{N[A]_{BR}}$  such that  $X$  is consistent, and  $\pi_1(X) \in \mathcal{F}(S_{a,A})$ . Call the set of such  $X$   $\mathcal{F}_a(2^{N[A]_{BR}})$ .

Then we can define a replacement function as follows for  $X \in \mathcal{F}^a(N \times N_A)$ :

$$\begin{aligned} R_a(X) &:= \{Y \in \mathcal{F}_a(2^{N[A]_{BR}}) \mid \pi_a(Y) = \pi_a(X) \\ &\quad \wedge (\forall Z \in \mathcal{F}_a(2^{N[A]_{BR}})(\pi_a(Z) = \pi_a(X) \rightarrow |Y \cap X| \geq |Z \cap X|))\} \end{aligned}$$

We say that  $S_a[A]_{BR}$  is defined as the simplicial complex whose facets come from  $\bigcup_{X \in \mathcal{F}^a(N \times N_A)} R_a(X)$ .

**Theorem 4.5.** *Let  $\mathcal{M}$  be a simplicial model for belief and  $A$  a simplicial action model for belief. Then the output model  $\mathcal{M}[A]_{BR}$  also satisfies the UCF property.*

*Proof.* Section 4.8 □

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<sup>37</sup> $N[A]_B$  is a subset of  $N \times N_A$ ,  $V[A]_{BR} : N[A]_{BR} \rightarrow Ag$ ,  $L[A]_{BR} : N[A]_{BR} \rightarrow 3^{\mathfrak{P}}$ , and  $S_a[A]_{BR}$  is a UCF subcomplex of  $\mathfrak{M}(N[A]_{BR}, V[A]_{BR}, L[A]_{BR})$ , just as in a regular simplicial model.

Now we need to show that simplicial actions for belief with revision are belief protocols. To do this, we need the following notion. Note that if  $Y \in \mathcal{F}(S_a[A]_{BR})$ , then  $R_a^{-1}(Y)$  is the preimage. We need to define this preimage for all faces in  $S_a[A]_{BR}$ . Given a face  $X \in S_a[A]_{BR}$ , say that  $R_a^{-1}(X)$  is the set of faces which subset of all facets

$$R_a^{-1}(X) := \{X' \mid \exists Y \in \mathcal{F}(S_a[A]_{BR})(X \subseteq Y \wedge \exists Y' \in R_a^{-1}(Y)(X' \subseteq Y')) \\ \wedge \forall a \in Ag(\exists x \in X(V[A]_{BR}(x) = a) \leftrightarrow \exists x' \in X'(V[A]_{BR}(x') = a))\}$$

**Theorem 4.6.** *Let  $I$  be a simplicial model for belief and  $A$  an action model. Define  $\Xi_{a,BR}$  similarly to  $\Xi_a$ :*

$$\Xi_{a,BR}(F) := \{X \in I_a[A]_{BR} \mid \forall X' \in R_a^{-1}(X)(\pi_0(X') \subseteq F)\}$$

*Then, the tuple  $(I, I[A]_{BR}, \{\Xi_{a,BR}\}_{a \in Ag})$  is a Belief Protocol.*

*Proof.* Section 4.8 □

Morally, this is true because the replacement function takes facets to sets of facets that, among other things, share  $a$ -colored nodes. Hence, the same moral argument we gave above still applies to show the intersection property.

## 4.6 Advantages of Action Models

In the introduction for this chapter, we suggested that it would be interesting to explore connecting action models and simplicial semantics since both have tight connections to distributed computing. Now that we have done so, we can talk more concretely about specific advantages afforded by modeling protocols using simplicial actions. The first concerns the epistemic intuitiveness of actions. In all of the examples above, we were able to create the formal object of an action for a specific protocol simply by carefully considering an agent's sender and receiver behavior. As a result, modifying actions becomes quite simple. For example, consider the action model drawn in Figure 52.<sup>38</sup> This action describes “ $a$  sends  $P_a$  to  $b$  with a chance of failure, and  $c$  is unaware that any message is sent.” This is easy to read off of the action. The perspective  $a_S$  is  $a$ 's perspective where they send the message. They do not know if  $b$  will receive it, hence they consider a facet possible where  $b$  is in perspective  $b_R$ , where the message is received, and another  $b_{\sim R}$ , where it is not. When the message is received,  $b$  knows this, and so there is only one facet in  $S_b$  containing  $b_R$ . Agent  $c$ , by contrast, only considers a single world possible, namely one where no message was sent. They also believe every other agent only considers this world possible. This is the facet  $\{a_{\sim S}, b_{\sim R}, c\}$ , which is in all three of  $S_a$ ,  $S_b$ , and  $S_c$ .

Suppose we wanted to modify this protocol so that  $c$  was aware that a message containing  $P_a$  may have been sent, but they are unsure, and if it was sent, they are unsure that it was received. Modifying Figure 52 to account for this is straightforward. Agent  $c$  simply considers more facets possible. Namely, the facets where  $P_a$  was sent and received, and also where it was sent but not received. This is done by adding  $\{a_S, b_R, c\}$  and  $\{a_S, b_{\sim R}, c\}$  to  $\mathcal{F}(S_c)$ . The result is drawn in Figure 53.

Another reason describing protocols as actions is advantageous is that protocols disentangle the “sender/receiver” behavior from the input model. In the usual definition of protocol, repeated above, the input/output complexes are specified, and the protocol is given as a morphism specifying which facets map where.[23] Hence, the protocol can't be disentangled from a particular input.<sup>39</sup> By contrast, actions can be applied to any input model with a sensible result as the output. For example, the actions drawn in Figure 52 and Figure 53 could be applied to any input whose vertices correspond to six perspectives where each of three agents has an

<sup>38</sup>Note that this is the simplicial translation of the action given on page 22 of [4].

<sup>39</sup>This is also true of the definition of protocol given in [12], though we will say more about this presentation below.

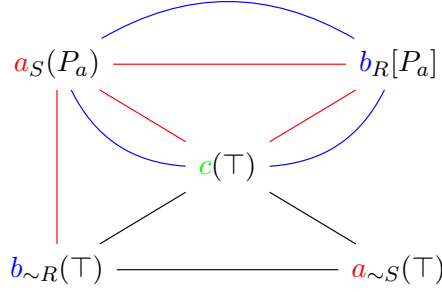


Figure 52: The simplicial action model describing “ $a$  sends  $P_a$  to  $b$  with a chance of failure, and  $c$  is unaware that any message is sent.”

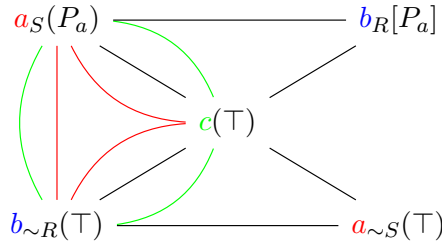


Figure 53: The simplicial action model describing “ $a$  sends  $P_a$  to  $b$  with a chance of failure, and  $c$  thinks it’s possible that the message was successfully sent, unsuccessfully sent, or no message was sent at all.”

assigned bit value. For example, these actions could be applied to the model drawn in Figure 23, or any UCF complex drawn on those same vertices.

Another advantage of action models is that, in addition to separating out “signals” from “actions”, they also compartmentalize information in a convenient way. Consider again the “Alternating Message Passing Protocol”. In the language of [23], this is a “layered” protocol, as it transforms over many steps. So, to represent it in the traditional way, we require an infinite sequence of complexes. We would start with the drawing in Figure 38, then the drawing in Figure 43, followed by another drawing with the branches extended a step further, and so on. This sequence is also drawn on page 30 of [23]. However, we are able to represent this protocol *entirely* with two simplicial models, rather than an infinite sequence. The input, given in Figure 38, and the action, given in 42, suffice. As we proved, any action specifies a protocol. So, we can describe at least some of these more complicated layered protocols with simply an input and an action, which we understand to be applied at each stage, rather than with an infinite sequence of models.

This disentangling of input and protocol affords more opportunities for testing protocols as well. In the near future, we are hoping to be able to implement the application of simplicial actions to inputs in a language like Python, in order to be able to consider more and larger examples more easily. Since the application of actions to inputs generates very large models very quickly, this would seem to be essential for further research. However, in such a setting, the separation of inputs and protocols would appear to be an advantage. Given an appropriate specification of vertices  $N$ ,  $V$ , and  $L$ , perhaps a protocol is “successful” on (given a concrete definition of successful) *most*, but not all, UCF subcomplexes of  $\mathfrak{M}(N, V, L)$ . We could then evaluate if cases where the protocol fails are sufficiently pathological as to be avoided or considered irrelevant. This testing environment would also allow for the ability to test belief revision itself in a philosophically exciting way. Given that applying a simplicial belief action with or without revision generates a belief protocol, we should be able to directly compare the performance of

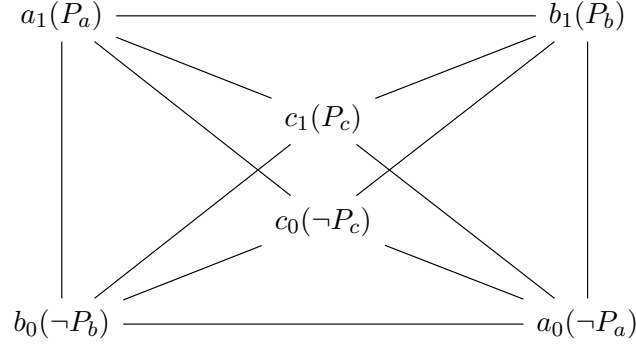


Figure 54: Knowledge complex for the initial configuration of three agents, each with a private bit value, with total uncertainty.

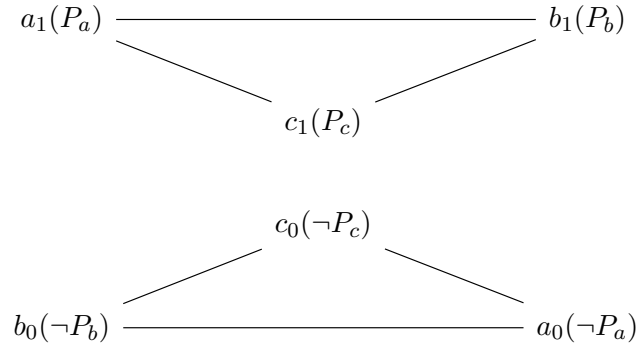


Figure 55: Action describing “ $a$  sends their bit value to both  $b$  and  $c$ .”

various protocols with and without revision. This would let us evaluate in an empirical sense when and how revision gives learners an advantage, which would be an exciting philosophical argument.

The last advantage of the action model presentation of protocols is that we might be able to leverage the topological properties of simplicial action models in the description of protocols. Leveraging such topological properties is a core part of the application of combinatorial topology to distributed computing. [23] However, ours is not the first example of a topological characterization of protocols themselves (not just the inputs and outputs). The paper “A topological perspective on distributed network algorithms”, for example, has already done this. [12] We will briefly recount a core example from that paper. Consider the knowledge complex drawn in Figure 54. Here, agents have a privately held bit value, and there is total uncertainty. The goal will be for agents to send messages in such a sequence that they could plausibly come to consensus. There is no need for the majority to win, either. If two agents initially have 0, but ultimately all agents agree to flip to 1, this is considered a success.<sup>40</sup> Hence, the desired output complex consists of two disconnected facets, drawn in Figure 55.

In [12], they consider two different protocols as potential solutions to this problem. The first involves a cycle of messages:  $a$  signals  $b$  her value,  $b$  signals  $c$  her value, and  $c$  signals  $a$  her value. It is very important to note that they consider protocols in the “know-all” setting. Hence, all agents are aware that this is the protocol, and there is no uncertainty. This is called the “structure awareness” assumption. Moreover, they effectively describe protocols geometrically as what we would call the application of a protocol to an action. That is, for an input  $I$  and

<sup>40</sup>However, if all agents have 0, they must agree on 0 in the end.

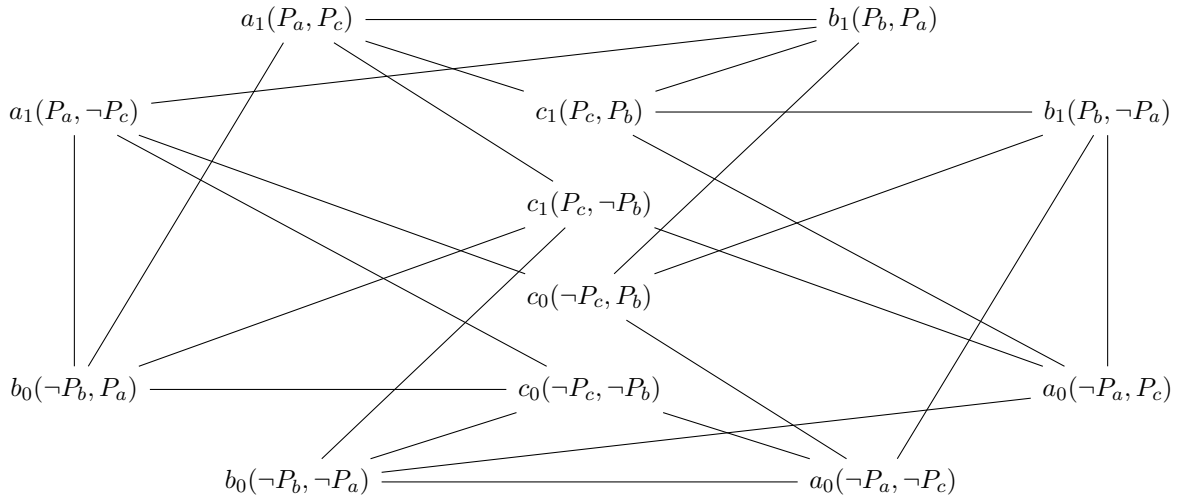


Figure 56: Knowledge complex for the protocol where  $a$  signals  $b$ ,  $b$  signals  $c$ , and  $c$  signals  $a$ .

an action  $A$ , their definition of a “protocol complex” is effectively what we would call  $I[A]$ , though they do not separately specify the actions. Hence, they define a “protocol complex” as the result of applying a sequence of signals to an input. Moreover, because of the “structural awareness” assumption, there must be a bijection between the facets of the input and the facets of the protocol complex. This is because they assume that there is common knowledge of what the input worlds are, what the messages are, who sends them, and who receives them. So every agent knows exactly what messages will be sent and received at any given world. Obviously, action model descriptions of protocols, as given above, have no such restrictions, even in merely the knowledge setting.

The result of applying the cyclical sequence of messages to Figure 54 is drawn in Figure 56. Each perspective is split, depending on whether they heard a 1 or a 0. However, crucially, the complex remains connected.

By contrast, consider instead the sequence of signals “ $a$  signals both  $b$  and  $c$ ”. Agent  $a$ ’s perspective would not duplicate, as they learn nothing. But agent  $b$ ’s and agent  $c$ ’s both do, as they learn  $a$ ’s bit value. This result of this is drawn in Figure 57. Crucially, this complex is disconnected. Because the target, too, is disconnected, it is necessary for the analysis given in [12] that the protocol complex be disconnected as well. Hence, the first cyclical sequence of signals does not solve the task of consensus, but  $a$  broadcasting does. This application of connectivity leverages this topological depiction of protocols.

As mentioned before, [12] is viewing “protocol complexes” as what we would call the application of an action to an input. So, let’s draw these two protocol complexes instead as actions. Consider first the cycle, where  $a$  signals  $b$ ,  $b$  signals  $c$ , and  $c$  signals  $a$ . Considering both sender and receiver behavior, each agent has 4 perspectives: they either send a 1 or 0, and they either receive a 1 or 0. In fact, there is no difference between our action and the protocol complex drawn in Figure 56, save for how we label postconditions. This is drawn in Figure 58. One can check that the application of this action to the input is trivial, as it just redraws the action. Moreover, the application of this action to the input in Figure 54 does yield the model given in Figure 57, which would be as desired.

The action for the other protocol is much more interesting by contrast. There are two  $a$  perspectives, depending on what  $a$  sends. For  $b$  and  $c$  each, there are also two perspectives, depending on what they receive. Since  $a$  does not receive anything, and  $b$  and  $c$  do not send anything, this fully characterizes the set of perspectives. This is drawn in Figure 59. Note that this complex is disconnected, just as the target is. So, possibly, we can leverage topology similar

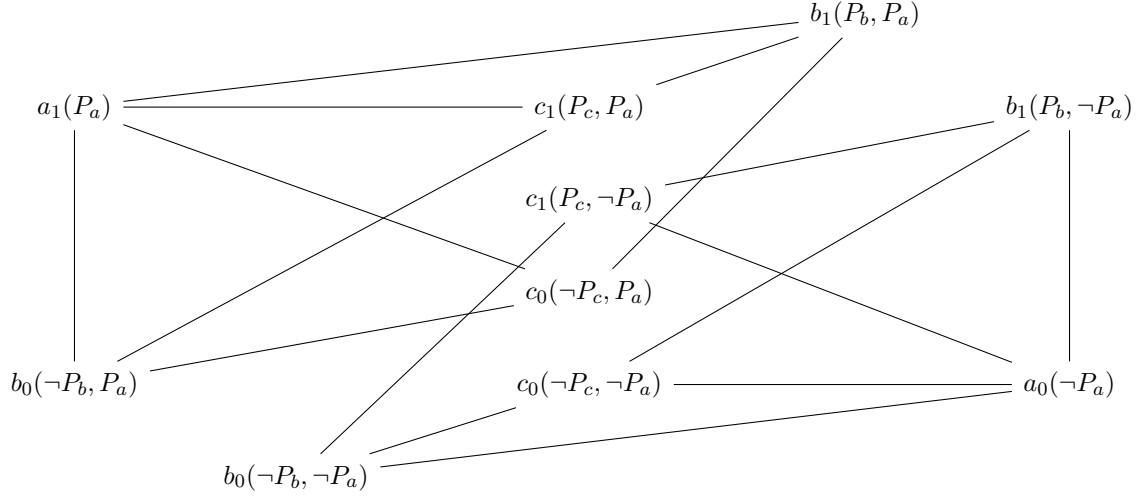


Figure 57: Knowledge complex for the protocol where  $a$  signals  $b$  and  $c$ .

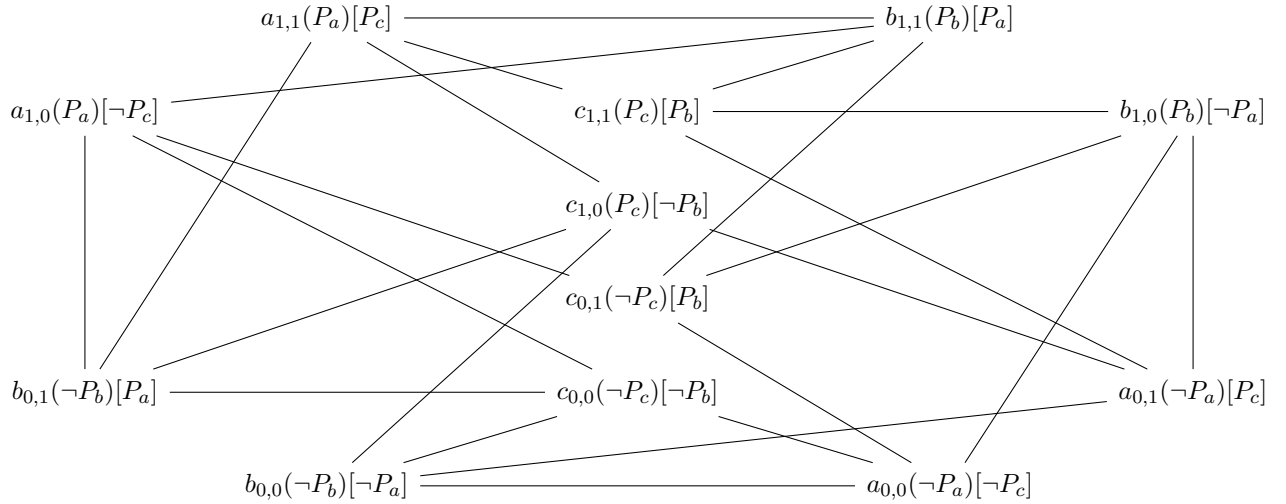


Figure 58: Action model for the protocol where  $a$  signals  $b$ ,  $b$  signals  $c$ , and  $c$  signals  $a$ .

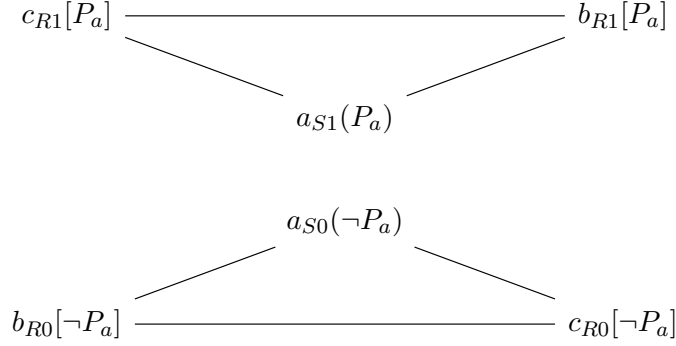


Figure 59: Action model for the protocol where  $a$  signals  $b$  and  $c$ .

to how it is done in [12]. However, much more work needs to be done to generalize this.

However, it should be noted that action models do seem to be far more general than the definition of protocol considered in [12]. As mentioned above, the facets of a protocol complex must be in bijection with the input.[12] As we have seen, our action model definition of protocol has no such restriction. This is in part because we are not beholden to the same “awareness assumption” as they are in that paper. Hence, agents can be uncertain of whether or not a particular message was sent, or of its contents, or of who the sender or receiver is, and so no. In belief protocols, agents can even have a false conception of what messages are being sent. These generalizations directly expand on the work in [12]. Consider the following passage from the conclusion of [12], where they consider future work:

“For instance, fully capturing the popular local model requires removing the structure awareness assumption, and studying the details of how the protocol complex evolves round after round.”[12]

It is possible that, by analyzing protocols using actions, and in particular considering belief protocols, we have presented a model with sufficient generality to capture the “local model”. However, more work would need to be done to check this. However, it is clear that we have generalized the conception of protocol given in [12]. Moreover, the action presentation of protocols still equips them with a topology, which might be able to be leveraged in familiar ways, such as what is briefly explored above, and possibly novel ways we are not yet aware of. Following up on this is an exciting dimension for future research.

## 4.7 Conclusion

In this chapter, we set out to give a new definition for action models in the setting of simplicial semantics. In particular, we wanted a notion of action which could apply to models for belief as in [26], so that we could incorporate the concept of revision from Chapter 3. As a result, we are able to, on the one hand, describe many existing, albeit simple, simplicial protocols as action models. Forthcoming work with the NASA Langley Research Center also uses these definitions to define a novel protocol. This protocol is a gossip protocol in the sense of [8], and using this we are able to explore the possibilities of using revision, as defined here, as a further means of fault-tolerance in distributed protocols. However, exploring the use of revision as a means of fault-tolerance is still ongoing work.

This leads us to directions of future work. The first major direction for future work would be to implement action models in simplicial complexes in a simulation. In particular, a program which could compute action updates would be very helpful. Even small examples get large very quickly, and being able to work with more examples would help us to answer many questions, some of which we have already posed.



Firstly, does belief revision actually help with fault tolerance? If so, how? In what contexts? Answering this would require comparing and contrasting many protocols, ideally being able to compare protocols who only differ in whether or not they use the revision updating procedure. Getting clear on exactly how revision affects a protocol, especially after many stages, would also be philosophically exciting and would be a means of empirically testing the revision in a way that could speak to its value as a model of learning.

Secondly, what are the limitations of representing protocols using actions? A question this chapter has not addressed is the actual, provable, scope of using action models to represent protocols. Are there protocols which cannot be represented using action models? If so, which ones? Why? A place to start would seem to be to take the definition of protocol from [23] and try to see if one can prove that all such protocols could be defined as actions.

## 4.8 Proofs

**Theorem 4.7.** *Let  $\mathcal{M}$  be a simplicial model for knowledge and  $A$  a simplicial action model. Then the output model  $\mathcal{M}[A]$  also satisfies the UCF property.*

*Proof.* Fix  $b \in Ag$ , and  $F \in \mathcal{F}(S[A])$ . We first want to show that there is  $(x, y) \in F$  such that  $V[A](x, y) = b$ . Note that  $\pi_0(F) \in \mathcal{F}(S)$ . Hence, since  $S$  is UCF, there is  $x \in \pi_0(F)$  such that  $V(x) = b$ . Fix  $y$  such that  $(x, y) \in F$ . Then, by the definition of  $V[A]$ ,  $V[A](x, y) = b$ , as desired.

Suppose  $(x, y), (x', y') \in F$  and  $V[A](x, y) = V[A](x', y') = b$ . We need to show that  $(x, y) = (x', y')$ . Note that by the definition of  $V[A]$ ,  $V(x) = V(x') = b$ . Similarly, by the definition of  $N[A]$ ,  $V_A(y) = V_A(y') = b$ . Because  $x, x' \in \pi_0(F) \in \mathcal{F}(S)$ , and  $S$  is UCF, we have that  $x = x'$ . Because  $y, y' \in \pi_1(F) \in \mathcal{F}(S_A)$ , and  $S_A$  is UCF, we have that  $y = y'$ . The result follows.  $\square$

**Theorem 4.8.** *Let  $I$  be a simplicial model for knowledge and  $A$  an action model. The tuple  $(I, I[A], \Xi)$  is a Simplicial Protocol.*

*Proof.* Suppose  $F \subseteq G$ . Then suppose  $X \in \Xi(F)$ . Then  $\pi_0(X) \subseteq F$ . Then  $\pi_0(X) \subseteq G$ , so  $X \in \Xi(G)$ , as desired.

Suppose  $X \in \Xi(F \cap G)$ . Then  $\pi_0(X) \subseteq F \cap G$ . So,  $\pi_0(X) \subseteq F$  and  $\pi_0(X) \subseteq G$ , and we get that  $X \in \Xi(F) \cap \Xi(G)$ , as desired. The reverse is identical.

In order to show that  $\Xi$  is chromatic we will first establish the result for singletons. Let  $n \in I$  be a node and fix  $a \in Ag$  such that  $V(n) = a$ . Then in particular, if  $X \in \Xi(\{n\})$ , then  $\pi_0(X) = n$ . It follows from the UCF property that  $|X| = 1$ , and fixing  $(n, y) \in X$ ,  $V[A]((n, y)) = V(n) = a$ .

Now we generalize this to larger faces. For each  $n \in F$  where  $F$  is a face in  $I$ , we know that  $V[A](\Xi(\{n\})) = V(n)$ . Suppose  $X \in \Xi(F)$ . Then, for each  $a \in Ag$  such that there's an  $(x, y) \in X$  such that  $V[A]((x, y)) = V(x) = a$ , because  $\pi_0(X) \subseteq F$ ,  $x \in F$  and thus there is an  $a$ -perspective in  $F$ . This suffices to show chromaticity.

By construction,  $\bigcup_{\sigma \in \mathcal{I}} \Xi(\sigma) \subseteq I[A]$ . Suppose that  $X$  is a face in  $I[A]$ . Take  $\sigma = \pi_0(X)$ . By the definition of  $I[A]$ ,  $\sigma$  is a face in  $I$ . Fix  $Y$  any facet containing  $X$ . This suffices to show  $\bigcup_{\sigma \in \mathcal{I}} \Xi(\sigma) = I[A]$ .  $\square$

**Theorem 4.9.** *Let  $\mathcal{M}$  be a simplicial model for belief and  $A$  a simplicial action model for belief. Then the output model  $\mathcal{M}[A]_B$  also satisfies the UCF property.*

*Proof.* Fix  $b \in Ag$ , and  $F \in \mathcal{F}(S_a[A]_B)$ . We first want to show that there is  $(x, y) \in F$  such that  $V[A]_B(x, y) = b$ . Note that  $\pi_0(F) \in \mathcal{F}(S_a)$ . Hence, since  $S_a$  is UCF, there is  $x \in \pi_0(F)$  such that  $V(x) = b$ . Fix  $y$  such that  $(x, y) \in F$ . Then, by the definition of  $V[A]_B$ ,  $V[A]_B(x, y) = b$ , as desired.

Suppose  $(x, y), (x', y') \in F$  and  $V[A]_B(x, y) = V[A]_B(x', y') = b$ . We need to show that  $(x, y) = (x', y')$ . Note that by the definition of  $V[A]_B$ ,  $V(x) = V(x') = b$ . Similarly, by the

definition of  $N[A]_B$ ,  $V_A(y) = V_A(y') = b$ . Because  $x, x' \in \pi_0(F) \in \mathcal{F}(S_a)$ , and  $S_a$  is UCF, we have that  $x = x'$ . Because  $y, y' \in \pi_1(F) \in \mathcal{F}(S_{a,A})$ , and  $S_{a,A}$  is UCF, we have that  $y = y'$ . The result follows.  $\square$

**Theorem 4.10.** *Let  $I$  be a simplicial model for belief and  $A$  an action model. The tuple  $(I, I[A]_B, \{\Xi_a\}_{a \in Ag})$  is a Belief Protocol.*

*Proof.* Suppose  $F$  and  $G$  are faces of  $S_a$  and  $F \subseteq G$ . Then suppose  $X \in \Xi_a(F)$ . Then  $\pi_0(X) \subseteq F$ . Then  $\pi_0(X) \subseteq G$ , so  $X \in \Xi_a(G)$ , as desired.

In order to show that each  $\Xi_a$  is chromatic we will first establish the result for singletons. Let  $n \in I$  be a node and fix  $a \in Ag$  such that  $V(n) = a$ . Then in particular, if  $X \in \Xi_a(\{n\})$ , then  $\pi_0(X) = n$ . It follows from the UCF property that  $|X| = 1$ , and fixing  $(n, y) \in X$ ,  $V[A]((n, y)) = V(n) = a$ .

Now we generalize this to larger faces. For each  $n \in F$  where  $F$  is a face in  $I$ , we know that  $V[A](\Xi_a(\{n\})) = V(n)$ . Suppose  $X \in \Xi_a(F)$ . Then, for each  $a \in Ag$  such that there's an  $(x, y) \in X$  such that  $V[A]((x, y)) = V(x) = a$ , because  $\pi_0(X) \subseteq F$ ,  $x \in F$  and thus there is an  $a$ -perspective in  $F$ . This suffices to show chromaticity.

Let  $F$  and  $G$  be two faces of  $S_a$ . To verify the intersection property, we can see that the only interesting case is when  $\pi_a(F \cap G) \neq \emptyset$ . If  $\pi_a(F \cap G) = \emptyset$ , then,  $\Xi_a(\pi_a(F \cap G)) = \pi_a(\Xi_a(F)) \cap \pi_a(\Xi_a(G)) = \emptyset$ . Suppose  $X \in \Xi_a(\pi_a(F \cap G))$ . Then  $\pi_0(X) \subseteq \pi_a(F \cap G)$ . Note that this implies that  $\pi_0(X)$  is a singleton containing the unique  $a$ -colored node in both  $F$  and  $G$ . So,  $\pi_0(X) \subseteq F$  and  $\pi_0(X) \subseteq G$ , and we get that  $X \in \pi_a(\Xi_a(F)) \cap \pi_a(\Xi_a(G))$ , as desired. Suppose now that  $X \in \pi_a(\Xi_a(F)) \cap \pi_a(\Xi_a(G))$ . Then  $X \in \pi_a(\Xi_a(F))$ . So, by the UCF property,  $X$  is an  $a$ -colored singleton in  $I_a[A]$ . By the definition of  $\Xi_a$ , there is  $X'$  such that  $X \subseteq X'$  and  $\pi_0(X') \subseteq F$ . Since  $X$  contains the unique  $a$ -colored node of  $X'$ , we can conclude that  $\pi_0(X) \subseteq F$  is the singleton set containing the unique  $a$ -colored node of  $F$ . The same argument will show that  $\pi_0(X)$  is the singleton set containing the unique  $a$ -colored node of  $G$ . So,  $\pi_0(X)$  is the singleton set containing the unique  $a$ -colored node of  $F \cap G$ . Hence,  $\pi_0(X) = \pi_a(F \cap G)$ . By definition, then,  $X \in \Xi_a(\pi_a(F \cap G))$ . This suffices to show the intersection property.

By construction,  $\bigcup_{\sigma \in \mathcal{I}} \Xi_a(\sigma) \subseteq I[A]$ . Suppose that  $X$  is a face in  $I[A]$ . Take  $\sigma = \pi_0(X)$ . By the definition of  $I[A]$ ,  $\sigma$  is a face in  $I$ . Fix  $Y$  any facet containing  $X$ . This suffices to show  $\bigcup_{\sigma \in \mathcal{I}} \Xi_a(\sigma) = I[A]$ .  $\square$

**Theorem 4.11.** *Let  $\mathcal{M}$  be a simplicial model for belief and  $A$  a simplicial action model for belief. Then the output model  $\mathcal{M}[A]_{BR}$  also satisfies the UCF property.*

*Proof.* This follows immediately from the fact that, by assumption, each facet in  $S_a[A]_{BR}$  is given by  $\mathcal{R}_a(X)$  for some  $X \in \mathcal{F}^a(\mathcal{M}[A]_B)$ . Because  $\mathcal{R}_a(X) \subseteq \mathcal{F}_a(2^{N[A]_{BR}})$  for all  $X \in \mathcal{F}^a(\mathcal{M}[A]_B)$ , and the facets in  $\mathcal{F}_a(2^{N[A]_{BR}})$  are assumed to be UCF, the result follows.  $\square$

**Theorem 4.12.** *Let  $I$  be a simplicial model for belief and  $A$  an action model. Define  $\Xi_{a,BR}$  similarly to  $\Xi_a$ :*

$$\Xi_{a,BR}(F) := \{X \in I_a[A]_{BR} \mid \forall X' \in R_a^{-1}(X)(\pi_0(X') \subseteq F)\}$$

*Then, the tuple  $(I, I[A]_{BR}, \{\Xi_{a,BR}\}_{a \in Ag})$  is a Belief Protocol.*

*Proof.* Suppose  $F$  and  $G$  are faces of  $S_a$  and  $F \subseteq G$ . Then suppose  $X \in \Xi_{a,BR}(F)$ . Then for all  $X' \in R_a^{-1}(X)$   $\pi_0(R_a^{-1}(X')) \subseteq F$ . Then  $\pi_0(R_a^{-1}(X')) \subseteq G$ , so  $X \in \Xi_{a,BR}(G)$ , as desired.

In order to show that each  $\Xi_{a,BR}$  is chromatic we will first establish the result for singletons. Let  $n \in I$  be a node and fix  $a \in Ag$  such that  $V(n) = a$ . Then in particular, if  $X \in \Xi_{a,BR}(\{n\})$ , then for all  $X' \in R_a^{-1}(X)$ ,  $\pi_0(X') = n$ . It follows from the UCF property and the fact that  $R_a^{-1}$  preserves colors that  $|X| = |X'| = 1$ . Fix  $y$  such that  $(n, y) \in X'$  and  $x$  and  $z$  such that  $(x, z) \in X$ . By the definition of  $R_a^{-1}$ ,  $V[A]_{BR}((x, z)) = V[A]_{BR}((n, y)) = V(n) = a$ .

Now we generalize this to larger faces. For each  $n \in F$  where  $F$  is a face in  $I$ , we know that  $V[A]_{BR}(\Xi_{a,BR}(\{n\})) = V(n)$ . Suppose  $X \in \Xi_{a,BR}(F)$ . Consider each  $a \in Ag$  such that

there's an  $(x, y) \in X$  such that  $V[A]_{BR}((x, y)) = V(x) = a$ . For all  $X' \in R_a^{-1}(X)$ , we have that  $\pi_0(X') \subseteq F$ . Fix  $(z, w) \in X'$ . Then  $z \in F$  and because  $a = V(x) = V[A]_{BR}(x, y) = V[A]_{BR}(z, w) = V(z)$ , there is an  $a$ -perspective in  $F$ . This suffices to show chromaticity.

Let  $F$  and  $G$  be two faces of  $S_a$ . To verify the intersection property, we can see that the only interesting case is when  $\pi_a(F \cap G) \neq \emptyset$ . If  $\pi_a(F \cap G) = \emptyset$ , then, by the singleton case of chromaticity, it follows that  $\Xi_{a, BR}(\pi_a(F \cap G)) = \pi_a(\Xi_{a, BR}(F)) \cap \pi_a(\Xi_{a, BR}(G)) = \emptyset$ .

It will be useful for the reasoning below to realize the following. If  $X$  is a singleton,  $a$ -colored node, then  $R_a^{-1}(X) = X$ . Suppose  $X' \in R_a^{-1}(X)$ . Then by definition there is  $Y \in \mathcal{F}(S_a[A]_{BR})$  such that  $X \subseteq Y$  and there exists  $Y' \in R_a^{-1}(Y)$  where  $X' \subseteq Y'$ . By the definition of  $R_a$ ,  $\pi_a(Y) = \pi_a(Y')$ . So, by UCF, and the fact that  $R_a^{-1}$  preserves colors,  $X' = \pi_a(Y')$  and  $X = \pi_a(X') = \pi_a(Y)$ . Suppose  $X \in \Xi_{a, BR}(\pi_a(F \cap G))$ . Then for all  $X' \in R_a^{-1}(X)$ ,  $\pi_0(X') \subseteq \pi_a(F \cap G)$ . Note that this implies that  $\pi_0(X')$  is a singleton containing the unique  $a$ -colored node in both  $F$  and  $G$ . So,  $\pi_0(X') \subseteq F$  and  $\pi_0(X') \subseteq G$ , and we get that  $X \in \pi_a(\Xi_{a, BR}(F)) \cap \pi_a(\Xi_{a, BR}(G))$ , as desired. Suppose now that  $X \in \pi_a(\Xi_{a, BR}(F)) \cap \pi_a(\Xi_{a, BR}(G))$ . Then  $X \in \pi_a(\Xi_{a, BR}(F))$ . So, by the UCF property,  $X$  is an  $a$ -colored singleton. By the fact that  $R_a^{-1}$  preserves color, for all  $X' \in R_a^{-1}(X)$ ,  $X'$  is an  $a$ -colored singleton. By the definition of  $\Xi_{a, BR}$ ,  $\pi_0(X') \subseteq F$ . Since  $X'$  contains the unique  $a$ -colored node of  $X$ , we can conclude that  $\pi_0(X') \subseteq F$  is the singleton set containing the unique  $a$ -colored node of  $F$ . The same argument will show that  $\pi_0(X')$  is the singleton set containing the unique  $a$ -colored node of  $G$ . So,  $\pi_0(X')$  is the singleton set containing the unique  $a$ -colored node of  $F \cap G$ . Hence,  $\pi_0(X') = \pi_a(F \cap G)$ . By definition, then,  $X \in \Xi_{a, BR}(\pi_a(F \cap G))$ . This suffices to show the intersection property.

By construction,  $\bigcup_{\sigma \in \mathcal{I}_1} \Xi_{a, BR}(\sigma) \subseteq I_a[A]_{BR}$ . Suppose that  $X$  is a face in  $I[A]_{BR}$ . We first need to show that  $R_a^{-1}(X)$  is nonempty. Fix  $Y \in \mathcal{F}(S_a[A]_{BR})$  such that  $X \subseteq Y$ . Then by definition, there is  $Y' \in R_a^{-1}(Y)$ . Fix  $X'$  to be the subset of  $Y'$  which shares the same colors as  $X$ . Then  $X' \in R_a^{-1}(X)$ , as desired. Because  $Y' \in \mathcal{F}^a(N \times N_A)$ ,  $\pi_0(Y') \in \mathcal{F}(I_a)$ . So, because  $X' \subseteq Y'$ ,  $\pi_0(X') \in I_a$ . It follows that  $X \in \Xi_{a, BR}(\pi_0(X'))$ . This suffices to show  $\bigcup_{\sigma \in \mathcal{I}} \Xi_{a, BR}(\sigma) = I[A]_{BR}$ .  $\square$

## 5 Conclusion

In this thesis, we extended both the conceptual foundations for and applications of simplicial semantics for epistemic modal logic. [11, 14, 18, 17, 21, 19, 20, 28] In Chapter 2, we established a novel approach for modeling belief in simplicial complexes. The key difference between modeling belief, and modeling knowledge, is that belief cannot be factive. That is, the axiom schema **T**, which states that, if an agent  $a$  believes that  $\varphi$ , then  $\varphi$  is true, e.g.,  $B_a\varphi \rightarrow \varphi$  for all  $a \in Ag$ , cannot be sound. In the usual simplicial semantics, the soundness of **T** is unavoidable. This is because an agent  $a$  “knows”  $\varphi$  at a facet  $X$ , that is,  $K_a\varphi$  is true at  $X$ , if and only if  $\varphi$  is true in all facets that contain  $a$ ’s perspective at  $X$ , e.g., true at all  $Y$  such that  $\pi_a(Y) = \pi_a(X)$ . Since, trivially,  $\pi_a(X) = \pi_a(X)$ , factivity is sound.

Our solution to this problem was to allow each agent to have their own simplicial complex, called a *belief-complex*. We denote  $a$ ’s belief complex as  $S_a$ . The truth condition for belief then is as follows:  $B_a\varphi$  is true at a facet  $X$ <sup>41</sup> if and only if, for all facets  $Y$  in  $S_a$  such that  $\pi_a(Y) = \pi_a(X)$ ,  $\varphi$  is true at  $Y$ . The appeal that the  $Y$  comes from  $S_a$ , rather than a complex shared by all the agents, allows for contradictions of factivity. This is easily seen when the facet we start at,  $X$ , is not a facet in  $S_a$ .

This is not the first attempt to model belief in the simplicial literature. [14, 11] However, it has an advantage over each of the previous attempts. Unlike the work in [14], our models validate the axiom “Knowledge Implies Belief”,  $K_a\varphi \rightarrow B_a\varphi$ . Furthermore, if one restricts the belief models of [14] to those which validate  $K_a\varphi \rightarrow B_a\varphi$ , one ends up validating  $K_a\varphi \leftrightarrow B_a\varphi$ , which is much too strong. Similarly, the work in [11] allows for models with more than one perspective for a given agent on a facet.[11] The UCF assumption, which contradicts this, is conceptually appealing, and a standard assumption across much of the literature.[14, 19] In [11], allowing that facets contain a plurality of perspectives for any given agent is crucial for ensuring that **T** is not sound. If one restricts the models in [11] to those which have exactly one node for any agent, one again ends up validating  $K_a\varphi \leftrightarrow B_a\varphi$ .

In Chapter 3, we leveraged the models for belief developed in Chapter 2 to gain a new conceptual insight. In the classic paper “Probabilities of Conditionals and Conditional Probabilities”, David Lewis introduced a novel semantics for the Stalnaker conditional using the notion of “conditionalization”. [24] Under this mechanism, when agents learn that  $\varphi$  is true, they do not simply erase the worlds where  $\varphi$  is false. Instead, they may replace some of the worlds where  $\varphi$  is false with “nearby” or “similar” worlds where  $\varphi$  is true. One issue with this idea, however, is that it is often quite unclear, in practice, which worlds should be “nearer” or “further” from any other.

We showed that Simplicial Semantics offers a potential solution to this problem. First, we modified the semantics from Chapter 2 so that atomic propositions are assigned directly to the nodes of the simplicial complex as opposed to the facets. This is the standard approach to simplicial semantics in the literature. [14] This is also more conceptually in-line with what we call the “perspective-perspective”. This is the epistemic foundation of simplicial semantics as we see it, and articulates that the primary epistemic “object” under consideration in simplicial semantics is an agent “point-of-view” or perspective. These “perspectives” correspond to the nodes of a simplicial complex, and we interpret the atomic propositions assigned to the nodes as the content of the agent’s perspective, which we often cache out as what an agent is seeing *directly* in that perspective.

Then, we observed that, if nodes are perspectives, then facets, which are the stand-in for worlds in simplicial semantics, are *made of* perspectives. If worlds are made of perspectives, this gives us a notion of “nearness” that we can appeal to: Two worlds are nearer, or more similar, if they share more perspectives. We formally specified this notion of revision, as well as independently conceptually motivated it.

Finally, in Chapter 4 we gave a novel application for this notion of revision. We first ac-

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<sup>41</sup> $X$  need not be a facet from  $S_a$ , or any  $S_i$  in particular. Rather,  $X$  is a facet in some background complex, of which each  $S_a$  is a subcomplex.

$$\mathcal{M}_0 : \quad a_0(P) \text{ — } b_0(P) \qquad \mathcal{M}_1 : \quad a_0 \text{ — } b_0(P)$$

Figure 60: Two models, with a pair of perspectives each, forming a single facet in each. In  $\mathcal{M}_0$ , both  $a_0$  and  $b_0$  have  $P$  assigned to them. In  $\mathcal{M}_1$ , only  $b_0$  is assigned  $P$ .

knowledge the history of attempting to use action models in distributed computing. [19, 14, 2, 25, 29, 13] In this literature, an action is applied to an input in order to model a single “step” of a distributed protocol. Seeing as simplicial semantics itself has its origins in combinatorial topological models of distributed protocols, our goal was to marry these two projects by defining actions for simplicial models for epistemic modal logic. [20, 28, 14, 19, 17, 18, 23] In fact, defining action models for simplicial semantics has already been attempted, on the basis of this natural connection. [14, 19] However, to our knowledge, no attempt has been made to use this connection to actually give an account of already existing protocols in the literature concerning applying combinatorial topology to distributed computing. Moreover, we sought to leverage the work we’ve already done in the Chapter 3 to give a novel way of understanding how agents learn in a distributed protocol.

We started by defining perspective-based models for both knowledge and belief. We then introduced definitions for actions in the knowledge setting, similar to the work done in [14]. We then proved that the pair of models given by an input, and the output of an action applied to that input, as well as an appropriate morphism on faces, specified a “protocol” in the sense of the textbook “Distributed Computing Through Combinatorial Topology”. [23] We then use this definition of action model to formally give examples of existing basic distributed protocols as actions.

We then gave a new definition for an action model in the context of belief models. However, the definition of “protocol” used in the previous proofs cannot apply here. That is because that definition presumes the input and output are models for knowledge, and, hence, have only a single complex, rather than one for every agent. So, we modified the definition of protocol to what we call a “belief protocol”, and proved that belief actions are belief protocols in the expected way. We concluded by defining a belief revision action, which specifies a different set of output facets after applying a belief action using the definition of revision from Chapter 3. We concluded by proving that belief revision actions are also belief protocols.

We will conclude with a discussion of future work. The first idea for future work is also mentioned in Chapter 3. Consider the pair of models drawn in Figure 60. Using the translation into Kripke models from Chapter 3, these correspond to the same model. This makes sense, as the single facet  $X := \{a_0, b_0\}$  satisfies exactly the same formulas in both cases.

However, conceptually, we understand these models to nevertheless be saying something different. At  $X$  in  $\mathcal{M}_0$ , we are saying that agent  $a$  knows/believes  $P$  *because it is in her perspective*. At  $X$  in  $\mathcal{M}_1$ , by contrast, we are saying that  $a$  knows/believes  $P$  because all of the collections of perspectives she considers possible are ones which entail  $P$ . Nothing in our language is able to cache out this difference. However, expanding the language to account for this seems to be simple. One could add a modal operator  $A_x$  for all  $x \in Ag$  which applies only to literals, and for a model  $\mathcal{M}$  and facet  $X$ , we could say that  $\mathcal{M}, X \models A_x P$  iff  $L(\pi_x(X))(P) = 1$ , and  $\mathcal{M}, X \models A_x \neg P$  iff  $L(\pi_x(X))(P) = 0$ . It would follow that  $\mathcal{M}_0, X \models A_b P$ , but  $\mathcal{M}_1, X \not\models A_b P$ . As such, this would help pull apart the intuitive difference between  $\mathcal{M}_0$  and  $\mathcal{M}_1$ .

It seems that this could be a fruitful connection between simplicial semantics and something like awareness logic.[15] One can read  $A_x(P)$  as stating that “ $x$  is aware of  $P$ ”, and in doing so, think of the literals true at a perspective as constituting what the agent is “aware” of at that perspective. Because what is and is not assigned to perspectives, and hence what agents are and are not aware of, plays such a major role in our conception of revision, this might also be a way of connecting awareness and revision.

A further area for future work, also mentioned in Chapter Two, is to further push the notion of revision into nearby bodies of research. For example, it would be nice to explore

more connections between this semantics for revision and other presentations of belief revision. In particular, future work should explore connections between our semantics and the “AGM” presentation of revision. [1, 22] As a first step, we should check if and how we can use our semantics to model multi-agent belief revision. [3] Another possibility would be to use the same perspective-based structure of worlds in simplicial semantics to model belief merging instead of belief revision. We can therefore phrase the question of merging as follows: Given a model, and two perspectives in the model, what is the “best” set of worlds one could create using the two sets of worlds containing each of the two perspectives, respectively. A simple answer is obviously the intersection. Given two perspectives, call them  $x_1$  and  $x_2$ , let  $S_i$  be the set of facets in a model  $\mathcal{M}$  containing  $x_i$ . Then merging the perspectives  $x_1$  and  $x_2$  could be given by the set of worlds  $S_1 \cap S_2$ . But we could obviously say something more nuanced. Perhaps  $X \in S_1$  but  $X \notin S_2$ . However, there is a world  $X'$  that differs from  $X$  in only one perspective, and  $X' \in S_1 \cap S_2$ . Perhaps the merge set should include worlds like  $X'$ . Similar to our discussion of announcements, one can also modify this so that the update is  $S_1 \cap S_2$ , unless this intersection is empty, in which case one selects some set of nearby worlds to  $S_1 \cup S_2$ .

A third idea I want to explore is the relationship between action models and other presentations of protocols. In particular, from an action model, it should be possible to define a “Runs-and-Systems” model as in the textbook “Reasoning About Knowledge”. [16] Recall the definition of such a model. Let  $L_e$  be the possible environment states, and for each  $i \in Ag$ , let  $L_i$  be the possible states for  $i$ . Then  $\mathcal{G} = L_e \times (\times_{i \in Ag} L_i)$ , and  $R$  is a function from time (in our case,  $\mathbb{N}$ ) to  $\mathcal{G}$ . So, given a Simplicial Model for Knowledge as input  $\mathcal{M}$ , and a Simplicial Action Model  $A$ , one can compute each iterated output of repeatedly applying  $A$  to  $\mathcal{M}$ . That is,  $\mathcal{M}[A]$ ,  $\mathcal{M}[A][A]$ , and so on. We abbreviate  $n$  applications of  $A$  as  $\mathcal{M}[A]^n$ . We can then think of each successive application of  $A$  as a “time slice” of the model, and perspectives/facets in each “time slice” tell us what the worlds are, and how they should be decomposed into the tuples. More specifically, we can define  $R$  so that  $R(n) = \mathcal{F}(\mathcal{M}[A]^n)$ . Working this out more carefully requires further work. More significantly, finding examples of runs-and-systems models which are not definable by an input and some action would seem to be possible, at least, intuitively, but has not been worked out yet.

Forthcoming work with the NASA Langley Research Center also uses these definitions to define a novel protocol. This protocol is a gossip protocol in the sense of [8], and using this we are able to explore the possibilities of using revision, as defined here, as a further means of fault-tolerance in distributed protocols. However, exploring the use of revision as a means of fault-tolerance is still ongoing work. The best way to test this would seem to be implementing the application of actions in a programming language like Python, as suggested in the conclusion for Chapter 4. Repeated application of actions tends to create very large models very quickly. As a result, such simulations are necessary for testing more complex examples. However, more excitingly, such a testing environment would allow us to ask many new and interesting questions. For example:

- Can we categorize initial configurations for which an agent that learns by revision tends to perform better?
- Can we identify protocols that perform better, under some specified metric, when the agent learns via revision instead of classically?
- How does revision affect or change many existing, traditional protocols?
- Comparing the same protocol across different initial configurations, can we find protocols which don’t succeed *always* but do well *typically*? Could this be a defense for some known but generally discarded protocols?

In the near future, we hope that beginning to answer all of these questions and others will be within reach.

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